

भारतीय मानक

IS 3854 : 2023

Indian Standard

घरेलु और समान कार्यों के लिए स्विच — विशिष्टि

(तीसरा पुनरीक्षण)

Switches for Domestic and Similar Purposes — Specification

(Third Revision)

ICS 29.120.40

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Electrical Wiring Accessories Sectional Committee, ETD 14

FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Wiring Accessories Sectional Committee had been approved by the Electrotechnical Division Council.

This standard was first published in 1966 and revised in 1986 and 1997. This revision has been undertaken to bring it in-line with IEC publication 60669-1 : 2017 'Switches for household and similar fixed electrical installations — Part 1: General requirements' issued by the International Electrotechnical Commission (IEC).

This edition includes the following significant technical changes with respect to the previous edition:

- a) Change of the scope for motor load switches;
- b) Change of the scope for lamp load switches;
- c) Change of the scope for applicability of standard on different types of switches;
- d) New definitions have been added;
- e) Additional marking requirements;
- f) Requirements concerning 13 A switches have been included;
- g) Requirements and test conditions for flexible conductors have been included in **12**;
- h) Requirements for pilot light units have been included;
- j) New test for self-ballasted lamp loads in **19.3**;
- k) Table 23 has been completely redrawn to cover normal, mini and micro-gap switches and renumbered Table 9;
- m) New informative Annex C about the circuit development for **19.3**; and
- n) New informative Annex D including additional requirements for insulation-piercing terminals.

The composition of the Committee, responsible for the formulation of this standard is given at Annex E.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the results of a test or analysis shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

SWITCHES FOR DOMESTIC AND SIMILAR PURPOSES — SPECIFICATION

(Third Revision)

1 SCOPE

1.1 This standard applies to manually operated general-purpose switches for alternating currents only, with a rated voltage not exceeding 440 V and a rated current not exceeding 63 A, intended for household and similar fixed-electrical installations either indoors or outdoors.

1.2 The rated current is limited to 16 A for switches provided with screw less terminals.

NOTE — The rated current is limited to 16 A for switches provided with insulation piercing terminals (IPT's) according to Annex D.

1.3 Switches covered by this document are, where applicable, intended for the control in normal use of all the following loads:

- A circuit for a tungsten filament lamp load;
- A circuit for an externally ballasted lamp load (for example LED, CFL, fluorescent lamp load);
- A circuit for a self-ballasted lamp load (for example LEDi or CFLi);
- A circuit for a substantially resistive load with a power factor not less than 0.95; and
- A single phase circuit for motor load with a rated current not exceeding 3 A at 250 V (750 VA) and a power factor not less than 0.8. This applies to both switches rated not less than 10 A that have not undergone additional tests and to momentary switches rated not less than 6 A that have not undergone additional tests.

1.4 This document also applies to boxes for switches, with the exception of flush mounting boxes for flush-type switches.

NOTE — General requirements for boxes for flush-type switches are given in IS 14772.

1.5 It also applies to switches such as:

- Switches incorporating pilot lights;
- Switches incorporating a time-delay device;
- Combinations of switches and other functions (with the exception of switches combined with fuses); and

- Switches having facilities for the outlet and retention of flexible cables (*see* Annex A).

Switches complying with this standard are suitable for use at ambient temperatures not normally exceeding 35 °C, but occasionally reaching 45 °C.

NOTE — Switches complying with this standard are suitable only in incorporating in equipment in such a way and in such a place that it is unlikely that the surrounding reaches a temperature exceeding 45 °C.

1.6 In locations where special conditions prevail, such as in ships, vehicles and the like and in hazardous locations, for example where explosions are liable to occur, special construction and/or additional requirements may be required.

1.7 This standard does not apply to circuit breakers for household and similar installations, to switches for appliances, in-line cord switches and switches incorporated in cable reels.

2 REFERENCES

The following standard contains provisions, which through reference in this text constitute provisions of the standard. At the time of publication, the edition indicated was valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of the standard indicated below:

<i>IS/IEC No.</i>	<i>Title</i>
632 : 1978	Specification for gamma-BHC (lindane) emulsifiable concentrates (<i>fourth revision</i>)
694 : 2010	Polyvinyl chloride insulated unsheathed and sheathed cables/cords with rigid and flexible conductor for rated voltages up to and including 450/750 v (<i>fourth revision</i>)

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<i>IS/IEC No.</i>	<i>Title</i>	<i>IS/IEC No.</i>	<i>Title</i>
1068 : 1993	Electroplated coatings of nickel plus chromium and copper plus nickel plus chromium — Specification (<i>third revision</i>)	9000 (Part 7/Sec 7) : 2006/IEC 60068-2-75 (1997)	Environmental testing — Part 2-75: Tests — Test Eh: Hammer tests
1401 : 2008	Protection of persons and equipment by enclosures — Probes for verification (<i>second revision</i>)	IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP Code)
1573 : 1986	Specification for electroplated coatings of zinc on iron and steel (<i>second revision</i>)	IS/IEC 60669-2-1 : 2002	Switches for household and similar fixed electrical installations — Part 2-1: Particular requirements — Electronic switches
2824 : 2007/ IEC 60112 : 2003	Method for the determination of the proof and the comparative tracking indices of solid insulating materials (<i>second revision</i>)	IS/IEC 60669-2-2 : 2006	Switches for household and similar fixed electrical installations — Part 2-2: Particular requirements — Electromagnetic remote-control switches (RCS)
9968-1 : 1988	Specification for elastomer insulated cables: Part 1 For working voltages up to and including 1100 volts (<i>first revision</i>)	IEC 60999-1 : 1999	Connecting devices — Electrical copper conductors — Safety requirements for screw-type and screwless-type clamping units — Part 1: General requirements and particular requirements for clamping units for conductors from 0.2 mm ² up to 35 mm ² (included)
10462-1 : 1983	Fictitious calculation method for determination of dimensions of protective coverings of cables: Part 1 Elastomeric and thermoplastic insulated cables	IEC 60669-2-4 : 2004	Switches for household and similar fixed electrical installations — Part 2-4: Particular requirements — Isolating switches
10462-2 : 1991	Fictitious calculation method for determination of dimensions of protective covering of cable: Part 2 Paper insulated cables	IEC 60669-2-5 : 2013	Switches for household and similar fixed electrical installations — Part 2-5: Particular requirements — Switches and related accessories for use in home and building electronic systems (HBES)
IS 11000 (Part 2/Sec 1) : 2018	Fire hazard testing: Part 2 Test methods, Section 1 Glow-wire apparatus and common test procedure (<i>second revision</i>)	IEC 60669-2-6 : 2012	Switches for household and similar fixed electrical installations — Part 2-6: Particular requirements — Fireman's switches for exterior and interior signs and luminaires
12360 : 1988	Voltage bands for electrical installations including preferred voltages and frequency	IS/ISO/IEC Guide 51 : 2005	Safety aspects — Guidelines for their inclusion in standards (<i>first revision</i>)
14772 : 2019	General requirements for enclosures for accessories for household and similar fixed electrical installations — Specification		

3 TERMINOLOGY

Where the terms voltage and current are used, they imply r.m.s values unless otherwise specified. For the purpose of this standard, the following definitions shall apply.

3.1 Switch — A device designed to make or break the current in one or more electric circuit.

3.2 One Operation — The transfer of the moving contacts from one operating positions to another.

3.3 Terminal — The conductive part of one pole composed of one or more clamping unit(s) and insulation if necessary.

3.4 Screw Type Terminals — Terminal intended for the connections, by clamping only, of (an) external rigid or flexible conductors.

3.5 Pillar Terminal — Screw type terminal in which the conductor(s) is (are) inserted into a hole or cavity, where it is clamped under the shank of the screw or screws.

NOTES

1 The clamping pressure can be applied directly by the shank of the screw or through an intermediate clamping part to which pressure is applied by the shank of the screw.

2 Examples of pillar terminals are shown in Fig. 1.

3.6 Screw Head Terminal — Screw-type terminal in which the conductor is clamped under the head of the screw.

NOTES

1 The clamping pressure may be applied directly by the head of the screw or through an intermediate part, such as a washer, a clamping plate or an anti-spread device.

2 Examples of screw head terminals are shown in Fig. 2.

3.7 Stud Terminal — Terminal with screw clamping in which the conductor is clamped under a nut.

NOTES

1 The clamping pressure may be applied directly by a suitable shaped nut or through an intermediate part, such as a washer, a clamping plate or an anti-spread device.

2 Examples of stud terminals are shown in Fig. 2.

3.8 Saddle Terminal — Terminal with screw clamping in which the conductor is clamped under a saddle by means of two or more screws or nuts.

NOTE — Examples of saddle terminals are shown in Fig. 3.

3.9 Lug Terminal — Screw terminal or a stud terminal, designed for clamping a cable lug or bar by means of a screw or nut.

NOTE — Examples of lug terminals are shown in Fig. 4.

3.10 Mantle Terminal — Screw-type terminal in which the conductor is clamped against the base of a slot in a threaded stud by means of a nut.

NOTES

1 The conductor is clamped against the base of the slot by a suitable shaped washer under the nut, by a central peg if the nut is a cap nut, or by equally effective means for transmitting the pressure from the nut to the conductor within the slot.

2 Examples of mantle terminals are shown in Fig. 5.

3.11 Screwless Type Terminal — Connecting device for the connection and subsequent disconnection of a rigid (solid or stranded) or flexible conductor or the interconnection of two conductors capable of being dismantled, the connection being made directly or indirectly, by means of springs, parts of angled, eccentric or conical form, etc without special preparation of the conductor concerned other than removal of insulation.

3.12 Mechanical Time-Delay Device — Device which, through a mechanical auxiliary, operates sometime after the instant at which the conditions which cause it to operate are established.

3.13 Base — Part of the switch retaining current-carrying parts and which normally retains the mechanism in position.

3.14 Rated Voltage — Voltage assigned to the switch by the manufacturer for a specific operating condition of the switch.

3.15 Rated Current — Current assigned to the switch by the manufacturer for a specific operating condition of the switch.

3.16 Push Button Switch — Switch having a single actuator which moves with a single motion when operated by an external, manual force usually applied by the finger or palm of the hand and having stored energy return.

3.17 Momentary Contact Switch — Switch returns its contacts automatically to the initial state after operation.

NOTE — Momentary contact switches are intended to operate bells, electromagnetic remote control switches or time-delay switches.

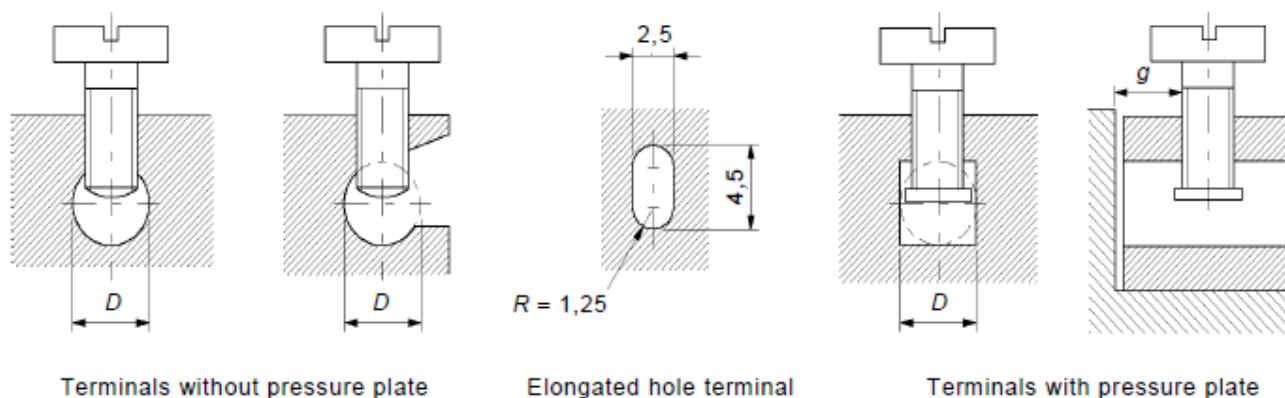
3.18 Momentary Push-Button Switch — Push button which returns automatically to the initial state after operation.

3.19 Cord-Operated Switch — Switch the operating means of which is a cord which has to be pulled in order to change its contact state.

3.20 Switch of Mini-Gap Construction — Switch construction having a clearance between the contacts which is less than 3 mm but is not less than 1.2 mm.

NOTE — Switches of mini-gap construction are for functional purposes and they are not intended to be used for safety isolation purposes.

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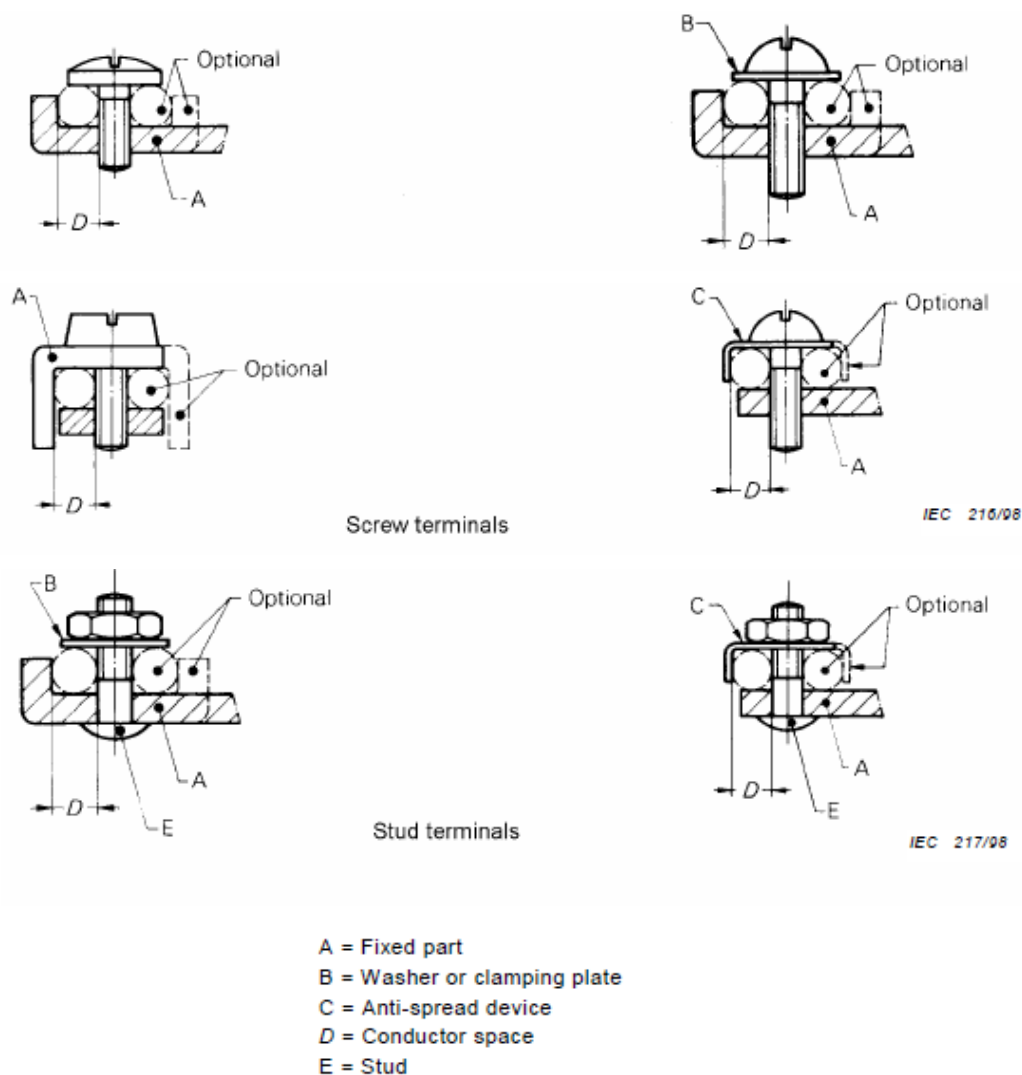
<i>Cross- Section of Conductor Accepted by the Terminal</i>	<i>Minimum diameter D (or minimum dimension) of conductor Space</i>	<i>Minimum distance g between clamping screw and end of conductor when fully inserted</i>		<i>Torque</i>					
mm ²	mm	mm		Nm					
(1)	(2)	(3)		(4)					
				1 ¹⁾		3 ¹⁾		5 ¹⁾	
		One screw	Two screws	One screw	Two screws	One screw	Two screws	One screw	Two screws
Up to 1.5	2.5	1.5	1.5	0.2	0.2	0.4	0.4	0.4	0.4
2.5 (circular hole)	3.0	1.5	1.5	0.25	0.2	0.5	0.4	0.5	0.4
2.5 (elongated hole)	2.5 × 4.5	1.5	1.5	0.25	0.2	0.5	0.4	0.5	0.4
4	3.6	1.8	1.5	0.4	0.2	0.8	0.4	0.8	0.4
6	4.0	1.8	1.5	0.4	0.25	0.8	0.5	0.8	0.5
10	4.5	2.0	1.5	0.7	0.25	1.2	0.5	1.2	0.5

¹⁾ The values specified apply to screws covered by the corresponding columns in Table 5.

The part of the terminal containing the threaded hole and the part of the terminal against which the conductor is clamped by the screw may be two separate parts, as in the case of terminals provided with a stirrup.

The shape of the conductor space may differ from those shown provided that a circle with a diameter equal to the minimum specified for D of the minimum outlines specified for the elongated hole accepting cross sections of conductors up to 2.5 mm² can be inscribed.

FIG. 1 PILLAR TERMINALS



The part which retains the conductor in position may be of insulating material provided the pressure necessary to clamp the conductor is not transmitted through the insulating material.

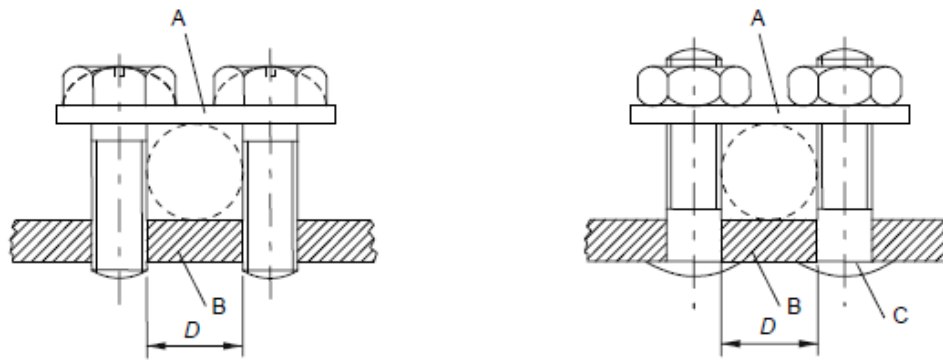
The second optional space for the terminal accepting cross-sections of conductors up to 2.5 mm² may be used for the connection of the second conductor, when it is required to connect two 2.5 mm² conductors.

Cross-section of conductor accepted by the terminal mm ²	Minimum diameter D of conductor space mm	Torque Nm			
		3 ^a		4 ^a	
		One screw	Two screws	One screw or stud	Two screws or studs
Up to 1.5	1.7	0.5	—	0.5	—
Up to 2.5	2.0	0.8	—	0.8	—
Up to 4	2.7	1.2	—	1.2	—
Up to 6	3.6	2.0	1.2	2.0	1.2
Up to 10	4.3	2.0	1.2	2.0	1.2
Up to 16	5.5	2.0	1.2	2.0	1.2
Up to 25	7.0	2.5	2.0	3.0	2.0

^a The values specified apply to the screws covered by the corresponding columns in Table 5.

FIG. 2 SCREW TERMINALS AND STUD TERMINALS

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A saddle

B fixed part

C stud

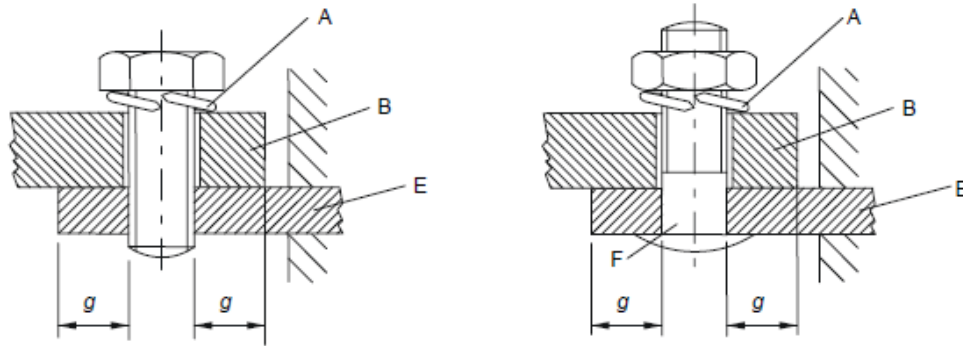
D conductor space

<i>Cross-section of conductor accepted by the terminal</i> mm ²	<i>Minimum diameter D of conductor space</i> mm	<i>Torque</i> Nm
Up to 4	3.0	0.5
Up to 6	4.0	0.8
Up to 10	4.5	1.2
Up to 16	5.5	1.2
Up to 25	7.0	2.0

The shape of the conductor space may differ from that shown in the figure, provided that a circle with a diameter equal to the minimum value specified for D can be inscribed.

The shape of the upper and lower faces of the saddle may be different to accommodate conductors of either small or large cross-sectional areas by inverting the saddle.

FIG. 3 SADDLE TERMINAL



A locking means
B cable lug or bar
E fixed part
F stud

Cross-section of conductor accepted by the terminal mm ²	Minimum distance <i>g</i> between edge of hole and side of clamping area mm	Torque Nm	
		3 ^a	4 ^a
Up to 16	7.5	2.0	2.0
Up to 25	9.0	2.5	3.0

^a The values specified apply to the screws covered by the corresponding columns in Table 5.

For this type of terminal, a spring washer or equally effective locking means shall be provided and the surface within the clamping area shall be smooth.
For certain types of switches, the use of lug terminals of sizes smaller than that specified is allowed.

FIG. 4 LUG TERMINAL

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A fixed part

D conductor space^a

<i>Cross-section of conductor accepted by the terminal</i> mm ²	^a <i>Minimum diameter D of conductor space^a</i> mm	<i>Minimum distance between fixed part and end of conductor when fully inserted</i> mm
Up to 1.5	1.7	1.5
Up to 2.5	2.0	1.5
Up to 4	2.7	1.8
Up to 6	3.6	1.8
Up to 10	4.3	2.0
Up to 16	5.5	2.5
Up to 25	7.0	3.0

^aThe bottom of the conductor space shall be slightly rounded in order to obtain a reliable connection.

The value of the torque to be applied is that specified in column 2 or 4 of Table 5, as appropriate.

FIG. 5 MANTLE TERMINAL

3.21 Clamping Unit — Part or parts of a terminal necessary for the mechanical clamping and the electrical connection of conductor(s).

3.22 Thread-Forming Screw — Tapping screw having an uninterrupted thread which by screwing in, forms a thread by displacing material.

NOTE — An example of thread-forming screw is shown in Fig. 6.

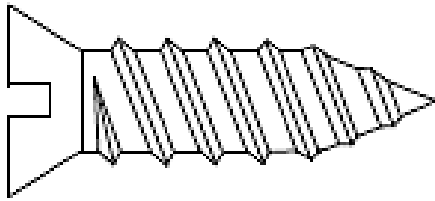


FIG. 6 EXAMPLE OF THREAD-FORMING SCREW

3.23 Thread-Cutting Screw — Screw having an uninterrupted thread which, by screwing in, forms a thread by displacing material in the cavity.

NOTE — an example of a thread-cutting screw is shown in Fig. 7.

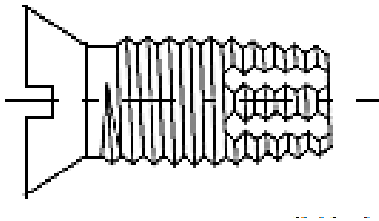


FIG. 7 EXAMPLE OF THREAD-CUTTING SCREW

3.24 Operating Member — Part of cord-operated switch which connects the internal mechanism with a pull cord and is usually attached to the actuating member of switch.

3.25 Pole of a Switching Device — A conducting path may be constituted by portions common to other conducting paths of the switch.

Portion of a switching device associated exclusively with one electrically separated conducting path of its main circuit and excluding those portions which provide a means for mounting and operating all poles together.

NOTE — a switching device is called single-pole if it has only one pole. If it has more than one pole, it may be called multipole (two-pole, three-pole, etc) provided the poles are or can be coupled in such a manner as to operate together.

3.26 Safety Extra-Low Voltage — Voltage not exceeding 32 V between conductors and between conductors and earth, the no load voltage not exceeding 38 V.

3.27 Routine Tests — ‘Tests carried out on each item to check the essential requirements which are likely to vary during production.

3.28 Acceptance Tests — Tests carried out on samples taken from a test for the purpose of acceptance of the test.

3.29 Type Tests — Tests carried out to prove conformity with the requirements of the standard. These are intended to prove the general qualities and design of a given type of appliances.

3.30 Dolly Type Switches — A switch having mechanism similar to Tumbler Switch except that the fixed and moving contacts are mounted in the direction of application of force by human finger.

3.31 Switch of Micro-gap Construction — Switch construction having a clearance between the contacts in the open position of less than 1.2 mm.

3.32 On/off Switch — Switch for alternatively closing and opening one or more electric circuits.

3.33 Actuating Member — Part which is pulled, pushed, turned or otherwise moved to cause an operation of the switch.

3.34 Pilot Light — Device incorporating a light source either integral or designed to be installed with the switch and intended to give for example an indication of the switch state or to indicate the switch location.

3.34 Main Part — Assembly consisting of the base and other parts which is not intended to be dismantled at any time after manufacture.

3.35 Grommet — Component used to support and protect the wires or cable or conduit at the point of entry.

NOTES

1 A grommet may also prevent the ingress of moisture or contaminants.

2 Examples are shown in Fig. 8.

3.36 Entry Membrane — Component or integral part of the switch used to protect the wires or cable which may be used to support the cable or conduit at the point of entry.

NOTES

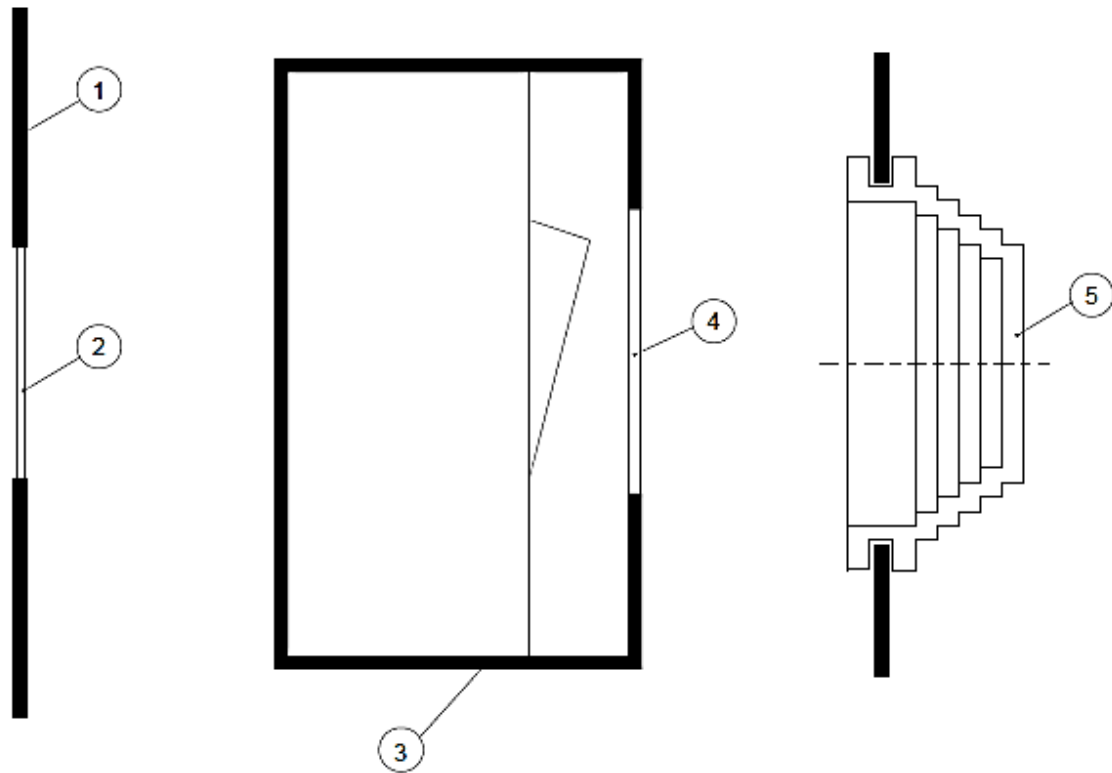
1 An entry membrane may also prevent the ingress of moisture or contaminants and may be part of a grommet.

2 Examples are shown in Fig. 8.

3.37 Protecting Membrane — Component or integral part of the switch that is not intended to be penetrated in normal use and is intended to provide protection against ingress of water or solid objects and/or to allow the operation of an accessory.

NOTE — Examples are shown in Fig. 8.

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Key

- 1 Box
- 2 Entry membrane
- 3 Envelope
- 4 Protective membrane
- 5 Grommet

FIG. 8 EXAMPLES OF MEMBRANES AND GROMMETS

3.38 Self-ballasted Lamp SBL — Unit which cannot be dismantled without being permanently damaged, which is provided with a lamp cap or caps and which incorporates a light source and any additional elements necessary for starting and stable operation of the light source.

NOTE — In the text, a self-ballasted lamp is also called CFLi or LEDi, where:

- a) CFL stands for Compact Fluorescent Lamp;
- b) LED stands for Light Emitting Diode; and
- c) “i” means that the control gear is incorporated in the lamp.

3.39 Externally Ballasted Lamp — Lamp other than an incandescent lamp which cannot be dismantled without being permanently damaged, which is provided with a lamp cap and which incorporates a light source to be controlled by a separate lamp control gear.

3.40 Covers or Cover Plates or Actuating Members or Parts of Them — Parts which are intended to ensure protection against electric shock (which cover the current carrying parts ensuring protection against electric shock like knob of switches used for making ON & OFF).

4 GENERAL REQUIREMENTS

Switches and boxes of surface mounting accessories shall be so designed and constructed that in normal use, their performance is reliable. In general, compliance is checked by carrying out all the relevant tests specified.

NOTE — Safety is achieved by reducing risk to tolerable level as defined in IS/ISO/IEC guide 51.

5 GENERAL NOTES ON TESTS

5.1 Tests according to this standard are type tests. Schedule of routine, acceptance and type tests are given in 26.

5.2 Unless otherwise specified, the specimens are tested as delivered and under normal conditions of use.

Switches having provision for pilot lights shall be tested with pilot lights fitted, unless otherwise stated. The results of the tests shall be considered to apply to switches of the same type which do not have provision for pilot lights.

Flush and semi-flush type switches shall be tested when installed in a box complying with the IS 14772. In case the switch is manufactured for a specific box, the tests shall be conducted when the switch is installed in the corresponding box as specified by the manufacturer.

5.3 Unless otherwise specified, the tests are carried out in the order of the clauses, at an ambient temperature between + 15 °C and + 35 °C.

In case of doubt, the tests are made at an ambient temperature of (27 ± 2) °C.

For each of the two combinations of rated voltage and rated currents marked on the switch, three specimens

are subjected to all the relevant tests except the test of **19.2** and **19.3**, where two further sets of three specimens are used.

For switches with a rated current up to and including 16 A the tests of **19.1**, **19.2** and **19.3** shall be carried out.

For switches with a rated current above 16 A and up to and including 20 A the tests of **19.1** and **19.3** shall be carried out.

For switches with a rated current above 16 A and up to 20 A inclusive the test of **19.2** shall be carried out only if the manufacturer has assigned the switch with an externally ballasted lamp current rating.

For switches with a rated current above 20 A, only the tests of **19.1** shall be carried out.

A switch marked 250/415 V is tested as a 415 V switch.

Momentary contact switches are not to be submitted to the tests of **18.3**, **19.2** and **19.3**.

The specimens are submitted to all the relevant tests and comply with this document if all the requirements of all the relevant tests are met.

If one specimen does not satisfy a test due to an assembly or a manufacturing fault, that test and any preceding one which may have influenced the results of the test shall be repeated and also the tests which follow shall be made in the required sequence on another full set of specimens, all of which shall comply with the requirements.

The applicant may submit, together with a number of specimens specified in Table 1, the additional set of specimens which may be required, should one specimen fail. The test laboratory should then, without further request, test additional specimens and will reject them only if a further failure occurs. If the additional set of specimens is not submitted at the same time, the failure of one specimen will result in non-compliance.

6 RATINGS

6.1 Switches shall preferably have rated voltage of 110 V, 220 V, 230 V, 240 V, 250 V, 400 V, 415 V or 440 V.

If any other voltage rating is used, it shall be in accordance with IS 12360.

6.2 Switches shall preferably have rated currents of 6 A, 10 A, 13A, 16 A, 20 A, 25 A, 32 A, 40 A, 45 A, or 63 A. The rated current of switches shall not be less than 6 A, except for momentary contact switches where a rated current of 1 A, 2 A or 4 A is allowed.

Switch with a rated current not exceeding 16 A, shall have an externally ballasted lamp current rating equal to the rated current. This requirement does not apply to switches of pattern numbers 3 and 03 or to momentary contact switches.

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Table 1 Number of Specimens Needed for the Tests

(Clause 5.3)

Sl. No	Clauses and Sub-Clauses		Number of Specimens	Number of Additional Specimens for Dual Current Rating
(1)	(2)	(3)	(4)	(5)
i)	6	Ratings	A	
ii)	7	Classification	A	
iii)	8	Marking	A	
iv)	9	Checking of dimensions	ABC	
v)	10	Protection against electric shock	ABC	
vi)	11	Provision for earthing	ABC	
vii)	12	Terminals ^{a, f, l}	ABC	JKL
viii)	13	Constructional requirements ^{b, m}	ABC	
ix)	14	Mechanism	ABC	
x)	15	Resistance to ageing, protection provided by enclosures of switches, and resistance to humidity	ABC	
xi)	16	Insulation resistance and electric strength ^c	ABC	
xii)	17	Temperature rise	ABC	JKL
xiii)	18	Making and breaking capacity	ABC ⁱ	JKL
xiv)	19	Normal operation	ABC ⁱ	JKL
xv)	20	Mechanical strength ^{d, g}	ABC	
xvi)	21	Resistance to heat ^h	ABC	
xvii)	22	Screws, current-carrying parts and connections	ABC	
xviii)	23	Creepage distances, clearances and distances through sealing compound	ABC	
xix)	19.2	Test for switches intended for externally ballasted lamp loads	DEF ^j	MNO
xx)	19.3	Test for switches intended for self-ballasted lamp loads	UVW ^k	XYZ
xxi)	24.1	Resistance to abnormal heat and to fire	GHI	
xxii)	24.2	Resistance to tracking ^e	GHI	
xxiii)	25	Resistance to rusting	GHI	
TOTAL			12	9
a	Five extra screwless terminals are used for the test of 12.3.11 and one extra set of specimens is used for the test of 12.3.12 .			
b	An extra set of membranes are needed for each of the tests of 13.15.1 and 13.15.2 .			
c	One extra set of specimens of switches fitted with pilot light may be used for the tests of Clause 16 .			
d	One extra set of specimens of cord-operated switches is needed for the test of 20.10 .			
e	One extra set of specimens may be used.			
f	Two extra set of specimens of terminals suitable for rigid and flexible conductors are required for 12.2.5 , 12.2.6 and 12.2.7 .			
g	One extra set of specimens is needed for the tests of 20.5.1 and 20.5.2 .			
h	One extra set of specimens may be used for the tests of 21.2 and 21.3 . In this case the specimens shall be subjected first to the tests of 15.1 .			
i	One extra set of specimens of switches of pattern number 7 are necessary for 18 and 19.1 .			
j	One extra set of specimens of switches of pattern number 7 are necessary for 19.2 .			
k	One extra set of specimens of switches of pattern number 7 are necessary for 19.3 .			
l	Number of specimens required for insulation-piercing terminals (IPTs) are shown in Table D-1			
m	For switches with pilot light units if the electronic circuitry is so enclosed that the short-circuiting or disconnecting of components is impossible or difficult, the manufacturer shall provide additional prepared test specimens.			

Switches with a rated current not exceeding 20 A shall be tested with self-ballasted lamp (SBL) circuits as given in 19.3.

Switches which are tested according to 19.3 are capable of switching the rated power of SBL circuits according to Table 2.

Table 2 Relationship Between Rated Current of the Switch and Rated Power of the SBL Circuit
(Clause 6.2)

SI No.	Rated Current of the Switch [A]	Distribution System [V]: 220/400 230/400 240/415 250/415 Rated power of the SBL circuit [W]
(1)	(2)	(3)
i)	Up to and including 10	100
ii)	Above 10 up to and including 13	150
iii)	Above 13 up to and including 16	200
iv)	Above 16 up to and including 20	250
NOTE — Higher values of rated power of SBL circuits can be declared by the manufacturer according to Table 2.		

6.3 Preferred Combinations of Number of Poles and Ratings

Preferred combinations of number of poles and ratings are shown in Table 3.

Table 3 Preferred Combinations of Numbers of Poles and Ratings
(Clause 6.3)

SI No.	Rated Current A	Number of Poles	
		Rated Voltage up to 250 V Inclusive	Rated Voltage Over 250 V
(1)	(2)	(3)	(4)
i)	1, 2 and 4	1	-
ii)	6	1 2	1 2
iii)	10 and 13	1 2	1 2 3 4
iv)	16, 20, 25, 32, 40, 45 and 63	1 2 3 4	1 2 3 4

7 CLASSIFICATIONS

7.1 Switches are classified.

7.1.1 According to the possible connections (see Fig. 9):

	Pattern Number
Single-pole switches	1
Double-pole switches	2
Three-pole switches	3
Three-pole plus switched neutral switches	03
Two-way switches	6
Two-circuit switches with a common incoming	5
Two-way switches with one off-position	4
Two-way double-pole switches	6/2
Two-way double-pole reversing switches	7

NOTES

1 Two or more switches having the same or different pattern numbers may be mounted on a common base.

2 For the pattern number for which an off-position is considered, the above classification refers also to push-button switches and momentary contact switches.

7.1.2 According to the contact opening:

- Switches of normal gap construction;
- Switches of mini-gap construction;
- Switches of micro-gap construction; and
- Switches without contact gap (semiconductor switching devices).

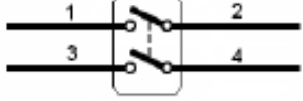
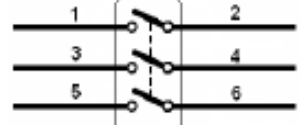
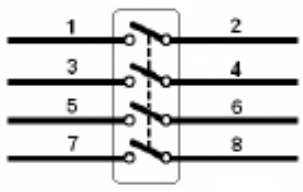
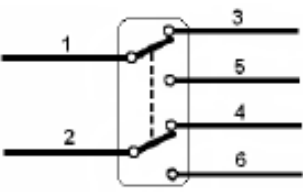
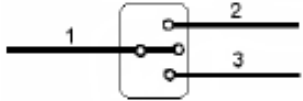
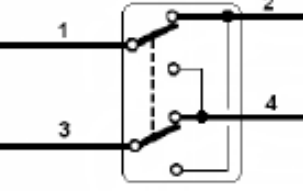
7.1.3 According to the degree of protection against access to hazardous parts and against harmful effects due to the ingress of solid foreign objects as described in IS/IEC 60529.

7.1.4 According to the degree of protection against harmful effects due to the ingress of water as described in IS/IEC 60529.

7.1.5 According to the method of activating the switch:

- Rotary;
- Tumbler;
- Rocker;
- Push-button;
- Cord-operated; and
- Dolly type.

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Pattern number	Number of poles	Possible connections	Pattern number	Number of poles	Possible connections
1	1		5	1	
2	2		6	1	
3	3				
03	4		6/2	2	
4	1				
			7	1	

The figures indicating the terminals are given for test purposes only and are not those required to be marked.

FIG. 9 CLASSIFICATION ACCORDING TO CONNECTIONS

7.1.6 According to the mounting of the switch:

- Surface-type;
- Flush-type;
- Semi flush type;
- Panel-type; and
- Architrave-type.

7.1.7 According to type of terminals:

- Switches with screw-type terminals,
- Switches with screwless terminals for rigid conductors only, and
- Switches with screwless terminals for rigid and flexible conductors only.

8 MARKING

8.1 Switches shall be Marked With:

- Rated currents:
 - In amperes (A) if the switch is tested according to 19.1 only;
 - In amperes (A) if the switch is tested according to 19.1 and 19.3 only;
 - In amperes (AX) if the switch is tested according to 19.1, 19.2 and 19.3 when the rated currents in amperes (A) and (AX) are equal; or
 - In amperes (A and AX) if the switch is tested according to 19.1, 19.2 and 19.3 when the rated currents in amperes (A) and (AX) are not equal.
- Rated voltage in volts;
- Symbol for nature of supply;
- Manufacturer's or responsible vendor's name trade-mark or identification mark;
- Type reference, which may be a catalogue number (optional);
- Symbol for switches of mini-gap construction, if applicable;
- Symbol for switches of micro-gap construction, if applicable;
- Symbol for switches without contact gap (semiconductor switching device), if applicable;
- First characteristic numeral for the degree of protection against access to hazardous parts and against harmful effects due to ingress of solid foreign objects, if declared to be higher than 4 in which case the second characteristic numeral shall also be marked;
- Second characteristic numeral for the degree of protection against harmful effects due to ingress of water, if declared to be higher than 2 in which case the first characteristic numeral shall also be marked;

- Length of insulation to be removed before the insertion of the conductor into the screw less type terminal; and
- Symbol for the suitability to accept rigid conductors only.

In addition, the following information shall be given in the manufacturer's documentation:

- For SBL loads: the rated power in watts and the type of load if the switch is tested according to 19.3; and
- Country of manufacture.

NOTES

1 Marking of the pattern number given in 7.1.2 is recommended if the connections are not clear from an inspection of the switch; this pattern number may be a part of the type reference.

2 If a base carries two more switches with separate operating devices, marking with the pattern numbers is recommended, for example 1 + 6 or 1 + 1 + 1.

3 If there is no marking on the terminals, they are suitable to accept both rigid and flexible conductors.

8.2 When symbols are used they shall be as follows:

Amperes (rated current for externally ballasted loads)	AX
(other current)	A
Volts	V
Alternating current	ac/~
Watt	W
Neutral	N
Line	L
Protective earth	
"open" position (off)	O
"closed" position (on)	I
Mini-gap construction	m
Micro-gap construction	μ
Without contact gap (semiconductor switching device)	ε
Degree of protection, when relevant	IPXX
Degree of protection for fixed accessories to be installed on rough surfaces (test wall of Fig. 10)	
Suitability to accept rigid conductors only	r

NOTES

1 In the IP code the letter "X", concerning protection against ingress of solid object replaced by the relevant number.

2 Lines formed by the constructions of the tool are not considered as part of the marking.

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The symbol “AX” may be replaced by the symbol “X”. For the marking with rated current and rated voltage the figures may be used alone.

The marking for the nature of supply shall be placed next to the marking for rated current and rated voltage.

NOTE — The marking for current, voltage and nature of supply may be, for example as follows:

10 AX 250 V ~	or	10 X/250 ~	or	$\frac{10 \text{ X}}{250} \sim$
20 A – 16 AX 250 V ~	or	20 -16 X/250 ~	or	$\frac{20 - 16 \text{ X}}{250} \sim$
10 AX 400 V ~	or	10 X/400 ~	or	$\frac{10 \text{ X}}{400} \sim$
25 AX 400 V ~	or	25 X/400 ~	or	$\frac{25 \text{ X}}{400} \sim$
25 A 250 V ~	or	25/250 ~	or	$\frac{25}{250} \sim$
25 AX 440 V ~	or	25 X/440 ~	or	$\frac{25 \text{ X}}{440} \sim$

8.3 Visibility of Markings

The marking shall be clearly visible with normal or corrected vision, without additional magnification, marked either on the front of the switch or on the inner part of its associated enclosure, or on the main part of the switch.

Markings as given in **8.1 a), b), c), d), e)** and, if applicable, **f), g), h), k)** and **m)** shall be placed on the main part of the switch.

Parts, such as cover plates, which are necessary for safety purposes and are intended to be sold separately, shall be marked with the maker's or responsible vendor's name, trade-mark or identification mark and type reference.

Markings **8.1 j)** and **k)**, when applicable, shall be marked to be easily discernible when the switch is mounted and wired as in normal use. These markings shall be placed on parts which cannot be removed without the use of a tool.

NOTE — Additional type references may be marked on the main part, or on the outside or the inside of the associated enclosure.

8.4 Marking on Terminals for Phase Conductors

Terminals intended for the connection of phase conductors (supply conductors) shall be identified unless the method of connection is of no importance, is self-evident or is indicated on a wiring diagram. Such identification may take the form of a letter L or, in the case of more than one such terminals, the letters L1, L2, L3, etc which may be accompanied by an arrow or arrows pointing to the relevant terminal or terminals.

Alternatively, the surface of such terminals shall be bare brass or copper, other terminals being covered with a metallic layer of another colour.

For switches of pattern numbers 2, 3, 03 and 6/2, terminals associated with any one pole shall have similar identification, if applicable, differing from that of the terminals associated with the other poles, unless the relationship is self-evident.

These indications shall not be placed on screws or any other easily removable parts.

NOTES

1 The wiring diagram may be an instruction sheet which accompanies the switch.

2 'Easily removable parts' are those parts which can be removed during the normal installation of the switch.

8.5 Marking on Terminals for Neutral and Earth Conductors

Terminals intended exclusively for the neutral conductor shall be indicated by the letter N.

Earthing terminals for the connection of the protective conductor shall be indicated by the symbol .

These markings shall not be placed on screws, or any other easily removable part.

NOTE — Easily removable parts" are those parts which can be removed during the normal installation of the switch.

Terminals provided for the connection of conductors not forming part of the main function of the switch shall be clearly identified unless their purpose is self-evident, or indicated in a wiring diagram which shall be fixed to the accessory.

The identification of switch terminals may be achieved by:

- their marking with graphical symbols according to IEC 60417 or colors and/or alphanumeric system; and
- their physical dimension or relative location.

Leads of pilot lights are not considered to be conductors for the purpose of **8.5**.

8.6 Marking of the Switch Position

If switches are marked to indicate the switch position, they shall be so marked that the direction of movement of the actuating member to its different positions or the actual position is clearly indicated. For switches having more than one actuating member, this marking shall indicate, for each of the actuating members, the effect achieved by its operation.

The marking shall be clearly visible on the front of the switch when fitted with its cover or cover plate.

If this marking is placed on the cover, cover plate, or removable actuating members, it shall not be possible to fix them in a position such that the marking is incorrect.

The symbols for 'ON' and 'OFF' shall not be used for the indication of the switch positions, unless, at the same time, they clearly indicate the direction of the movement of the actuating members.

NOTE — Other suitable means for the indication of the switch position can be used, for example pilot lights.

The short-straight line indicating the 'on' position shall be radial for rotary switches, perpendicular to the axis of rotation of the dolly for tumbler switches and rocker switches and vertical for push-button switches when mounted vertically.

These requirements do not apply to cord-operated switches and to switches of pattern number 6, 6/2, 7 and it is not necessary to have such indication for push button switches.

Compliance with the requirements of **8.1** to **8.6** is checked by inspection.

8.7 A push-button shall be colored red only if it serves to open the circuit to be controlled and in addition, may serve to close auxiliary contacts for control circuits, pilot lamps, etc.

8.8 Additional Requirements for Marking

If it is necessary to take special precautions when installing the switch, details of these shall be given in an instruction sheet which accompanies the switch.

Compliance with the requirements of **8.7** and **8.8** is checked by inspection.

NOTES

1 Special precautions may, for example, be necessary for unenclosed switches and for panel mounting.

2 After installation, the conditions necessary to meet the requirements of this standard are achieved, when the instruction sheet should include clear information with regard to the following:

- a) Dimensions of the space to be provided for the switch;
- b) Dimensions and position of the means for supporting and fixing the switch within this space;
- c) Minimum clearances between the various parts of the switches and the surrounding parts where fitted; and
- d) Minimum dimensions of ventilating openings, if needed, and their correct arrangement.

8.9 Durability

Marking shall be durable and easily legible.

Compliance is checked by inspection and by the following test:

The test is made by rubbing the marking by hand for 15 s with a piece of cloth soaked with water and again for 15 s with a piece of cloth soaked with petroleum spirit.

NOTES

1 Marking made by impression, moulding, pressing or engraving is not subjected to this test.

2 It is recommended that the petroleum spirit used consist of a solvent hexane with an aromatic content of maximum 0.1 percent in volume percentage, a kauributanol value of approximately 29, an initial boiling point of approximately 65°C, a dry point of approximately 69°C and density of approximately 0.68 g/cm³.

8.10 The switches may also be marked with the Standard Mark.

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

9 CHECKING OF DIMENSIONS

Switches and boxes shall comply with the manufactures drawings.

Compliance is checked by the measurement.

10 PROTECTION AGAINST ELECTRIC SHOCK

10.1 Prevention of Access to Live Parts

Switches shall be so designed that when they are mounted and wired as in normal use, live parts are not accessible even after removal of parts which can be removed without the use of a tool.

Switches which are designed to be fitted with pilot lights supplied at voltage other than ELV shall have means to prevent direct contact with the lamp.

Compliance is checked by inspection and if necessary by the following test:

The specimen is mounted as in normal use and fitted with conductors of the smallest cross-sectional areas specified in **12**; the test is repeated using conductors of the largest cross-sectional areas specified in **12**.

The standard test finger, is applied in every possible position, an electrical indicator, with a voltage not less than 40 V and not more than 50 V, being used to show contact with the relevant part.

Switches, having enclosures or covers in thermoplastic or elastomeric material are subject to the following additional test which is carried out at an ambient temperature of $(35 \pm 2)^\circ\text{C}$, the switches being at this temperature.

During this additional test, the switches are subjected for 1 min to a force of 75 N, applied through the tip of test probe 11 of IS 1401.

This finger, with an electrical indicator as described above, is applied to all places where yielding of insulating material could impair the safety of the switch, but is not applied to membranes or the like and is applied to thin walled knock outs with a force of only 10 N.

During this test, switches with their associated mounting means shall not deform to such an extent that live parts can be touched through the tip of test probe 11 of IS 1401.

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NOTE — Membranes or the like are tested according to 13.15.1 only.

10.2 Knobs, operating levers, push-buttons, rockers and the like shall be of insulating material, unless their accessible metal parts are separated from the metal parts of the mechanism by double insulation or reinforced insulation or, as an alternative, they are reliably connected to earth.

Compliance is checked by inspection and by the tests of 16 and 23.

This requirement does not apply to removable keys or intermediate keys on intermediate parts, such as chains or rods.

10.3 Requirements for Accessible Metal Parts

10.3.1 Accessible parts of switches when in normal use shall be made of insulating material with the exception of the following:

- a) Small screws and the like which are isolated from live parts and which are used for fixing bases and covers or cover plates;
- b) Actuating members complying with 10.2; and
- c) The covers and cover plates of metal which comply with the requirements of 10.3.2 or 10.3.3.

10.3.2 Covers or cover plates of metal shall be protected by additional insulation made by insulating linings or insulating barriers. The insulating linings or insulating barriers shall either:

- a) Be fixed to covers or cover plates or the body of the switches in such a way that they cannot be removed without being permanently damaged, or
- b) Be so designed that:
 - 1) They cannot be replaced in an incorrect position;
 - 2) If they are omitted, the accessories are rendered inoperable or manifestly incomplete;
 - 3) There is no risk of accidental contact between live parts and metal covers or cover plates, for example, through their fixing screws, even if a conductor should come away from its terminals; and
 - 4) Precautions are taken in order to prevent creepage distances or clearances becoming less than the values specified in 23.

Compliance is checked by inspection.

The above linings or barriers shall comply with the tests of 16 and 23.

NOTE — Insulating coating sprayed on the inside or on the outside of the metal cover or cover plates is not deemed to be an insulation lining or barrier for the purpose of 10.3.2.

10.3.3 The earthing of metal covers or cover plates can be made with fixing screws or other integral means, the resulting connection shall be of low resistance.

Compliance is checked by inspection and by the tests of 11.4.

10.4 Requirements for Insulation of the Mechanism

Metal parts of the mechanism, such as the spindle or the pivot of the dolly or rocker that are not insulated from live parts shall not protrude from the enclosure.

However, for switches operated by means of a removable key or similar device, such metal parts of the mechanism shall be insulated from live parts.

Compliance is checked by inspection, if necessary after the actuating member has been removed or broken.

If the actuating member has to be broken, compliance is checked by the test of 23.

10.5 Requirements for Insulation of the Mechanism with Respect to the Surrounding Environment

Metal parts of the mechanism, such as the spindle or the pivot of the dolly or rocker, shall not be accessible when the switch is fixed as in normal use.

In addition, they shall be insulated from accessible metal parts, including metal frames supporting the base of flush-type switches, liable to be mounted in a metal box, and from screws for fixing the base to its support.

These requirements do not apply if the metal parts of the mechanism are separated from live parts in such a way that the creepage distances and clearances have at least twice the values specified in 23 or as an alternative, if they are reliably connected to earth.

Compliance is checked by inspection and, if necessary, by measurement and by the tests of 10 and 16.

NOTE — When checking the inaccessibility of metal parts of the mechanism of unenclosed switches or architrave type switches, the protection provided by the normal way of mounting the switch is taken into account.

For unenclosed stack-type switches having a metal spindle pivoting in a metal base plate, the additional requirement means that the creepage distances and clearances between live parts and the spindle and between metal parts of the mechanism and the base plate, must have at least twice the values specified in 23.

10.6 Requirements for Switches Operated Indirectly

Switches operated by means of a removable key or by means of an intermediate part, such as a cord, a chain or a rod, shall be so designed that the key or intermediate part can only touch parts which are insulated from live parts.

The key or intermediate part shall be insulated from metal parts of the mechanism, unless the creepage distances and clearances between live parts and metal parts of the mechanism have at least twice the values specified in 23.

Compliance is checked by inspection and, by the test of 16.3 and if necessary, by measurement.

NOTE — Lacquer or enamel is not considered to be insulating material for the purpose of 10.1 to 10.6.

10.7 Requirements for Switches with Replaceable Pull Cord

Where cord-operated switches are provided with a pull cord, which can be fitted or replaced by the user, it shall be so designed that it is impossible to touch live parts when fitting or replacing the pull cord in the normal way.

Compliance is checked by inspection.

11 PROVISION FOR EARTHING

11.1 General

Accessible metal parts, which may become live in the event of an insulation fault, shall be provided with, or permanently and reliably connected to, an earthing terminal.

This requirement does not apply to the metal cover plates mentioned in 10.3.1.

For the purpose of this requirement, small screws and the like, isolated from live parts, for fixing bases, covers or cover plates, are not considered as accessible parts which may become live in the event of an insulation fault.

11.2 Earthing Terminals

Earthing terminals shall be terminals with screw clamping or screwless terminals and shall comply with the appropriate requirements of 12.

11.3 Requirements of Surface Type Switches

Surface-type switches with an enclosure of insulating material having an IP code higher than IPX0 and more than one cable inlet, shall be provided with an internal earthing terminal allowing the connection of an incoming and outgoing conductor for the continuity of the earthing circuit.

Clause 12 does not apply to floating terminals which shall comply with IS/IEC 60998-1 and the relevant sub-part of IEC 60998-2.

Compliance with the requirements of 11.1 to 11.3 is checked by inspection and by the test of 12.

11.4 The connection between the earthing terminal and accessible metal part to be connected thereto shall be of low resistance.

Compliance is checked by the following test:

A current derived from an a.c. source having a no-load voltage not exceeding 12 V and equal to 1.5 times rated current or 25 A, whichever is the greater is passed between the earthing terminals and each of the accessible metal parts in turn.

The voltage drop between the earthing terminal and the accessible metal part is measured and the resistance calculated from the current and this voltage drop.

In no case, the resistance shall exceed 0.05 ohm.

NOTE — Care should be taken that the contact resistance between the tip of the measuring probe and the metal part under test does not influence the test results.

12 TERMINALS

12.1 General

Switches shall be provided with terminals having screw clamping or with screwless terminals.

The means for clamping the conductors in the terminals shall not serve to fix any other component, although they may hold the terminals in place or prevent them from turning.

All the tests on terminals, with the exception of 12.3.11 shall be carried out after the test of 15.1, Rigid solid conductors shall be of class 1, rigid stranded conductors shall be of class 2 and flexible conductors shall be of class 5 according to IS 694.

Compliance is checked by inspection and by the tests of 12.2 or 12.3 as applicable.

12.2 Terminals with screw clamping for external copper conductors.

12.2.1 Terminals with screw clamping may be of the type suitable for rigid copper conductors only or of the type suitable for both rigid and flexible copper conductors having cross-sectional areas as shown in Table 4.

The conductor space shall be at least that specified in Fig. 1, 2, 3, 4 and 5.

Compliance is checked by inspection, and by fitting conductors of the smallest and largest cross-sectional areas specified.

12.2.2 Terminals with screw clamping shall allow the conductor to be connected without special preparation.

Compliance is checked by inspection.

NOTE — The term “special preparation” covers soldering of the wires of the conductor, use of cable lugs, formation of eyelets, etc., but not the reshaping of the conductor before its introduction into the terminal or the twisting of a flexible conductor to consolidate the end.

12.2.3 Terminals with screw clamping shall have adequate mechanical strength.

Screws and nuts for clamping the conductors shall have a metric ISO thread or a thread comparable in pitch and mechanical strength.

Screws shall not be of metal which is soft or liable to creep, such as zinc or aluminum.

Compliance is checked by inspection and by the tests of 12.2.6 and 12.2.8.

NOTE — Provisionally SI, BA and UN threads are considered to be comparable in pitch and mechanical strength to metric ISO thread.

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**Table 4 Relationship Between Rated Currents and Connectable
Cross-sectional Areas of Copper Conductors**

(Clauses 12.2.1, 12.2.5, 12.2.7, 12.2.8, 12.2.11, 13.2 and 15.2.2.1)

Sl No.	Ranges of Rated Currents Nominal	Rigid Conductors (Solid or Stranded) ^c	
		Cross-sectional Areas mm ²	Diameter of Largest Conductor mm
(1)	(2)	(3)	(4)
i)	up to and including 4 ^a	—	—
ii)	above 4 up to and including 6	From 0.75 up to 1.5 inclusive	1.45
iii)	above 6 up to and including 13 ^b	From 1 up to 2.5 inclusive	2.13
iv)	above 13 up to and including 16 ^b	From 1.5 up to 4 inclusive	2.72
v)	above 16 up to and including 25	From 2.5 up to 6 inclusive	3.34
vi)	above 25 up to and including 32	From 4 up to 10 inclusive	4.34
vii)	above 32 up to and including 50	From 6 up to 16 inclusive	5.46
viii)	above 50 up to and including 63	From 10 up to 25 inclusive	6.85
^a For special purposes such as ELV applications, where flexible conductors are used (from 0.5 mm ² up to 1 mm ² inclusive).			
^b Each supply terminal of switches, other than those of pattern numbers 3, 03 and 7 shall allow the connection of two 2.5 mm ² conductors. For switches having a rated voltage not exceeding 250 V a round hole is sufficient for the connection of two 2.5 mm ² conductors.			
^c The use of flexible conductors is permitted.			

12.2.4 Terminals with screw clamping shall be resistant to corrosion.

Terminals, the body of which is made of copper or a copper alloy as specified in **22.5** are considered as complying with this requirement.

12.2.5 Screw-type terminals shall be so designed that they clamp the conductor without undue damage to the conductor(s).

Compliance is checked by the following test:

Terminals suitable for rigid conductors only are checked with rigid solid conductors and on a new set of specimens with rigid stranded conductors, if existing.

NOTE — “If existing” means that the conductor is available in the market where the product is sold and installed.

Terminals suitable for rigid and flexible conductors are checked with rigid conductors as above and on a new set of samples with flexible conductors.

The terminal is placed in the test apparatus according to Fig. 11. The terminal is fitted first with a conductor of the smallest and then with a conductor of the largest cross-sectional area according to Table 4, the clamping screw(s) or nut(s) being tightened with the torque according to Table 5.

Where a screw has a hexagonal head with a slot for tightening with a screwdriver and the values of columns 3 and 5 are different, the test is made twice, first applying to the hexagonal head the torque specified in column 5 and then applying the torque specified in

column 3 by means of a screwdriver. If the values of columns 3 and 5 are the same, only the test with the screwdriver is made.

For mantle terminals, the specified nominal diameter is that of the slotted stud.

The shape of the blade of the test screwdriver shall suit the head of the screw to be tested.

The screws and nuts shall be tightened in one smooth and continuous motion.

The length of the test conductor shall be 75 mm longer than the height (H) specified in Table 6.

The end of the conductor is passed through an appropriate sized bushing in a platen positioned at a height (H) below the equipment as given in Table 6. The bushing shall be positioned in a horizontal plane such that its centre line describes a circle of 75 mm diameter, concentric with the centre of the clamping unit in the horizontal plane. The platen is then rotated at a rate of (10 ± 2) r/min.

The distance between the mouth of the clamping unit and the upper surface of the bushing shall be within ± 15 mm of the height of Table 6. The bushing may be lubricated to prevent binding, twisting, or rotation of the insulated conductor.

A mass as specified in Table 6 is to be suspended from the end of the conductor. The duration of the test shall be 15 min.

Table 5 Tightening Torque for the Verification of the Mechanical Strength of Screw-type Terminals

(Clauses 12.2.5, 12.2.6, 12.2.7, 12.2.8, 15.2.2.1, 15.2.2.3, 17.1, 20.2, Fig 1, Fig 4, and Fig 5)

Nominal Diameter of Thread (mm)	Torque (Nm) A					
	1	2	3	4	5	6
Up to and including 2.8	0.2	—	0.4	—	0.4	—
Over 2.8 up to and including 3.0	0.25	—	0.5	—	0.5	—
Over 3.0 up to and including 3.2	0.3	—	0.6	—	0.6	—
Over 3.2 up to and including 3.6	0.4	—	0.8	—	0.8	—
Over 3.6 up to and including 4.1	0.7	1.2	1.2	1.2	1.2	—
Over 4.1 up to and including 4.7	0.8	1.2	1.8	1.8	1.8	—
Over 4.7 up to and including 5.3	0.8	1.4	2.0	2.0	2.0	—
Over 5.3 up to and including 6.0	—	1.8	2.5	3.0	3.0	—
Over 6	—	—	—	—	—	0.8
<i>Column 1</i> — Applies to screws without heads if the screw when tightened does not protrude from the hole, and to other screws which cannot be tightened by means of a screwdriver with a blade wider than the diameter of the screw. <i>Column 2</i> — Applies to nuts of mantle terminals which are tightened by means of a screwdriver. <i>Column 3</i> — Applies to other screws which are tightened by means of screwdriver. <i>Column 4</i> — Applies to nuts of mantle terminals in which the nut is tightened by means other than a screwdriver. <i>Column 5</i> — Applies to screws or nuts, other than nuts of mantle terminals, which are tightened by means other than a screwdriver. <i>Column 6</i> — Applies to nuts of switches with central hole fixing.						
NOTES 1 the above values are provisional. 2 For Dolly type switches, applicable values for row 6 are as given below: Over 4.1 up to and including 4.3 0.8 1.2 1.4 1.4 1.4 — Over 4.3 up to and including 4.7 0.8 1.2 1.8 1.8 1.8 —						

Table 6 Test Values for Flexion and Pull Out for Copper Conductors

(Clauses 12.2.5, 12.3.10 and 12.4.10)

Conductor Cross-Sectional Area mm ²	Diameter of Bushing Hole ^a mm	Height H ^b mm	Mass for Conductor kg
0.50	6.5	260	0.3
0.75	6.5	260	0.4
1.0	6.5	260	0.4
1.5	6.5	260	0.4
2.5	9.5	280	0.7
4.0	9.5	280	0.9
6.0	9.5	280	1.4
10.0	9.5	280	2.0
16.0	13.0	300	2.9
25.0	13.0	300	4.5
^a If the bushing hole diameter is not large enough to accommodate the conductor without binding, a bushing having the next larger hole size may be used.			
^b Tolerance for height H = ± 15 mm.			

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During the test, the rigid conductor, or any wire of rigid stranded or flexible conductor, shall neither slip out of the clamping unit nor break near the clamping unit, nor shall the conductor be damaged in such a way as to render it unfit for further use.

In the case of the flexible conductor, the breakage of a few wires shall not be taken into account provided it does not exceed 15 percent of the original number of wires.

12.2.6 Terminals with screw clamping shall be so designed that they clamp the conductor reliably and between metal surfaces.

Compliance is checked by inspection and by the following test:

Terminals suitable for rigid conductors only are checked with rigid solid conductors and on a new set of specimens with rigid stranded conductors, if existing.

NOTE — “If existing” means that the conductor is available in the market where the product is sold and installed.

Terminals suitable for rigid and flexible conductors are checked with rigid conductors as above and on a new set of specimens with flexible conductors.

The terminals are fitted with rigid stranded conductors of the smallest and largest cross-sectional areas specified in Table 4, the terminals screws being tightened with a torque equal to two-thirds of the torque shown in the appropriate column of Table 5.

If the screw has a hexagonal head with a slot, the torque applied is equal to two-thirds of the torque shown in column 3 of Table 5.

The conductor is then subjected to a pull as specified in Table 7, applied without jerks, for 1 min, in the direction of the axis of the conductor space.

If the clamp is intended for two conductors, the appropriate pull is applied consecutively to each conductor.

During the test, the conductor shall not move noticeably in the terminal.

If the clamp is intended for connection of more than two conductors, reference is made for the testing to the requirements given in the appropriate part of IEC 60998.

12.2.7 Terminals with screw clamping shall be so designed or placed that neither a rigid solid conductor

nor a wire of a stranded conductor can slip out while the clamping screws or nuts are tightened.

This requirement does not apply to lug terminals.

Compliance is checked by the following test:

Terminals suitable for rigid conductors only are checked with rigid solid conductors and on a new set of specimens with rigid stranded conductors, if existing.

NOTE — “If existing” means that the conductor is available in the market where the product is sold and installed.

Terminals suitable for rigid and flexible conductors are checked with rigid conductors as above and on a new set of specimens with flexible conductors

The terminals are fitted with conductors having the largest cross-sectional area specified in Table 4.

Terminals intended for the looping-in of two or three conductors are checked, being fitted with the permissible number of conductors.

The terminals are fitted with conductors having the composition shown in Table 8.

Before insertion into the clamping means of the terminal, wires of rigid (solid or stranded) conductors are straightened; rigid stranded conductors may, in addition, be twisted to restore them approximately to their original shape and flexible conductors are twisted in one direction so that there is a uniform twist of one complete turn in a length of approximately 20 mm.

The conductor is inserted into the clamping means of the terminal for the minimum distance prescribed or, where no distance is prescribed, until it just projects from the far side of the terminal and in the position most likely to allow the wire to escape. The clamping screw is then tightened with a torque equal to two-thirds of the torque shown in the appropriate column of Table 5.

After the test, no wire shall have escaped from the clamping unit. Thus reducing creepage distances and clearances to values lower than those indicated in Table 9.

12.2.8 Terminals with screw clamping shall be so fixed or located within the accessory that, when the clamping screws or nuts are tightened or loosened the terminals shall not work loose from their fixing to the switch.

These requirements do not imply that the terminals must be so designed that their rotation or displacement is prevented, but any movement must be sufficiently

Table 7 Test Values for Pulling Out Test

(Clauses 12.2.5, 12.2.6, 12.3.10, and 12.4)

Sl No.	Cross Section of Terminal Accepted by Terminals (mm ²)	0.5 up to 0.75 inclusive	0.75 up to 1.5 inclusive	1.5 up to 2.5 inclusive	2.5 up to 4.0 inclusive	4.0 up to 6.0 inclusive	6.0 up to 10 inclusive	10.0 up to 16.0 inclusive	16.0 up to 25.0 inclusive
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Pull (N)	30	40	50	50	60	80	90	100

limited so as to prevent noncompliance with this standard.

The use of sealing compound or resin is considered to be sufficient preventing a terminal from working loose provided that:

- a) the sealing compound or resin is not subject to stress during normal use, and
- b) the effectiveness of the sealing compound or resin is not impaired by temperatures attained by the terminal under the most unfavorable conditions specified in this standard.

Compliance is checked by inspection, by measurement and by the following test:

A solid rigid copper conductor of the largest cross sectional area specified in Table 4 is placed in the terminal.

Screw and nuts are tightened and loosened five times by means of a suitable test screwdriver or spanner, the torque applied when tightening being equal to the torque shown in the appropriate column of Table 5 or in the table of the appropriate Fig. 1 to Fig. 4, whichever is the highest.

During the test, terminals shall not work loose and there shall be no damage, such as breakage of screw or damage to the head, slots, threads, washers or stirrups that will impair the further use of the terminals.

NOTES

- 1 For mantle terminals the specified nominal diameter is that of the slotted stud.
- 2 The shape of the blade of the test screwdriver must suit the head of the screw to be tested.
- 3 The screw and nuts must not be tightened in jerks.

12.2.9 Clamping screws or nuts of earthing terminals with screw clamping shall be adequately locked against accidental loosening and it shall not be possible to loosen them without the aid of a tool. Compliance is checked by manual test.

In general, the designs of terminals shown in Fig. 1 to Fig. 5, provide sufficient resiliency to comply with this requirement for other designs, special provisions, such as the use of an adequately resilient part, which is not likely to be removed inadvertently, may be necessary.

12.2.10 Earthing terminals with screw clamping shall be such that there is no risk of corrosion resulting from contact between these parts and the copper of the earthing conductor, or any other metal that is in contact with these parts.

The body of earthing terminals shall be of brass or other metal no less resistant to corrosion, unless it is a part of the metal frame or enclosure, when the screw or nut shall be of brass or other metal no less resistant to corrosion.

If the body of the earthing terminal is a part of a frame or enclosure of aluminum alloy, precautions shall be

taken to avoid the risk of corrosion from contact between copper and aluminum or its alloys.

Compliance is checked by inspection.

NOTE — Screws or nuts of plated steel withstanding the corrosion test are considered to be of a metal no less resistant to corrosion than brass.

12.2.11 For pillar terminals, the distance between the clamping screw and the end of the conductor, when fully inserted, shall be at least that specified in Fig. 1.

NOTE — The minimum distance between the clamping screw and the end of the conductor applies only to pillar terminals in which the conductor cannot pass right through.

For mantle terminals, the distance between the fixed part and the end of the conductor, when fully inserted, shall be at least that specified in Fig. 5.

Compliance is checked by measurement, after a solid conductor of the largest cross-sectional area specified, for the appropriate rated current in Table 4, has been fully inserted and fully clamped.

12.2.12 Lug terminals shall be used only for switches having a rated current of 40 A or greater. If such terminals are provided they shall be fitted with spring washers or equally effective locking means.

Compliance is checked by inspection.

12.3 Screwless Terminals for External Copper Conductors

12.3.1 Screwless terminals may be of the type suitable for rigid copper conductors only or of the type suitable for both rigid and flexible copper conductors.

For the latter type, the tests are carried out with rigid conductors first and then repeated with flexible conductors.

NOTE — This sub-clause is not applicable to switches provided with:

- a) Screwless terminals requiring fixing of the special devices to the conductors before clamping them in the screwless terminals. For example flat push-on connectors.
- b) Screwless terminals requiring twisting of the conductors for example those with twisted joints.
- c) Screwless terminals providing direct contact to the conductors by means of edges or points penetrating the insulation.

Screwless terminals requiring fixing of the special devices to the conductors before clamping them in the screwless terminals. For example flat push-on connectors.

Screwless terminals requiring twisting of the conductors for example those with twisted joints.

Screwless terminals providing direct contact to the conductors by means of edges or points penetrating the insulation.

12.3.2 Screwless terminals shall be provided with clamping units which allow the proper connection of rigid or of rigid and flexible copper conductors having nominal cross-sectional areas as shown in Table 10.

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Table 8 Composition for Terminals Fitted with Conductors

(Clause 12.2.7)

Sl No.	Cross-Sectional Area mm ²	Number of Wires and Nominal Diameter of Wires Trim	
		Solid Conductor	Stranded Conductor
1.	1	1 × 1.13	7 × 0.42
2.	1.5	1 × 1.38	7 × 0.52
3.	2.5	1 × 2.78	7 × 0.67
4.	4	1 × 2.25	7 × 0.86
5.	6	1 × 2.76	7 × 1.05
6.	10	1 × 3.57	7 × 1.35
7.	16		7 × 1.70
8.	25		7 × 2.14

Table 9 Creepage Distances, Clearances and Distances through Insulating Sealing Compound

(Clauses 12.2.7, 12.3.8, 13.3.1, 13.3.3 and 23.1)

Description		mm		
		normal gap	mini gap	micro gap
Creepage distances				
1	Between internal live parts which are separated when the contacts are open.	3	3 ^a	3 ^a
2	Between live parts of different polarity including all terminals for external wiring.	4 ^{c, e}	4 ^{c, e}	4 ^{c, e}
3	Between live parts and			
	– accessible surfaces of parts of insulating material;	3	3	3
	– earthed metal parts, including the earthing circuit;	3	3	3
	– metal frames supporting the base of flush-type switches;	3	3	3
	– screws or devices for fixing main parts, covers or cover-plates; and	3	3	3
	– metal parts of the mechanism, if required to be insulated from live parts (<i>see 10.4</i>).	3	3	3
4	Between metal parts of the mechanism, if required to be insulated from accessible metal parts (<i>see 10.5</i>), and			
	– screws or devices for fixing main parts, covers or cover-plates;	3	3	3
	– metal frames supporting the base of flush-type switches; and	3	3	3
	– Accessible metal parts.	3	3	3
5	Between live parts and accessible unearthed metal parts, with the exception of screws and the like.	6 ^d	6 ^d	6 ^d
Clearances				
6	Between internal live parts which are separated when the contacts are open.	3	1.2 ^b	< 1.2 ^b
7	Between live parts of different polarity including all terminals for external wiring.	3 ^e	3 ^e	3 ^e
8	Between live parts and			
	– accessible surfaces of insulating material;	3	3	3
	– earthed metal parts, including the earthing circuit, not mentioned under items 9 and 11;	3	3	3
	– metal frames supporting the base of flush-type switches;	3	3	3
	– screws or devices for fixing main parts, covers, or cover-plates; and	3	3	3
	– metal parts of the mechanism, if required to be insulated from live parts (<i>see 10.4</i>).	3	3	3
9	Between live parts and			
	– exclusively earthed metal boxes (<i>see NOTE</i>) with the switch mounted in the most unfavourable position; and	3	3	3
	– unearthed metal boxes, without insulating lining, with the switch mounted in the most unfavourable position.	4.5	4.5	4.5

Table 9 (Concluded)

Description		mm		
		normal gap	mini gap	micro gap
Clearances				
10	Between metal parts of the mechanism, if required to be insulated from accessible metal parts (<i>see 10.5</i>), and			
	– screws or devices for fixing main parts, covers or cover-plates;	3	3	3
	– metal frames supporting the base of flush-type switches; and	3	3	3
	– accessible metal parts when the base is fixed directly on the wall.	3	3	3
11	Between live parts and the surface on which the base of a surface-type switch is mounted when the base is fixed directly on the wall.	6	6	6
12	Between live parts and the bottom of the space, if any, for external conductors, for surface-type switches.	3	3	3
13	Between live parts and accessible unearthed metal parts with the exception of screws and the like.	6 ^d	6 ^d	6 ^d
Distances through insulating sealing compound				
14	Between live parts covered with at least 2 mm of sealing compound and the surface on which the base of a surface-type switch is mounted.	4 ^e	4 ^e	4 ^e
15	Between live parts covered with at least 2 mm of sealing compound and the bottom of the space, if any, for external conductors, for surface-type switches.	2.5	2.5	2.5
NOTE — Exclusively earthed metal boxes are those suitable only for use in installations where earthing of metal boxes is required.				
^a For switches with rated voltage up to and including 250 V the creepage distances may be reduced to 1.2 mm if the insulating material has at least a comparative tracking index (CTI) value of 600 according to IEC 60112 or to 1.8 mm if the insulating material has at least a CTI value of 400 according to IEC 60112.				
^b The clearances between internal live parts shall be greater than the switch contact gap.				
^c The 4 mm value is reduced to 3 mm for switches having a rated voltage up to and including 250 V.				
^d The 6 mm value is reduced to 4.5 mm for accessories having a rated voltage up to and including 250 V.				
^e Clearances and creepage distances between live parts of different polarity are reduced to 1 mm between the lead wires in the pinch of a neon lamp or an LED with external resistor.				

Table 10 Relationship between Rated Currents and Connectable Cross-sectional Areas of Copper Conductors for Screwless Terminals
(Clause 12.3.2)

Rated current A	Conductor		
	Nominal Cross Sectional area mm ²	Diameter of Largest Rigid Conductor mm	Diameter Largest Flexible Conductor mm
Up to and including 4	0.75-1	1.19	—
Above 4 and including 6	1-1.5	1.45	1.73
Above 6 and including 16 ¹⁾	1.5-2.5	2.13	2.21
The diameters shown in the table are 5 percent larger than the diameters according to IS 10462 (Part 1), for the conductors with the largest nominal cross-sectional area			
¹⁾ Each supply terminals of switches other than those of pattern numbers 3, 03 and 7 shall allow the connection of two 2.5 mm ² conductors. In such a case terminal for rated current 10 A with separate independent clamping means for each conductor shall be used.			

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Compliance is checked by inspection and by fitting conductors of the smallest and largest cross-sectional areas specified.

12.3.3 Screwless terminals shall allow the conductor to be connected without special preparation.

Compliance is checked by inspection.

NOTE — The term 'special preparation' covers soldering of the wires of the conductors, use of terminals ends, etc, but not the reshaping of the conductor before its introduction into the terminals or the twisting of a flexible conductor to consolidate the end.

12.3.4 Parts of screwless terminals mainly intended for carrying current shall be of materials as specified in **22.5**.

Compliance is checked by inspection and by chemical analysis.

NOTE — Springs, resilient units, clamping plates and the like are not considered as parts mainly intended for carrying current.

12.3.5 Screwless terminals shall be so designed that they clamp the specified conductors with sufficient contact pressure and without undue damage to the conductor.

The conductor shall be clamped between metal surfaces.

Compliance is checked by inspection and by the test of **12.3.10**.

12.3.6 It shall be clear how the connection and disconnection of the conductors is intended to be effected.

The disconnection of a conductor shall require an operation, other than a pull on the conductor, such that it can be made manually with or without the help of a general purpose tool.

It shall not be possible to confuse the opening for the use of a tool to assist the connection or disconnection with the opening intended for the insertion of the conductor.

Compliance is checked by inspection and by the test of **12.3.10**.

12.3.7 Screwless terminals which are intended to be used for the interconnection of two or more conductors shall be so designed that:

- during the insertion, the operation of the clamping means of one of the conductors is independent of the operation of that of the other conductors(s);
- during the disconnection, the conductors can be disconnected either at the same time or separately;
- each conductor shall be introduced in a separate clamping unit (not necessarily in separate hole; and

- It shall be possible to clamp securely any number of conductors up to the maximum as designed.

Compliance is checked by inspection and by tests with the appropriate conductors.

12.3.8 Screwless terminals shall be so designed that undue insertion of the conductor is prevented and adequate insertion is obvious.

Screwless terminals of switches shall be so designed that undue insertion of the conductor is prevented by a stop if further insertion is liable to reduce the creepage distances and/or clearances required in Table 9, or to influence the mechanism of the switch.

Compliance is checked by inspection and by the test of **12.3.10**.

12.3.9 Screwless terminal shall be properly fixed to the switch.

They shall not work loose when the conductors are connected or disconnected during installations.

Compliance is checked by inspection and by the tests of **12.3.10**.

Covering of sealing compound without other means of locking is not sufficient. Self-hardening resins may, however, be used to fix terminals which are not subject to mechanical stress in normal use.

12.3.10 Screwless terminals shall withstand the mechanical stresses occurring in normal use.

Compliance is checked by the following test, which is carried out with uninsulated conductors on one screwless terminal of each specimen, using a new specimen for each test.

The test is carried out with solid copper conductors, first with conductors having the largest cross-sectional area, and then with conductors having the smallest cross-sectional area specified in **12.3.2**.

Conductors are connected and disconnected five times, new conductors being used each time, except for the fifth time, when the conductors used for the fourth connection are clamped at the same place. For each connection the conductors are either pushed as far as possible into the terminal or are inserted so that adequate connection is obvious.

After each connection, the conductor is subjected to a pull of 30 N; the pull is applied without jerks, for 1 min, in the direction of the longitudinal axis of the conductor space.

During the application of the pull, the conductor shall not come out of the screwless terminal.

The test is then repeated with rigid stranded copper conductors having the largest and smallest cross-sectional areas specified in **12.3.2**; these conductors are, however, connected and disconnected only once.

Screwless terminals, intended for both rigid and flexible conductors, shall also be tested with flexible conductors, applying five connections and disconnections.

Each conductor of screwless terminals is subjected for 15 min to a circular motion with (10 ± 2) r/min using an apparatus, an example of which is shown in Fig. 11. The conductor is subjected to a pull having a value shown in Table 7.

During the test, the conductors shall not move noticeably in the clamping unit.

After these tests, neither the terminals nor the clamping means shall have worked loose and the conductors shall show no deterioration impairing their further use.

12.3.11 Screwless terminals shall withstand the electrical and thermal stresses occurring in normal use.

Compliance is checked by the following tests: (a) and (b), which are carried out on five screwless terminals which have not been used for any other test.

Both tests are carried out with new copper conductors.

- a) The test is carried out loading the screwless terminals for 1 h with an alternating current as specified in Table 11, and connecting rigid solid conductors 1 m long having the cross-sectional area as specified in the same table.

Table 11 Test Current for the Verification of Electrical and Thermal Stresses in Normal Use of Screwless Terminals

(Clause 12.3.11)

Rated Current	Test Current	Cross-sectional Area of the Conductor
A	A	mm ²
Up to including 4	9	0.75
Above 4 and including 6	13.5	1
Above 6 and including 13	17.5	1.5
Above 13 and including 16	22	2.5

The test is carried out on each clamping unit.

During the test the current is not passed through the switch, but only through the terminals.

Immediately after this period, the voltage drop across each screwless terminals is measured with rated current flowing.

In no case shall the voltage drop exceed 15 mV.

The measurements are made across each screwless terminals and as near as possible to the place of contact.

If the back connection of the terminal is not accessible, the second connecting point in the case of two-way switches may be used for the returning wire; in the case of one-way switches the samples may be adequately prepared by the manufacturer, care shall be taken not to affect the behavior of the terminal.

During the period of the test, including the measurements, the conductors and the measurement means shall not be moved so that the result of the test is affected.

The screwless terminals already subjected to the determination of the voltage drop specified in the previous test a) are tested as follows:

- a) During the test, a current equal to the test current value given in Table 11 is passed.

The whole test arrangement, including the conductors, shall not be moved until the measurements of the voltage drop have been completed.

The terminals are subjected to 192 temperature cycles, continuously each cycle having a duration of approximately 1 h and being carried out as follows:

- a) the current is flowing for approximately 30 min; and
- b) for a further 30 min approximately no current is flowing.

The voltage drop in each screwless terminal is determined as prescribed for the test of a) and is done at the following moments:

- a) after the first 24 temperature cycles and after the 192 temperature cycles; and
- b) additional measurements to be done after any three of the following temperature cycles: after the 48th, 72nd, 96th, 120th, 144th or 168th temperature cycles.

In no case shall the voltage drop exceed 22.5 mV or two times the value measured after the 24th cycle, whichever is the smaller.

After this test an inspection by the normal or corrected vision without additional magnification shall show no changes evidently impairing further use such as cracks, deformations or the like.

In addition, the mechanical strength test according to **12.3.10** is repeated and all samples shall withstand this test.

12.3.12 Screwless terminals shall be so designed that the connected rigid solid conductor remains clamped, even when it has been deflected during normal installation, for example during mounting in a box, and the deflecting stress is transferred to the clamping unit.

Compliance is checked by the following test which is made on three samples of switches which have not been used for any other test.

The test apparatus, the principle of which is shown in Fig. 12 A, shall be so constructed that:

A specified conductor properly inserted into a terminal is allowed to be deflected in any of the 12 directions differing from each other by 30° with a tolerance referred to each direction of ± 5 .

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The starting point can be varied by 10° and 20° from the original point.

A reference point need not to be specified.

The deflection of the conductor from its straight position to the testing positions shall be effected by means of a suitable device applying a specified force to the conductor at a certain distance from the terminal.

The deflecting device shall be so designed that:

- the force is applied in the direction perpendicular to the un-deflected conductor;
- the deflection is attained without rotation or displacement of the conductor within the clamping unit; and
- the force remains applied while the prescribed voltage drop measurement is made.

Provision shall be made so that the voltage drop across the clamping unit under test can be measured when the conductor is connected, as shown for example in Fig. 12 B.

The specimen is mounted on the fixed part of the test apparatus in such a way that the specified conductor inserted into the clamping unit under test can be freely deflected.

To avoid oxidation, the insulation of the wire shall be removed immediately before starting the test.

NOTES

1 If necessary, the inserted conductor may be permanently bent around obstacles, so that these do not influence the results of the test.

2 In some cases, with the exception of the case of guidance for the conductor, it may be advisable to remove those parts of the samples which do not allow the deflection of the conductor corresponding to the force to be applied.

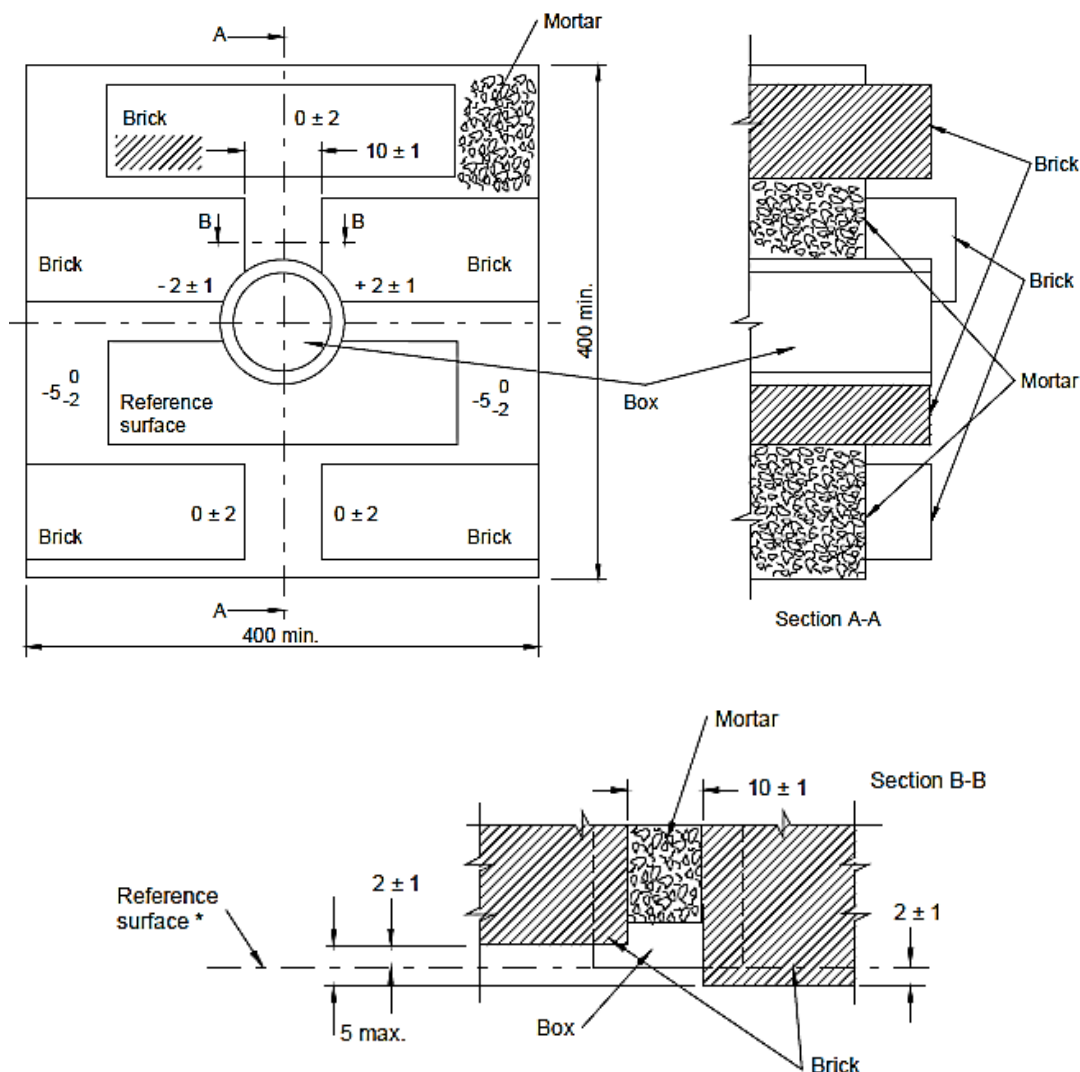
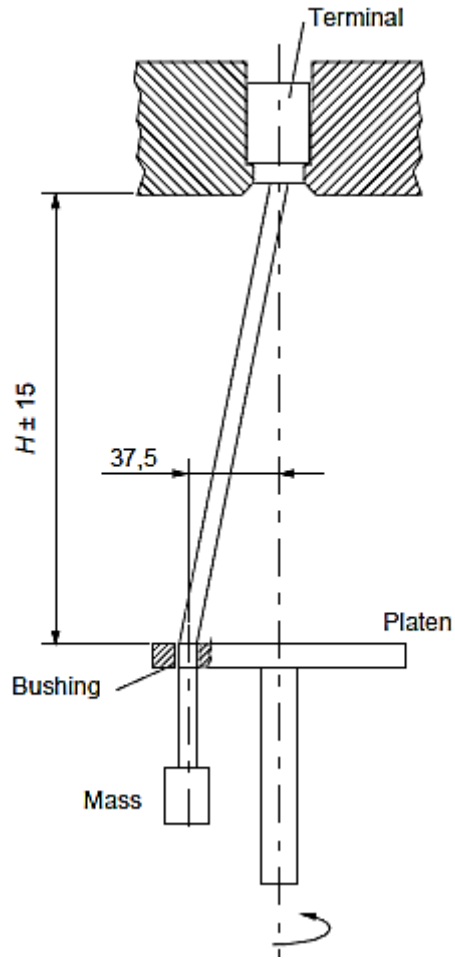


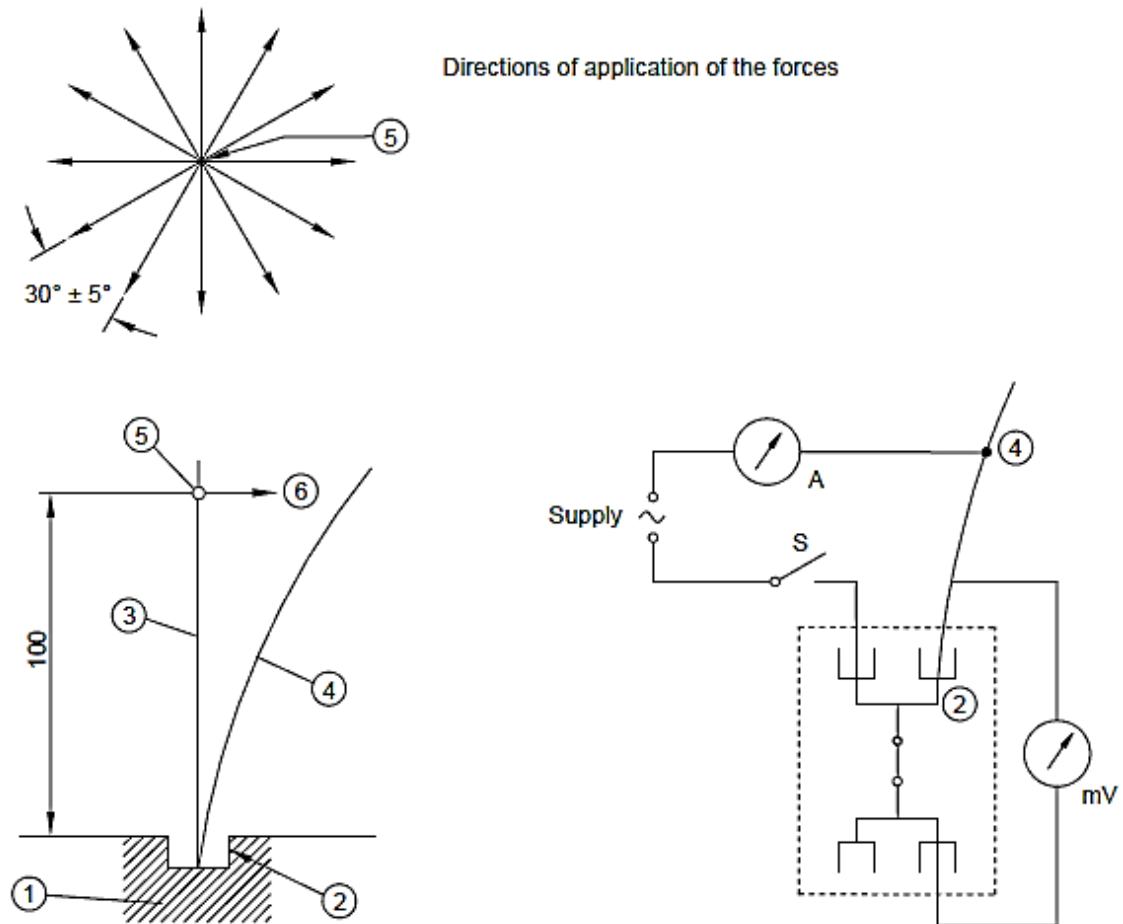
FIG. 10 TEST WALL IN ACCORDANCE WITH THE REQUIREMENTS OF 15.2.3



NOTE — Care should be taken that the bushing hole is made in a way which ensures that the force extended to the cable is pure pulling force and that the transmission of any torque to the connection in the clamping means is avoided.

FIG. 11 ARRANGEMENT FOR CHECKING DAMAGE TO CONDUCTORS

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Key

- A = Ammeter
- mV = Millivoltmeter
- S = Switch
- 1 = Specimen
- 2 = Clamping unit under test
- 3 = Test conductor
- 4 = Test conductor, deflected
- 5 = Point of application of the force for deflecting the conductor
- 6 = Deflection force (perpendicular to the straight conductor)

12 A PRINCIPLE OF THE TEST APPARATUS FOR
DEFLECTION ON SCREWLESS TERMINALS
DEFLECTION

12 B EXAMPLE OF TEST ARRANGEMENT TO
MEASURE THE VOLTAGE DROP DURING TEST
ON SCREWLESS TERMINALS

FIG. 12 INFORMATION FOR DEFLECTION TEST

A clamping unit is fitted as for normal use with a rigid solid copper conductor having the smallest cross-sectional area specified in Table 12 and is submitted to a first test sequence; the same clamping unit is submitted to a second test sequence using the conductor having the largest cross-sectional area, unless the first test sequence has failed.

Table 12 Cross-sectional Areas of Rigid Copper Conductors for Deflection Test of Screwless Terminals

(Clause 12.3.12)

Rated Current of the Switch A	Cross-sectional Area of the Test Conductor mm ²	
	1 st test sequence	2 nd test sequence
≤ 6	1.0 ^a	1.5
Above 6 up to and including 16	1.5	2.5

^aOnly for countries allowing the use of 1.0 mm² conductors in fixed installations.

The force for deflecting the conductor is specified in Table 13, the distance of 100 mm being measured from the extremity of the terminal, including the guidance, if any, for the conductor, to the point of application of the force to the conductor.

Table 13 Deflection Test Forces

(Clause 12.3.12)

Sl No.	Cross-sectional of the Test Conductor Area mm ²	Force for Deflecting the Test Conductor ¹⁾ N
(1)	(2)	(3)
i)	1.0	0.25
ii)	1.5	0.5
iii)	2.5	1.0

¹⁾ The forces are chosen so that they stress the conductors close to the limit of elasticity

The test is made with continuous current (that is the current is not switched on and off during the test). A suitable power supply should be used and an appropriate resistance should be inserted in the circuit so that the current variations are kept within ± 5 percent during the test.

A test current equal to the rated current of the switch is passed through the clamping unit under test. A force according to Table 13 is applied to the test conductor inserted in the clamping unit under test in the direction of one of the 12 directions shown in Fig. 12 A and the voltage drop across this clamping unit is measured. The force is then removed.

The force is then applied successively in each of the remaining 11 direction shown in Fig. 12 A following the same test procedure.

If for any of the 12 test directions the voltage drop is greater than 25 mV, the force is maintained in this direction until the voltage drop is reduced to a value below 25 mV, but for not more than 1 min. After the voltage drop has reached a value below 25 mV, the force is maintained in the same direction for a further period of 30 s, during which period the voltage drop shall not have increased.

The other two samples of switches of the set are tested following the same test procedure, but moving the 12 directions of the force so that they differ by approximately 10 for each specimen. If one specimen has failed at one of the directions of application of the test force, the tests are repeated on another set of specimens, all of which shall comply with the repeated tests.

13 CONSTRUCTIONAL REQUIREMENTS

13.1 Insulating linings, barriers and the like, shall have adequate mechanical strength and shall be secured in a reliable manner. Compliance is checked by inspection after the tests of 20.

13.2 Installation Requirements

Switches shall be so constructed as to permit:

- Easy introduction into the terminal and reliable connection of the conductors in the terminals, except for lead wires of pilot lights;

NOTE — Screw terminals shown in Fig. 1 to Fig. 5 are considered suitable for reliable connection of the conductors.

- Correct positioning of the conductors;
- Easy fixing of the switch to a wall as in a box; and
- Adequate space between the underside of the main part and the surface on which the main part is mounted or between the sides of the main part and the enclosure (cover or box) so that, after installation of the switch, the insulation of the conductors is not necessarily pressed against live parts of different polarity or against moving parts of the mechanism, such as the spindle of a rotary switch.

NOTE — This requirement does not imply that the metal parts of the terminals are necessarily protected by insulation barriers or insulating shoulders, to avoid contact, due to incorrect installation of the terminal metal parts, with the insulation of the conductor.

Surface type switches shall be constructed so that the fixing means do not damage the insulation of the cables during the installation.

Compliance is checked by inspection and by an installation test using conductors of the largest cross-sectional area specified, for the relevant ranges of rated currents, in Table 4.

Surface-type switches shall be constructed so that the fixing means do not damage the insulation of the cables during the installation.

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Compliance is checked by inspection and by an installation test using conductors of the largest cross-sectional area specified, for the relevant ranges of rated currents, in Table 4.

NOTE — For surface-type switches, mounted on a mounting plate, a wiring channel may be needed to comply with this requirement.

In addition, for switches comprising screwless terminals, the switches shall be so constructed that the connecting and/or disconnecting means of the screwless terminals cannot be activated by the conductors during and after installation of the switch in a box or on a wall.

NOTES

1 This requirement does not imply that the connecting and/or disconnecting means cannot be touched by the conductors.

2 This requirement can be met by the placement of the connecting and/or disconnecting means and/or the use of protective barriers or shoulders placed around the connecting and/or disconnecting means.

Compliance is checked by inspection and in case of doubt by the following test.

The test is carried out with a solid copper conductor having the smallest cross-sectional area specified in **12.3.2**.

The conductor is pushed as far as possible into the terminal under test or is inserted so that adequate connection is obvious.

A test probe 1 of IS 1401 is pushed against the connecting or disconnecting means with a force of 120 N in the direction opposite to the mounting direction as described in Fig 13 A.

During the application of the force, the conductor, except for lead wires of pilot lights, is subjected to a pull of 30 N; the pull is applied in one smooth and continuous motion, for 1 min, in the direction of the longitudinal axis of the conductor space.

During the application of the pull, the conductor shall not come out of the screwless terminal.

The force of 120 N has to be applied before the force of 30 N is applied. The force of 30 N is maintained on the conductor during the complete test.

Care should be taken that the test probe does not touch the conductor during the application of the forces.

Where the axis between the application force and the axis through the force necessary to operate the

connecting/disconnecting means deviates by more than 20°, it is allowed to exert the calculated resulting force directly onto the connecting/disconnecting means using the test probe. An example is shown in Fig 13 b.

If the angle is greater than 60° no test is necessary and the product is deemed to comply with the requirements without further tests.

If it is not possible to exert a force onto the connecting/disconnecting device, the product is deemed to comply with the requirements without further tests.

NOTE — This requirement does not imply that the connecting and/or disconnecting means cannot be touched by the cover or cover plate.

13.3 Fixing of Covers, Cover Plates and Actuating Members

13.3.1 Covers, cover plates and actuating members or parts of them, which are intended to ensure protection against electric shock (which cover the current carrying parts ensuring protection against electric shock like knob of switches used for making ON & OFF), shall be held in place at two or more points by effective fixing.

Covers, cover plates and actuating members (knobs) or parts of them may be fixed by means of a single fixing, for example, by a screw, provided that they are located by another means.

NOTES

1 The most appropriate method for fixings covers or cover plates or actuating members is to ensure that they are captive. The use of tight fitting washers of cardboard or the like is deemed to be an adequate method for securing screws intended to be captive.

2 Non-earthed metal part separated from live parts in such a way that creepage distances and clearances have values specified in Table 9, are not considered as accessible if the requirements of **13.3** are met.

Compliance is checked according to **13.3.2**, **13.3.3** or **13.3.4**.

13.3.2 For covers, cover plates or actuating members whose fixing is of the screw-type, by inspection only.

13.3.3 For covers, cover plates or actuating members whose fixing is not dependent on screws and whose removal is obtained by applying a force in a direction approximately perpendicular to the mounting or supporting surface (see Table 14).

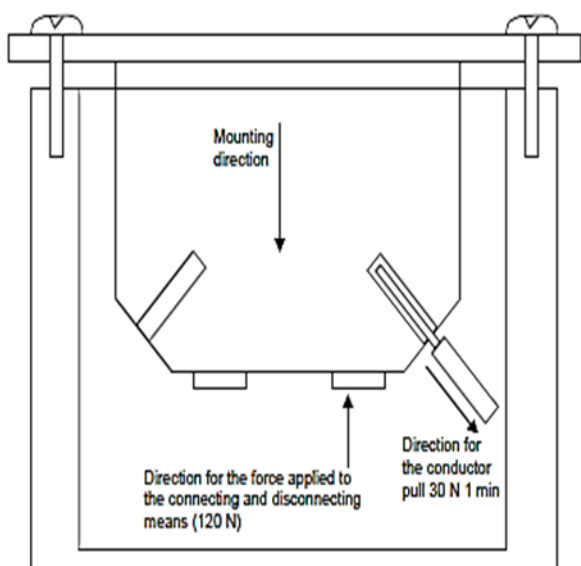
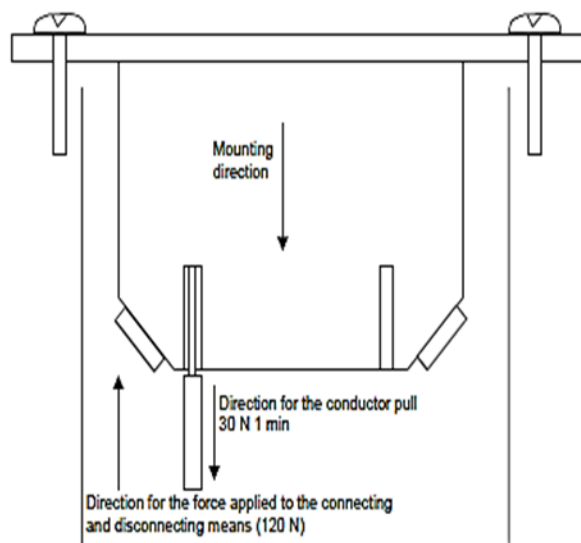


FIG. 13A DETERMINATION OF THE DIRECTION OF THE FORCES TO BE APPLIED

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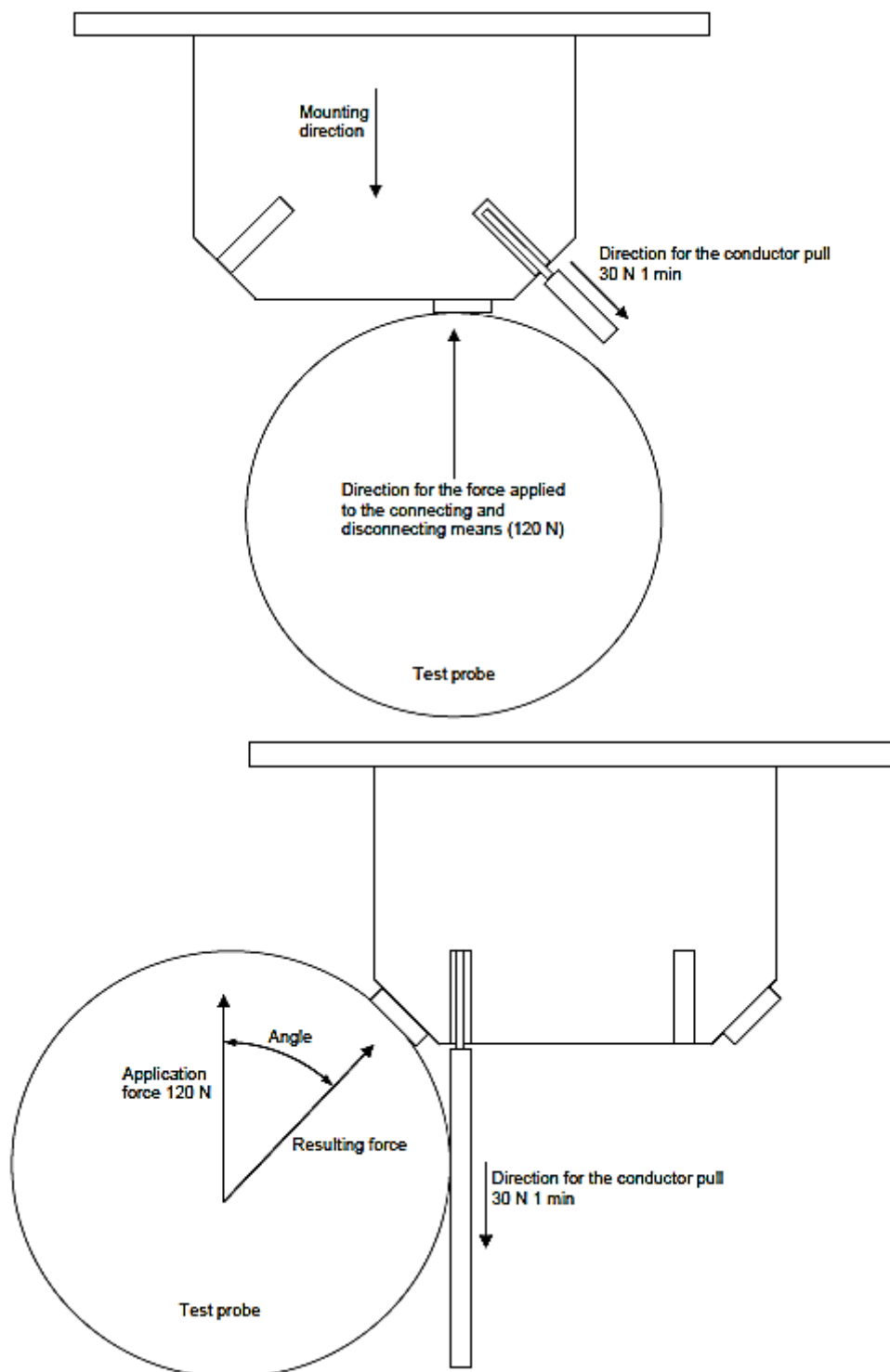


FIG. 13B TEST SET UP

FIG. 13 VERIFICATION OF THE REQUIREMENTS

**Table 14 Forces to be Applied to Covers, Cover-plates or Actuating Members
whose Fixing is not dependent on Screws**

(Clause 13.3.3)

Accessibility with the test Probe B of IS 1401 after Removal of Covers, Cover Plates or Parts of Them	Test according to...	Force to be Applied			
		N			
		Switches complying with 20.8 and 20.9		Switches not complying with 20.8 and 20.9	
		Shall not come off	Shall come off	Shall not come off	Shall come off
To live parts	20.5	40	120	80	120
To non-earthed metal parts separated from live parts by creepage distances according to 23.1	20.6	10	120	20	120
To insulating parts, earthed metal parts, the live parts of SELV ≤ 25 V AC or metal parts separated from live parts by creepage distances and by clearances twice those according to 23.1	20.7	10	120	10	120

- a) when their removal may give access, with the test probe B of IS 1401, to live parts
by the test of **20.5**;
- b) when their removal may give access, with the test prob B of IS 1401, to non-earthed metal parts separated from live parts in such a way that creepage distances and clearances have the values at least equal to those shown in Table 9.
by the test of **20.6**; and
- c) when their removal may give access, with the test probe B of IS 1401, only to
- insulating parts,
 - earthed metal parts,
 - metal parts separated from live parts in such a way that creepage distances, and clearances have at least twice the values shown in Table 9, or
 - live parts of SELV circuits not greater than 25 V AC and 60 V DC:
by the test of **20.7**.

13.3.4 For covers, cover plates or actuating members whose fixing is not dependent on screws and whose removal is obtained by using a tool, in accordance with the manufacturer's instructions given in an instruction sheet or catalogue:

by the same tests of **13.3.3** except that the covers, cover plates, actuating members or parts of them need not come out when applying a force not exceeding 120 N in directions perpendicular to the mounting/supporting surface.

13.4 Openings in Normal Use

Switches shall be so constructed that, when they are fixed and wired as in normal use, there are no free

openings in their enclosures according to their IP classification.

Compliance is checked by inspection and by an installation test with conductors of the smallest cross-sectional area specified in Table 4.

NOTE — Drain holes, small gaps between enclosures and conduits or cables, or between enclosures and operating means can be neglected if they comply with the relevant IP classification.

13.5 Attachment of knobs

Knobs of rotary switches shall be securely attached to the shaft or part operating the mechanism.

Compliance is checked by the following test.

Where it is possible to apply an axial pull in normal use, an axial pull shall be applied for 1 min to try to pull off the actuating member.

If the shape of the actuating member is such that an axial pull is likely to be applied in normal use, the force is 30 N.

If the shape of the actuating member is such that an axial pull is unlikely to be applied in normal use, the force is 15 N.

After this, knobs of switches having only one direction of operation are turned, if possible, without undue force, 100 times in the reverse direction.

During the test, the knob shall not become detached.

13.6 Mounting Means

Screws or other means for mounting the switch on a surface or in a box or enclosure shall be easily accessible from the front. These means shall not serve any other fixing purpose.

NOTE — Switch mounting means, if used to lock the rotation of the box, do not constitute a use for "any fixing purpose".

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13.7 Combinations of Switches

Combinations of switches, or of switches and socket-outlets, comprising separate bases shall be so designed that the correct position of each base is ensured.

The fixing of each base shall be independent of the fixing of the combination to the mounting surface

Compliance with the requirements of **13.6** and **13.7** is checked by inspection.

13.8 Accessories Combined with Switches

Accessories combined with switches shall comply with the relevant standards, if any, unless a standard exist for the combination.

13.9 Surface-type Switches Having an IP Code Higher than IP20

Surface-type switches that have an IP code higher than IP20 shall be according to their IP classification when fitted with conduits or with sheathed cables as for normal use.

Surface-type switches that have degrees of protection IPX4, IPX5 and IPX6 shall have provisions for opening a drain hole.

If a switch is provided with a drain hole, it shall be not less than 5 mm in diameter, or 20 mm² in area with a width and a length not less than 3 mm.

If the design of the switch is such that only one mounting position is possible, then drain hole shall be effective in that position. Alternatively, the drain hole shall be effective in at least two positions of the switch when this is mounted on a vertical wall, one of these with the conductors entering at the top and the other with the conductors entering at the bottom.

Lid springs, if any, shall be of corrosion resistant material, such as bronze or stainless steel.

Compliance is checked by inspection, by measurement and by the relevant tests of **15.2**.

NOTE — A drain hole in the back of the enclosure is deemed to be effective only if the design of the enclosure ensures a clearance of at least 5 mm from the wall, or provides a drainage channel of at least the size specified.

13.10 Installation in a Box

Switches to be installed in a box shall be so designed that the conductor ends can be prepared after the box is mounted in position, but before the switch is fitted in the box.

In addition, the base shall have adequate stability when mounted in the box.

Compliance is checked by inspection and by an installation test with conductors of the largest cross-sectional area specified, for the relevant current rating, in Table 4.

13.11 Connection of a Second Current-carrying Conductor

Surface type switches, other than ordinary, of pattern numbers 1, 5 and 6 with an enclosure having more than one inlet opening shall be provided for maintaining the continuity of a second current-carrying conductor either with a fixed additional terminal complying with the appropriate requirements of **12** or with an adequate space for a floating terminal.

Compliance is checked by inspection and by the relevant tests of **12**.

13.12 Inlet Openings

Inlet openings shall allow the introduction of the conduit or the sheath of the cable so as to afford complete mechanical protection.

Surface-type switches shall be so constructed that the conduit or protective covering can enter at least 1 mm into the enclosure.

In surface-type switches, the inlet opening for conduit entries, or at least two of them if there are more than one, shall be capable of accepting conduit sizes of 16, 20, 25 or 32 or a combination of at least two of any of these sizes not excluding two of the same size.

Compliance is checked by inspection during the test of **13.10** and by measurement.

In surface type switches the inlet opening for cable entries shall preferably be capable of accepting cables having the dimensions specified in Table 15 or be so as specified by the manufacturer.

NOTE — Inlet openings of adequate size may also be obtained by the use of knock-outs or of suitable insertion pieces.

13.13 Provision for Back Entry from a Conduit

If surface-type switches are intended for back entry from a conduit they shall be so designed that they have provision for back entry from a conduit perpendicular to the mounting surface of the switch.

Compliance is checked by inspection.

13.14 Switch Provided with Membranes or the Like for Inlet Openings

If the switch is provided with membranes or the like for inlet openings they shall be replaceable. Compliance is checked by inspection.

13.15 Requirements for Membranes in Inlet Openings

13.15.1 Membranes shall be reliably fixed and shall not be displaced by the mechanical and thermal stresses occurring in normal use.

Compliance is checked by the following test:

Table 15 External Cable Diameter Limits for Surface Type Switches

(Clause 13.12)

Rated Current A	Cross-Sectional Area of Conductors mm ²	Number of Conductors	Limits of External Diameters or Dimensions of Cables	
			Minimum mm	Maximum mm
6	0.75 up to and including 1.5	2	3.2 × 5.2 5.2	3.9 × 6.4 11.5
10	1 up to and including 2.5	2	6.4	13.5
		3		14.5
		4		15.5
16	1.5 up to and including 4	2	7.4	15
		3		16
		4		18
20-25	2.5 up to and including 6	2	8.9	18.5
		3		20
		4		22
32	4 up to and including 10	2	10.5	24
		3		25.5
		4		28
40	6 up to and including 16	2	11.5	27.5
45		3		29.5
50		4		32
63	10 up to and including 25	2	14.5	31.5
		3		34
		4		37.5

NOTE — The limits of external dimensions of cables specified in this table are based on IS 694 and IS 9968 (Part 1).

Membranes are tested when assembled in the switches.

First the switches are fitted with membranes which have been subjected to the treatment specified in **15.1**.

The switches are then placed for 2 h in a heating cabinet as described in **15.1**, the temperature being maintained at $(40 \pm 2) ^\circ\text{C}$.

Immediately after this period, a force of 30 N is applied for 5 s to various parts of the membranes by means of the tip of a straight unjointed test finger of the same dimensions as the standard test finger shown in IS 1401.

During these tests, the membranes shall not deform to such an extent that the live parts become accessible.

For membranes likely to be subjected to an axial pull in normal use, an axial pull of 30 N is applied for 5 s. During this test, the membranes shall not come out.

The test is then repeated with membranes which have not been subjected to any treatment.

13.15.2 It is recommended that membranes be so designed and; made of such material that the

introduction of the cables into the switch is permitted when the ambient temperature is low.

Compliance is checked by the following test:

The switches are fitted with membranes which have not been subjected to any ageing treatment, those without opening being suitably pierced.

The switches are then kept, for 2 h in a refrigerator at a temperature of $(-15 \pm 2) ^\circ\text{C}$.

After this period, the switches are removed from the refrigerator and immediately afterwards, while the switches are still cold, it shall be possible to introduce, without undue force, cables of the heaviest type through the membranes.

After the tests in **13.15.1** and **13.15.2**, the membranes shall show no harmful deformation, cracks or similar damage which would lead to non-compliance with this standard.

13.16 Pilot Light Units

Pilot light units shall comply with **101.1.1.1** and **102** of IS/IEC 60669-2-1.

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14 MECHANISM

14.1 Indication of the Position

The actuating member of a switch, when released, shall automatically take up the position corresponding to that of the moving contacts except that for cord-operated switches and for those with a single push-button, the actuating member may take up a single rest position.

14.2 Rest and Intermediate Position

Switches shall be so constructed that the moving contacts can come to rest only in the “on” and “off” positions, an intermediate position being, however, permissible if it corresponds to the intermediate position of the actuating member, and if the insulation between the fixed and the moving contacts is then adequate.

If necessary, the insulation between the fixed and the moving contact, when in an intermediate position, is checked by an electric strength test as specified in 16.3, the test voltage being applied between the relevant terminals without removing the cover or cover plate of the switch.

Compliance with the requirements of 14.1 and 14.2 is checked by inspection and by manual test.

14.3 Undue Arcing

Switches shall be so constructed that undue arcing cannot occur when the switch is operated slowly.

Compliance is checked at the end of the test of 19, breaking the circuit a further ten times, the actuating member being, however, moved steadily by hand over a period of 2 s and moving contacts being stopped, if possible, in an intermediate position, the actuating member being then released.

During the test, no sustained arcing shall occur nor there do any spark when the switch is in ‘ON’ and/or ‘OFF’ position.

14.4 Making and Breaking

Switches of pattern No. 2, 3, 03 and 6/2 shall make and break all poles substantially simultaneously except that for switches of pattern No. 03, the neutral shall not make after or break before the other poles. Compliance is checked by inspection and by manual test when the switch is mounted with the cover, cover plates and actuating members installed as for normal use.

14.5 Action of the Mechanism without Cover or Cover Plate

The action of the mechanism if the cover plates are removable for installation purposes shall be independent of the presence of the cover or cover plate.

NOTE — The actuating member, in some constructions, may constitute the cover.

Compliance is checked by connecting the switch, without cover or cover plate fitted, in series with a lamp and by normally pressing the actuating member without undue force as in normal use.

During the test the lamp shall not flicker.

14.6 Pull Force for Cord-operated Switches

Cord-operated switches shall be capable of effecting a change from the “off” position by application and removal of a steady pull not exceeding 45 N applied vertically and 65 N applied at $45^\circ \pm 5^\circ$ to the vertical and in a plane perpendicular to the mounting surface when the switches are mounted as in normal use as specified by the manufacturer.

Compliance is checked by manual test.

15 RESISTANCE TO AGEING, TO HARMFUL INGRESS OF WATER AND TO HUMIDITY

15.1 Resistance to Ageing

Switches shall be resistant to ageing.

Parts intended for decorative purposes only, such as certain lids, shall be removed if possible and these parts are not subjected to the test.

Compliance is checked by the following test:

Switches and boxes, mounted as for normal use, are subjected to a test in a heating cabinet with an atmosphere having the composition and pressure of the ambient air and ventilated by natural circulation.

Switches having an IP code higher than IPX0 are tested after having been mounted and assembled as prescribed in 15.2.2.

The temperature in the cabinet is $(70 \pm 2)^\circ\text{C}$.

The specimens are kept in the cabinet for 7 days (168 h).

The use of an electrically heated cabinet is recommended.

Natural circulation may be provided by holes in the wall of the cabinet.

After the treatment, the specimens are removed from the cabinet and kept at room temperature and at a relative humidity between 45 percent and 55 percent for at least four days (96 h).

The specimens shall show no crack visible with normal or corrected vision without additional magnification, nor shall the material have become sticky or greasy, this being judged as follows:

With the forefinger wrapped in a dry piece of rough cloth the specimen is pressed with a force of 5 N.

No traces of the cloth shall remain on the specimen and the material of the specimen shall not stick to the cloth.

After the test, the specimens shall show no damage which would lead to non-compliance with this standard.

NOTE — The force of 5 N can be obtained in the following way.

The specimen is placed on one of the pans of a balance and the other pan is loaded with a mass equal to the mass of the specimen plus 500 g.

Equilibrium is then restored by pressing the specimen with the forefinger wrapped in a dry piece of rough cloth.

15.2 Protection Provided by Enclosures of Switches

15.2.1 General

The enclosure of the switch shall provide protection against access to hazardous parts, against harmful effect due to ingress of solid foreign objects and against effects due to ingress of water in accordance with the IP classification of the switch.

Compliance is checked by the tests of 15.2.2 and 15.2.3.

15.2.2 Protection Against Access to Hazardous parts and Against Harmful Effects due to Ingress of Solid Foreign Objects

15.2.2.1 General

Enclosures of switches shall provide a degree of protection against access to hazardous parts and against harmful effects due to ingress of solid foreign objects in accordance with the IP classification of the switch.

Compliance is checked by the appropriate tests of IS/IEC 60529 under the conditions specified below.

The switches are mounted as in normal use.

Flush-type and semi-flush-type switches are mounted in an appropriate box according to the manufacturer's instructions.

Switches with screwed glands or membranes are fitted and connected with cables within the connecting range specified in Table 4. Glands are tightened with a torque equal to two-thirds of that applied during the test of 20.4.

Screws of the enclosure are tightened with a torque equal to two-thirds of the values given in Table 5.

Parts which can be removed without the aid of a tool are removed.

If a switch passes the test successfully, then this test is deemed to be passed for combination of such single switches.

Glands are not filled with sealing compound or the like.

15.2.2.2 Protection against access to hazardous parts

The appropriate test specified in IS/IEC 60529 is performed (*see also* 10).

15.2.2.3 Protection against harmful effects due to ingress of solid foreign bodies

The appropriate test according to IS/IEC 60529 is performed.

Test probes are not applied to drain holes.

For the test of the first characteristic numeral 5, enclosures of switches are considered to be of category 2 (*see* IS/IEC 60529) dust shall not penetrate in a quantity to interfere with satisfactory operation or impair safety.

For the test of the first characteristic numeral 6, enclosures of switches are considered to be of category 1 (*see* IS/IEC 60529); no dust shall penetrate.

15.2.3 Protection against Harmful Effects due to Ingress of Water

Enclosures of switches shall provide a degree of protection against harmful effects due to ingress of water in accordance with their IP classification.

Compliance is checked by the appropriate tests of IS/IEC 60529 under the conditions specified below.

Flush-type and semi flush-type switches are fixed in a test wall representing the intended use of the switch using an appropriate box in accordance with the manufacturer's instructions.

Where the manufacturer's instructions specify particular types of walls, these walls as well as any special installation requirements for the switch, shall be described in sufficient detail (*see* 8.7).

Where the manufacturer's instructions do not specify a type of wall, the test wall according to Fig. 10 is used. It is made with bricks having flat smooth surfaces. When the box is mounted in the test wall, it shall fit tight against the wall so that water cannot enter between the box and the wall.

If sealing material is used in order to seal the box into the wall, it shall not influence the sealing properties of the specimen to be tested.

NOTE — Fig. 10 shows an example where the edge of the box is positioned in the reference plane. Other positions are possible according to the manufacturer's instructions.

The test wall is placed in a vertical position.

Surface-type switches are mounted as in normal use on a vertical surface and fitted with cables with conductors whose largest and smallest cross-sectional area is given in Table 4, as appropriate for their rating.

Screws of the enclosure operated when mounting the switch are tightened with a torque equal to two-thirds of the appropriate value given in Table 5.

Glands are tightened with a torque equal to two-thirds of the appropriate value given in Table 16.

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Table 16 Torque for the Verification of the Mechanical Strength of Glands
(Clauses 15.2.3 and 20.4)

Diameter of Test Rod mm	Torque Nm	
	Metal glands	Glands of moulded material
Up to and including 14	6.25	3.75
Above 14 and up to and including 20	7.5	5.0
Above 20	10.0	7.5

Glands are not filled with sealing compound or the like.

Parts which can be removed without the aid of a tool are removed.

If the enclosure of a switch that has an IP code less than IPX5 is designed with drain holes, one drain hole is opened and in the lowest position. If an enclosure of a switch that has an IP code equal to or greater than IPX5 is designed with drain holes, they shall not be opened.

Care should be taken not to disturb, for example knock or shake, the assembly to such an extent as to affect test results.

If the switch has drain holes which have been opened, it shall be proved by inspection that any water which enters does not accumulate and that it drains away without doing any harm to the complete assembly.

The specimens shall withstand an electric strength test as specified in 16.3, which shall be started within 5 min of completion of the tests according to 15.2.

Immediately after the test specified in 15.2.2 and 15.2.3 of IS 12063, the specimen shall withstand an electric strength test as specified in 16.2 and inspection shall show that water has not entered the samples to any appreciable extent and has not reached live parts.

15.3 Resistance to Humidity

Switches shall be proof against humidity which may occur in normal use.

Compliance is checked by the humidity treatment described in this sub-clause, followed immediately by the measurement of the insulation resistance and by the electric strength test specified in 16.

Inlet openings, if any, are left open; if knock outs are provided, one of them is opened.

Parts which can be removed without the aid of a tool, are removed and subjected to the humidity treatment with the main part; spring lids are open during this treatment.

The humidity treatment is carried out in a humidity cabinet containing air with a relative humidity maintained between 91 percent and 95 percent.

The temperature of the air in which the specimens are placed is maintained within ± 1 K of any convenient value between 20 °C and 30 °C.

Before being placed in the humidity cabinet, the specimens are brought to a temperature between t and $t + 4$ °C.

The specimens are kept in the cabinet for:

- 2 days (48 h) for switches that have an IP code IPX0; and
- 7 days (168 h) for switches that have an IP code higher than IPX0

NOTES

1 In most cases, the specimens may be brought to the specified temperature by keeping them at this temperature for at least 4 h before the humidity treatment.

2 A relative humidity between 91 percent and 95 percent can be obtained by placing in the humidity cabinet a saturated solution of sodium sulphate (Na_2SO_4) or potassium nitrate (KNO_3) in water having a sufficient large contact surface with the air.

3 In order to achieve the specified conditions within the cabinet, it is necessary to ensure constant circulation of the air within and, in general, to use a cabinet which is thermally insulated.

4 For the purpose of acceptance test, the humidity treatment is subjected for 24 h.

After this treatment, the specimens shall show no damage within the means of this standard.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH

16.1 General

The insulation resistance and the electric strength of switches shall be adequate.

One pole of any pilot light is disconnected for the tests of 16.

Compliance is checked by the following tests, which are made immediately after the test of 15.3 in the humidity cabinet or in the room in which the specimens were brought to the prescribed temperature, after reassembly of those parts which can removed without the aid of tool and were removed for the test.

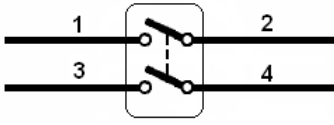
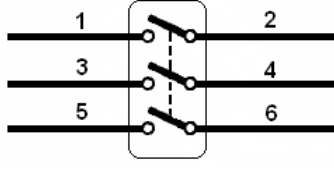
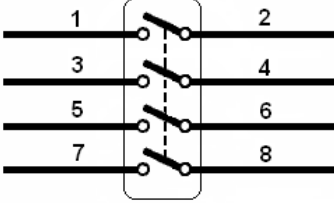
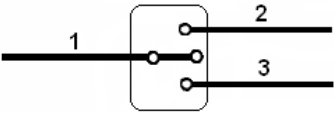
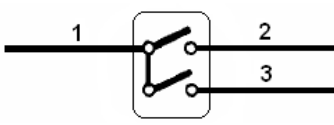
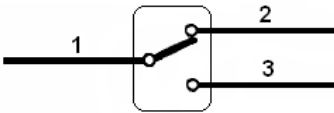
16.2 Test for Measuring the Insulation Resistance

The insulation resistance is measured with a d.c. voltage of approximately 500 V, the measurement being made 1 min after application of the voltage.

The measurements are made consecutively as indicated in Table 17 the switch positions and the connections necessary for items 1, 2 and 3 (see Table 18) being as shown in Table 17.

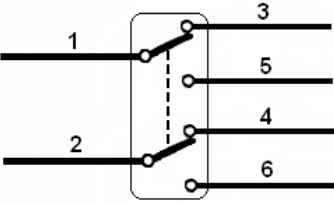
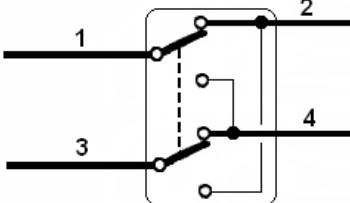
**Table 17 Points of Application of The Test Voltage for the Verification of
Insulation Resistance and Electric Strength**

(Clauses 16.2 and 16.3)

Pattern Number	Diagrams of Connections	Position	Number of Poles	Application of the voltage a	
				Between Terminal No.	Between Body (B) together with Terminal No.
1	 Single-pole switch	Off	1	1	B + 2
		On		2	B + 1
2	 Two-pole switch	Off	2	1 + 3 2 + 4	B + 2 + 4 B + 1 + 3
		On		1 - 2 1 - 2 + 3 - 4	B + 3 - 4 B
3	 Three-pole switch	Off	3	1 + 3 + 5 2 + 4 + 6	B + 2 + 4 + 6 B + 1 + 3 + 5
		On		1 - 2 3 - 4 5 - 6	B + 3 - 4 + 5 - 6 B + 1 - 2 + 5 - 6 B + 1 - 2 + 3 - 4
03	 Three-pole plus switched neutral	Off	4	1 + 3 + 5 + 7 2 + 4 + 6 + 8	B + 2 + 4 + 6 + 8 B + 1 + 3 + 5 + 7
		On		1 - 2 + 5 - 6 1 - 2 + 7 - 8	B + 3 - 4 + 7 - 8 B + 3 - 4 + 5 - 6
4	 Two-way switch with one off position	Off	1	1	B + 2 + 3
		On		1 - 2 1 - 3	B + 3 B + 2
5	 Two-circuit switch with a common incoming line	Off	1	2 + 3 1	B + 1 B + 2 + 3
		On		1 - 3 1 - 2 - 3	B + 2 B
6	 Two-way switch	—	1	1 - 3 1 - 2	B + 2 B + 3

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Table 17 (Concluded)

Pattern Number	Diagrams of Connections	Position	Number of Poles	Application of the voltage a	
				Between Terminal No.	Between Body (B) together with Terminal No.
6/2	 Two-way double-pole switch	—	2	1 – 3 + 2 – 4 1 – 5 + 2 – 6	B + 5 + 6 B + 3 + 4
7	 Two-way reversing switch (or intermediate switch)	—	1	1 – 2 3 – 4 1 – 4 2 – 3	B + 3 – 4 B + 1 – 2 B + 2 – 3 B + 1 – 4
^a – denotes an existing electrical connection.					
+ denotes an electrical connection made for the test.					

In addition, if electrically independent pattern numbers are combined in a common base the measurements are made between all terminals of one pattern number connected together and to the body and to all terminals of the other pattern numbers connected together. This is repeated for each contact position and for each pattern number of the combination. The minimum insulation resistance shall be 5 MΩ.

The term “body” includes accessible metal parts, metal frames supporting the base of flush-type switches operating keys, metal foil in contact with the outer surface of accessible external parts and operating keys of insulating material, the point of anchorage of the cord, chain or rod for switches operated by such means, fixing screws of bases or covers and cover plates, external assembly screws, earthing terminals and any

metal part of the mechanism if required to be insulated from live parts (*see 10.4*).

For the measurements according to items 1 and 2, the metal foil is applied in such a way that sealing compound is effectively tested.

The test according to 5 is only made if any insulating lining is necessary to provide insulation.

The insulation resistance shall be not less than the values shown in Table 18.

While wrapping the metal foil around the outer surface or placing it in contact with the inner surface of parts of insulating material, it is pressed against holes or grooves without any appreciable force, by means of a straight unjointed test finger having the same dimensions as the standard test finger shown in IS 1401.

Table 18 Test Voltage, Points of Application and Minimum Values of Insulating Resistance for the Verification of Electric Strength
(Clauses 16.2 and 16.3)

Insulation to be Tested	Minimum Value of Insulation Resistance MΩ	Test Voltage V	
		Switches having a Rated Voltage not Exceeding 130 V	Switches having a Rated Voltage Exceeding 130 V
1. Between all poles connected together and the body, with the switch in the “on” position.	5	1 250	2 000
2. Between each pole in turn and all other poles connected to the body, with the switch in the “on” position.	2	1 250	2 000
3. Between the terminals which are electrically connected together when the switch is in the “on” position, the switch being in the “off” position:			
a) normal/mini-gap construction;	2	1 250	2 000
b) micro-gap construction; and	2	500 (Note 1)	1 250 (Note 2)
c) Semiconductor switching devices.	(Note 3)	(Note 3)	(Note 3)
4. Between metal parts of the mechanism, when insulated from live parts, and			
a) live parts;	5	1 250	2 000
b) metal foil in contact with the surface of the knob or a similar actuating member;	5	1 250	2 000
c) the key of key-operated switches, if insulation is required (see 10.6);	5	1 250	2 000
d) the point of anchorage of the cord, chain or rod of switches operated by such means, if insulation is required (see 10.6);	5	1 250	2 000
e) accessible metal parts, including fixing screws, of the main part, if insulation is required (see 10.5).	5	1 250	2 000
5. Between any metal enclosure and metal foil in contact with the inner surface of its insulating linings, if any (Note 4).	5	1 250	2 000
6. Between live parts and accessible metal parts, if the metal parts of the mechanism are not insulated from live parts.	—	2 000	3 000
7. Between live parts and parts of the mechanism			
f) if the latter parts are not insulated from accessible metal parts (see 10.5);	—	2 000	3 000
g) if the latter parts are not insulated from the point of contact with a removable key or operating cord, chain or rod (see 10.6).	—	2 000	3 000
8. Between live parts and metal knobs, push-buttons and the like (see 10.2).	—	2 500	4 000

NOTES

1 This value also applies for the electric strength test after normal operation.

2 For switches having a rated voltage up to and including 250 V, this value is reduced to
– 750 V for the electric strength test after resistance to humidity;
– 500 V for the electric strength test after normal operation.

3 Tests for the verification to the “off” position of semiconductor switching devices according to item 3 are under consideration.

4 This test is only made if any insulation is necessary.

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16.3 Electric Strength Test

The insulation is subjected for 1 min to a voltage of substantially sine wave form, having a frequency of 50 Hz or 60 Hz. The value of the test voltage and the points of application are shown in Table 18.

The tests are made consecutively as indicated in Table 17 at points of application as described in Table 18, the switch position and the necessary connections for items 1, 2 and 3 of Table 18 being as shown in Table 17.

In addition, if electrically independent pattern numbers are combined in a common base the test is made between all terminals of the one pattern number connected together and to the body and to all terminals of the other pattern numbers connected together. This is repeated for each contact position and for each pattern number of the combination. The test voltage applied shall be 1 250 V for ratings not exceeding 130 V or 2 000 V for ratings greater than 130 V.

Initially, not more than half the prescribed voltage is applied, then it is raised rapidly to the full value.

No flashover or breakdown shall occur during the test.

NOTES

1 The high-voltage transformer used for the test shall be so designed that, when the output terminals are short-circuited after the output voltage has been adjusted to the appropriate test voltage, the output current is at least 200 mA.

2 The over current relay shall not trip when the output current is less than 100 mA.

3 Care is taken that the r.m.s. value of the test voltage applied is measured within ± 3 percent.

4 Glow discharges without drop in voltage are neglected.

17 TEMPERATURE RISE

17.1 Switches shall be so constructed that the temperature rise in normal use is not excessive.

The metal and the design of the contacts shall be such that the operation of the switch is not adversely affected by oxidation or any other deterioration.

Compliance is checked by the following test.

The switches are mounted vertically as in normal use fitted with PVC insulated rigid solid or stranded copper conductors as specified in Table 19, the terminal screws or nuts being tightened with a torque equal to two-thirds of that specified in Table 5.

To ensure normal cooling of the terminals, the conductors connected to them shall have a length of at least 1 m.

The switches are loaded for 1 h with an alternating current having the value shown in Table 19.

Table 19 Temperature-rise Test Currents and Cross-sectional Areas of Copper Conductors
(Clause 17.1)

Rated current	Test current	Nominal cross-sectional area of conductors
A	A	mm ²
1	1.5	0.5
2	3	0.75
4	5	1
6	8	1.5
10	13.5	2.5
13	17	2.5
16	20	4 ^a
20	25	4
25	32	6
32	38	10
40	46	16
45	51	16
50	57.5	16
63	75	25

^a For switches having a rated voltage not exceeding 250 V, other than those of pattern numbers 3 and 03, provided with screwless terminals, the test shall be carried out with conductors having a cross-sectional area of 2.5 mm².

NOTE — The test currents for switches having other rated currents are determined by interpolation between the next lower and higher ratings.

For switches of pattern numbers 4, 5, 6, 6/2 and 7, only one circuit is loaded.

Flush-mounted switches are mounted in flush-mounted boxes. The box is placed, in a block of pinewood filled around the box with plaster, so that the front edge of the box does not protrude and is not more than 5 mm below the front surface of the pinewood block.

The test assembly should be allowed to dry for at least 7 days when first made.

The size of the pinewood block, which may be fabricated from more than one piece, shall be such that there is at least 25 mm of wood surrounding the plaster, the plaster having a thickness between 10 mm and 15 mm around the maximum dimensions of the sides and rear of the box.

NOTE — The sides of the cavity in the pinewood block may have a cylindrical shape.

The cables connected to the switch shall enter through the top of the box, the point(s) of entry being sealed to prevent the circulation of air. The length of each conductor within the box shall be 80 mm $\pm$ 10 mm.

Surface-type switches shall be mounted centrally on the surface of a wooden block, which shall be at least 20 mm thick, 500 mm wide and 500 mm high.

Other types of switches shall be mounted according to the manufacturer's instructions or, in the absence of such instructions, in the position of normal use considered to give the most onerous conditions.

The test assembly shall be placed in a draught-free environment for the test.

The temperature is determined by means of melting particles, color changing indicators or thermocouples, so chosen and positioned that they have negligible effect on the temperature being determined.

The temperature rise of the terminals shall not exceed 45 K.

During the test the temperature rise necessary to perform the test of **21.3** shall be determined.

NOTE — Undue oxidation of the contacts may be prevented by sliding action or by the use of silver or silver-faced contacts.

In case of combination of switches, the test is carried out separately on each switch.

17.2 Switches Incorporating Pilot Lights

Switches incorporating pilot lights or intended to incorporate pilot lights shall be so designed that in normal use the temperature of accessible surfaces is not excessive.

Compliance is checked by the following test:

The switch is mounted and connected as in **17.1** with the pilot light supplied at rated voltage.

so that it is constantly illuminated for 1 h. The switch is loaded at its rated current.

The temperature rise of the external surface of the switch shall not exceed:

- 45 K for knobs, handles, sensing surfaces, etc., of non-metallic material;
- 55 K for other external parts of non-metallic material;
- 35 K for knobs, handles, sensing surfaces, etc., of metallic material; and
- 45 K for other external parts of metallic material.

Pilot lights using neon lamps or LEDs consuming no more than 3 mA are not tested.

18 MAKING AND BREAKING CAPACITY

18.1 General

Switches shall have adequate making and breaking capacity.

For the purpose of this test, pilot lights are disconnected.

Compliance is checked by the test of **18.2** and for, switches having a rated current not exceeding 16 A and

having a rated voltage up to and including 250 V, and for switches of pattern number 3 or number 03 having a rated voltage over 250 V, by the additional tests of **18.3**.

Cord-operated switches shall be tested mounted as in normal use and with a pull of a value adequate to operate the cord-operated switch, but not exceeding 50 N, on the cord throughout the test, at $30^\circ \pm 5^\circ$ to the vertical in a plane perpendicular to the mounting surface.

The tests are made by means of an apparatus the principle of which is as shown in Fig. 14.

The connections are as shown in Fig. 15.

For switches of pattern number 7, one side is tested as pattern number 6. The test is repeated on the other side of a new set of specimens if the other side is not identical.

For switches of pattern numbers 6, 6/2 and 7, the selector switch S shown in Fig. 15 is moved after the fraction of the total number of operations indicated in Table 20.

Switches are fitted with conductors as for the test of **17**.

Table 20 Fractions of Total Number of Operations

(Clauses 18.1, 18.2 and 18.3)

Pattern Number	Type of Switch	Fraction for Switch S
1, 2, 4 or 5	Rotary, both directions	—
	Other types	—
3 or 03	Rotary, both directions	—
	Other types	—
6, 6/2 or 7	Rotary, both directions	1/4 and 3/4
	Other types	1/2

18.2 Overload

Switches are tested at 1.1 times rated voltage and 1.25 times rated current.

The tolerance for the test voltage and test current is $^{+5}_0$ percent.

They are subjected to 200 operations at a uniform rate of;

- 30 operations per minute, if the rated current does not exceed 13 A;
- 15 operations per minute, if the rated current exceeds 13 A, but is less than 25 A; and
- 7.5 operations per minute, if the rated current is 25 A or more.

The on period shall be 25^{+5}_0 percent of the total cycle and the off period 75^{+5}_0 percent.

For rotary switches intended to be operated in either direction, the actuating member is turned in one direction for half the total number of operations and in the reverse direction for the remainder.

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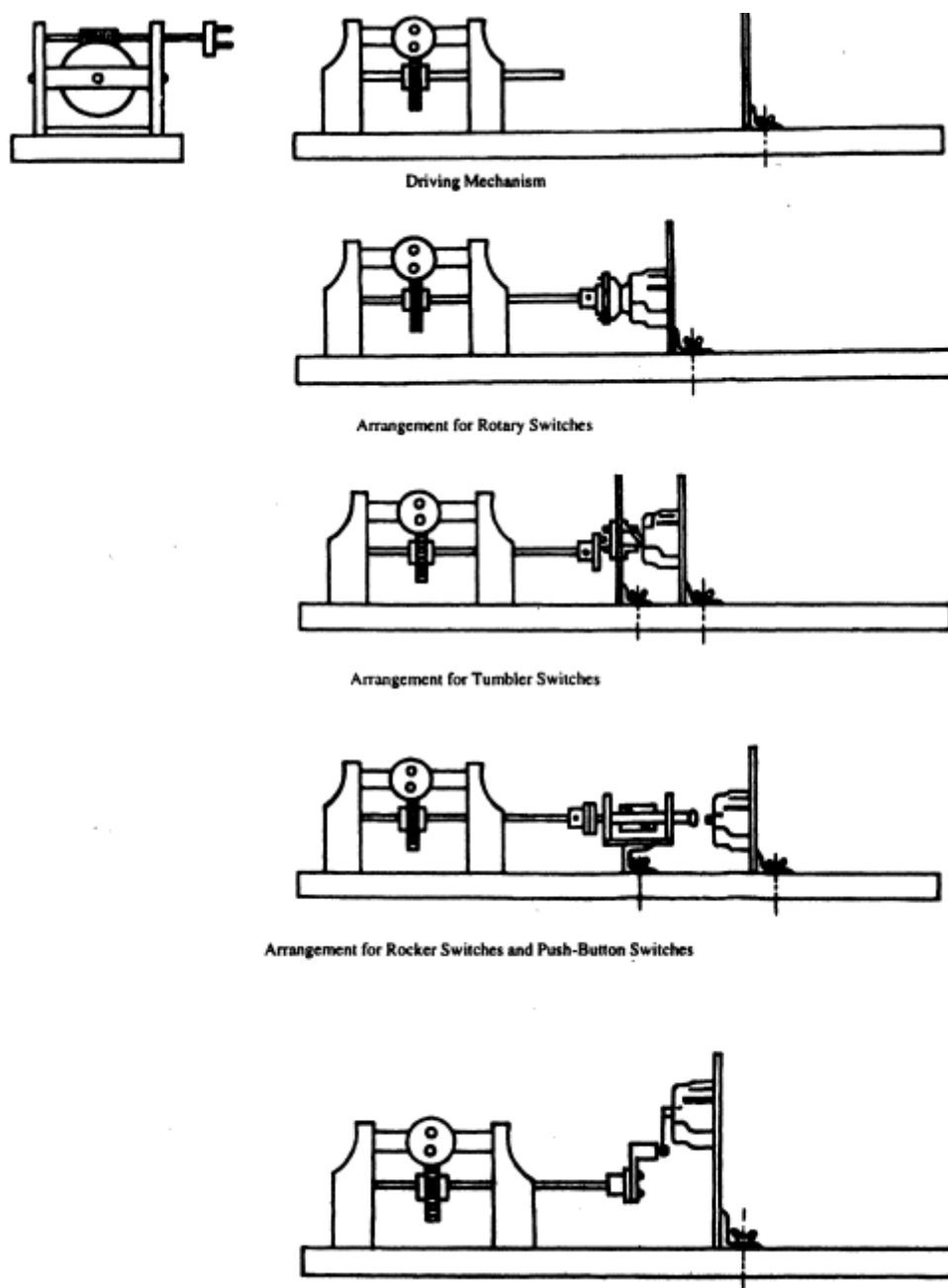
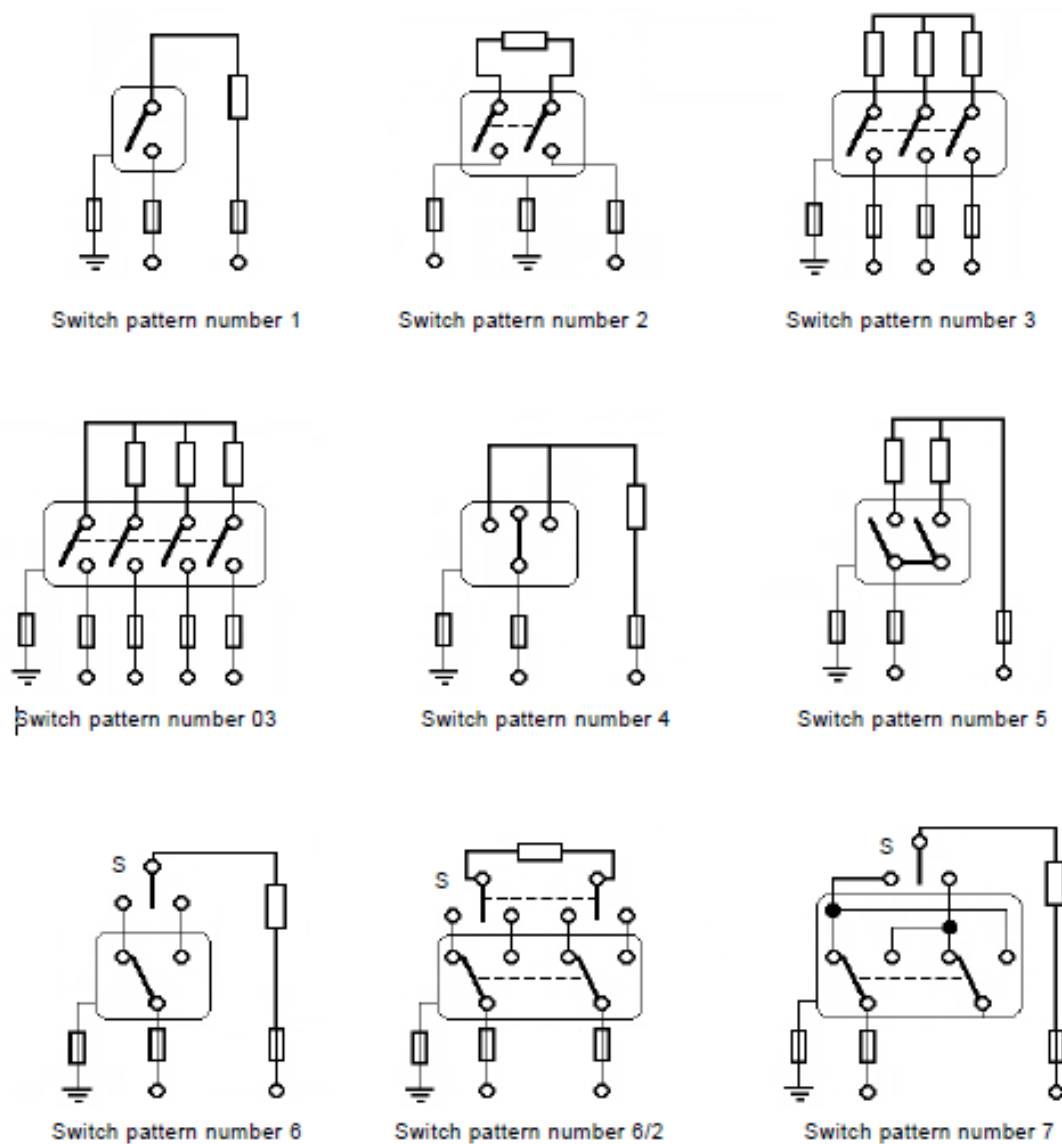


FIG. 14 APPARATUS FOR BAKING AND MAKING CAPACITY AND NORMAL OPERATION TEST



When the marking made by the manufacturer indicates other connections, this marking shall be followed.

FIG. 15 CIRCUIT DIAGRAMS FOR MAKING AND BREAKING CAPACITY AND NORMAL OPERATION

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Switches are tested with a.c. ($\cos \phi$ of 0.3 ± 0.05). Resistors and inductors are not connected in parallel, except that, if an air-core inductor is used a resistor taking approximately 1 percent of the current through the inductor is connected in parallel with it.

Iron-core inductors may be used, provided the current has substantially sinewave form.

For three-phase tests, three-core inductors are used.

The metal support of the switch, if any, on which the switch is mounted and the accessible metal parts of the switch, if any, shall be earthed through a fuse which shall not blow during the test. The fuse shall consist of a copper of 0.1 mm diameter and not less than 50 mm in length.

Switches of pattern number 5 with a single mechanism are operated 200 times with one circuit loaded with rated current (I_n) and the other with $0.25 I_n$ and 200 times with each circuit loaded with $0.625 I_n$.

Switches of pattern number 5 with two independent mechanisms are tested as two switches of pattern number 1, the tests being made consecutively. If both circuits are identical, only one circuit need be tested.

While testing one part, the other part is in the 'off' position.

Switches of pattern number 4 shall be tested for half the number of operations on one way and half the number of operations on the other way.

During the test no sustained arcing shall occur.

Sticking of the contacts which does not prevent further operation of the switch is not regarded as welding.

Sticking of contacts is permitted if the contacts can be separated with a force applied to the actuator of a value which does not damage the switch mechanically.

For the purpose of this test, blows which are intended to separate welded contacts are not permitted.

After the test, the samples shall show no damage which may impair their further use.

Breakage of the replaceable pull cord, not involving the part entering the cord operated switch, shall not be considered a failure to pass the test.

Care is taken that the test apparatus causes the actuating member of the switch to operate smoothly and does not interfere with the normal action of the switch mechanism and the free movement of the actuating member.

During the test, the specimens are not lubricated.

18.3 Overload Test with Filament Lamps

Switches are normally tested at rated voltage and 1.2 times the rated current.

Test is carried out using a number of tungsten filament lamps at a load current not less than 1.2 times the rated current of the switch at a voltage not less than 95 percent of the voltage.

The test voltage shall be the rated voltage of the lamps. The number of lamps shall be the smallest number giving a test current not less than 1.2 times the rated current of the switch.

Available short-circuit current shall be at least 1 500 A. The other conditions shall be as specified in **18.2**.

During the test no sustained arcing shall occur.

NOTE — Sticking of the contacts, which does not prevent the next operation of the switch is not considered as welding.

Sticking of contacts is permitted if the contacts can be separated with a force applied to the actuator of a value which does not damage the switch mechanically.

For the purpose of this test, blows which are intended to separate welded contacts are not permitted.

After the test, the sample shall show no damage which may impair its further use.

NOTE — Example: 10 A 250 V switches have to be tested.

The largest available rated voltage of 200 W tungsten filament lamps is 240 V.

The test voltage shall then be 240 V and the number of lamps:
 $240 \times 1.2 \times 10/200 = 14.4$, equivalent to 15.

19 NORMAL OPERATION

19.1 Test for Switches Intended for Inductive Loads

Switches shall withstand, without excessive wear or other harmful effect, the mechanical, electrical and thermal stresses occurring in normal use.

For the purpose of this test, pilot lights are disconnected.

Compliance is checked by the following test:

The switches are tested at rated voltage and rated current in the apparatus and with the connections specified in **18**.

The tolerances for the test voltage is $^{+5}_0$ percent.

The circuit details and the manner of operation of the selector switches S are as described in **18.2** unless otherwise specified.

The number of operations is as shown in Table 21.

Table 21 Number of Operations for Normal Operation Test

(Clauses 19.1 and 19.3)

Rated current	Number of operations
Up to and including 16 A, for switches having a rated voltage not exceeding 250 V, except for pattern numbers 3 and 03	40 000
Up to and including 16 A, for switches having a rated voltage exceeding 250 V and for pattern numbers 3 and 03	20 000
Over 16 A up to and including 50 A	10 000
Over 50 A	5 000

The rate of operation is as specified in **18.2**.

The 'ON' period shall be 25^{+5}_0 percent of the total cycle and the 'OFF' period 75^{+5}_0 percent.

For rotary switches of pattern number 5 intended to be operated in either direction, the actuating member is turned in one direction for half the total number of operations and in the reverse direction for the remainder.

For rotary switches of pattern numbers 1, 2 and 4 intended to be operated in either direction, the selector switch S_1 is moved after $3/8$ and $7/8$ of the total number of operations.

For other rotary switches intended to be operated in either direction, $3/4$ of the total number of operations is effected in the clockwise direction, and the remainder in the reverse direction.

Cord operated switches shall be tested mounted as in normal use and with a pull of a value adequate to operate the cord-operated switch but not exceeding 50 N, on the cord throughout the test at 30 ± 5 to the vertical and in a plane perpendicular to the surface.

Switches are tested with a.c. having a $\cos \phi$ of 0.6 ± 0.05 .

Switches of pattern number 2 are tested with the poles connected in series.

The two poles of switches of pattern No. 4 and No. 5 are tested as two switches of pattern No. 1. If the two poles are identical, only one pole needs to be tested.

For switches of pattern No. 5 with a single mechanism, each circuit is loaded with 0.5 times rated current.

Switches of pattern 4 and No. 6 shall be tested for half the number of operations on one pole and half the number of operations for the other pole.

Switches of pattern No. 6/2 are tested as one switch of pattern No. 6, if the two pairs of poles are identical. Otherwise, as two switches of pattern No. 6.

Switches of pattern No. 7 are tested as a double switch of pattern No. 6 while testing one part, the other part is in the 'off' position. The second set of specimens used

for the test of **18** is submitted to this test on the side already tested.

The test specimens shall be connected to the test circuit with cables of length (1 ± 0.1) m.

During the test, the specimens shall function correctly.

After the test, the specimens shall withstand an electric strength test as specified in **16** and a terminal temperature rise test as specified in **17**.

For the electric strength test, the test voltage of 4 000 V shall be reduced by 1 000 V and the other test voltages by 500 V. For the terminal temperature rise test, the test current shall be reduced to the value of the rated current and the only measurement made shall be at the terminals.

Specimen shall then not show:

- Wear impairing their further use;
- Discrepancy between the position of the actuating member and that of the moving contacts, if the position of the actuating member is indicated;
- Deterioration of enclosures, insulating linings or barriers to such an extent that the switch cannot be further operated or that it no longer complies with the requirements of **10**;
- Seepage of sealing compound;
- Loosening of electrical or mechanical connections; and
- Relative displacement of the moving contacts of switches of pattern numbers 2, 3, 03 or 6/2.

NOTE — The humidity treatment as per **15.3** is not repeated before the electric strength test of this sub-clause.

During the test, the samples are not lubricated.

The test is followed by the test of **14.3**.

Sticking of the contacts which does not prevent further operation of the switch is not regarded as welding.

Sticking of contacts is permitted if the contacts can be separated with a force applied to the actuator of a value which does not damage the switch mechanically.

During the test, no sustained arcing nor welding of the contacts shall occur.

For the purpose of this test, blows which are intended to separate welded contacts are not permitted.

Breakage of the replaceable pull cord not involving the part entering the cord-operated switch shall not be considered a failure to pass the test.

19.2 Test for Switches Intended for Externally Ballasted Lamp Loads

Switches intended for fluorescent lamp loads shall withstand, without excessive wear or other harmful effect, the electrical and thermal stresses occurring when controlling externally ballasted lamp circuits.

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Compliance is checked by the following test. For the test, new specimens are used.

Compliance is checked by the test circuit indicated in Fig. 16 A with the following test conditions

The prospective short-circuit current of the supply shall be between 3 kA and 4 kA at $\cos \phi = 0.9 \pm 0.05$ (lagging).

F is a copper-wire fuse of 0.1 mm nominal diameter having a length not less than 50 mm.

R1 is a resistor limiting the current to about 100 A.

The twin-core cable shall have a suitable length to give a resistance R_3 equal to 0.25Ω in the test circuit to the load. It shall have a cross-sectional area of 1.5 mm^2 when switches with rated current up to and including 13 A are being tested and 2.5 mm^2 when switches with rated current over 13 A up to and including 20 A are being tested.

Load A shall consist of

- a) C1, a capacitor bank having a capacitance of $70 \mu\text{F} \pm 10$ percent for 6 A switches and $140 \mu\text{F} \pm 10$ percent for other switches. The capacitors shall be connected with 2.5 mm^2 conductors having the shortest possible length; and
- b) L_1 and R_2 , an inductor and a resistor adjusted to give the power factor of 0.9 ± 0.05 (lagging) and the test current I_n^{+5} percent through the specimen.

The switches are tested at rated voltage and at the rated current assigned by the manufacturer for the externally ballasted lamp loads in the test apparatus and with the connections specified in 18.2.

The tolerance for the test voltage is ± 5 percent and for test current is $^{+5}_0$ percent. The circuit details and the manner of operations of the selector switch S are as described in 18.2.

The number of operations is as follows:

For switches with a rated externally ballasted lamp current of 6 A up to and including 13A:

10 000 operations with 30 operations per minute.

For switches with rated current above 13 A up to and including 16 A: 5 000 operations with 15 operations per minute.

For rotary switches of pattern No. 5 intended to be operated in either direction, the actuating member is turned in one direction for half the total number of operations and in the reverse direction for remainder.

For other rotary switches intended to be operated in either direction, 3/4 of the total number of operations is effected in the clockwise direction, and the remainder in the reverse direction.

Cord-operated switches shall be tested mounted as in normal use and with a pull of a value adequate to operate the cord-operated switch but not exceeding 50 N, on the cord throughout the test at $30^\circ \pm 5^\circ$ to the vertical and in a plane perpendicular to the mounting surface.

Switches of pattern No. 2 are tested with the poles connected in series.

The two circuits of switches of pattern number 5 are tested as two switches of pattern number 1. If the circuits are identical, only one circuit need be tested.

Switches of pattern numbers 4 and 6 shall be tested for half the number of operations on one way and half of the number of operations on the other way.

Switches of pattern No. 6/2 are tested as one switch of pattern No. 6, if the two pairs of poles are identical. Otherwise, as two switches of pattern No. 6.

For switches of pattern number 7, one side is tested as pattern number 6. The test is repeated on the other side of a new set of specimens if the other side is not identical.

The test specimens shall be connected to the test circuit with cables of length $(1 \pm 0.1) \text{ m}$.

The load shall be as specified in Fig. 16 A, load A.

The metal support of the switch, if any, on which the switch is mounted, and the accessible metal parts of the switch, if any, shall be earthed through a wire fuse which shall not blow during the test. The fuse element shall consist of a copper wire of 0.1 mm in diameter not less than 50 mm in length.

During this test, the switch shall be operated so that the test apparatus does not interfere with the normal action of the switch mechanism and the free movement of the actuating member.

During this test, the switch shall be operated so that the test apparatus does not interfere with the normal action of the switch mechanism and the free movement of the actuating member.

There shall be no forced actuation. The on-period shall be $25 \left(\begin{smallmatrix} +5 \\ 0 \end{smallmatrix} \right)$ (percent of the total cycle and the off-period $75 \left(\begin{smallmatrix} 0 \\ -5 \end{smallmatrix} \right)$ percent

During the no sustained arcing nor welding of the contacts shall occur.

Sticking of the contacts which does not prevent further operation of the switch is not regarded as welding.

Sticking of contacts is permitted if the contacts can be separated with a force applied to the actuator of a value which does not damage the switch mechanically.

For the purpose of this test, blows which are intended to separate welded contacts are not permitted.

During the test, the specimens shall function correctly.

After the test, the specimens shall withstand an electric strength test as specified in 16 and a terminal temperature rise test as specified in 17.

For the electric strength test, the test voltage of 4 000 V shall be reduced by 1 000 V and the other test voltage by 500 V. For the terminal temperature rise test, the test current shall be reduced to the value of the rated current and the only measurement made shall be at the terminals.

After the tests, it shall also be possible to make and break the switch by hand in the test circuit and the specimen shall not show:

- Wear impairing further use;
- Discrepancy between the position of the actuating member and that of the moving contacts, if the position of the actuating member is indicated;
- Deterioration of the enclosures, insulating lining or barriers to such an extent that the switch cannot be further operated or that the requirements of 10;
- Loosening of electrical or mechanical connections;
- Seepage of sealing compound; and
- Relative displacement of the moving contacts of switches of pattern No. 2, 3 or 6/2.

Breakage of the replaceable pull cord, not involving the part entering the cord-operated switch, shall not be considered a failure to pass the test

19.3 Test for Switches Intended for Self-ballasted Lamp Loads

Switches intended for self-ballasted lamp (SBL) loads shall withstand, without excessive wear or other harmful effect, the electrical and thermal stresses occurring when controlling self-ballasted lamp circuits.

Compliance is checked by connecting the load as given in Fig. 16 B *via* the switch under test to a power supply.

The calculations are based on the following parameters in order to have the required values for inrush current and I^2t :

- a prospective short-circuit current (rms) of the supply of 3 kA at $\cos \phi = 0.9$ (lagging); and
- a resistance R_3 equal to 0.25Ω in the test circuit to the load. This value is obtained with a wire having a cross-sectional area of 1.5 mm^2 when switches with rated current up to and including 13 A are being tested and 2.5 mm^2 when switches with rated current over 13 A up to and including 20 A are being tested.

The load shall be in accordance with Fig.16 B. The values for the maximum peak value and the maximum I^2t of the inrush current are given in Table 22.

The values of R_2 and C in load B shall be chosen in order to reach the values (± 5 percent) for I_{peak} and I^2t as given in Table 22 when the switching contact closes at $90^\circ \pm 5^\circ$ phase-angle.

The value of R_4 shall be chosen to reach the current in amperes to reach the corresponding SBL power as given in Table 22.

Table 22 Values for I_{peak} and I^2t Depending on the type of Distribution System
(Clause 19.3)

Rated current [A]	Distribution system [V]:	Distribution system [V]:	Distribution system [V]:	Distribution system [V]:	Distribution system [V]:	Distribution system [V]:
	220/380	220/380	220/380	220/380	220/380	220/380
	230/400	230/400	230/400	230/400	230/400	230/400
	240/415	240/415	240/415	240/415	240/415	240/415
	I_{peak}	I^2t	I_{peak}	I^2t	Rated power of the SBL circuit	Rated power of the SBL circuit
	[A]	[A ² s]	[A]	[A ² s]	[W]	[W]
Up to and including 10	108	2.8	162	5.9	100	60
Above 10 up to and including 13	142	5.5	162	5.9	150	60
Above 13 up to and including 16	170	9	200	11.5	200	100
Above 16 up to and including 20	192	13	231	18.5	250	150

NOTE — R_2 is the total series resistance in the lamp circuit including the ESR (equivalent series resistance) value of the capacitor.

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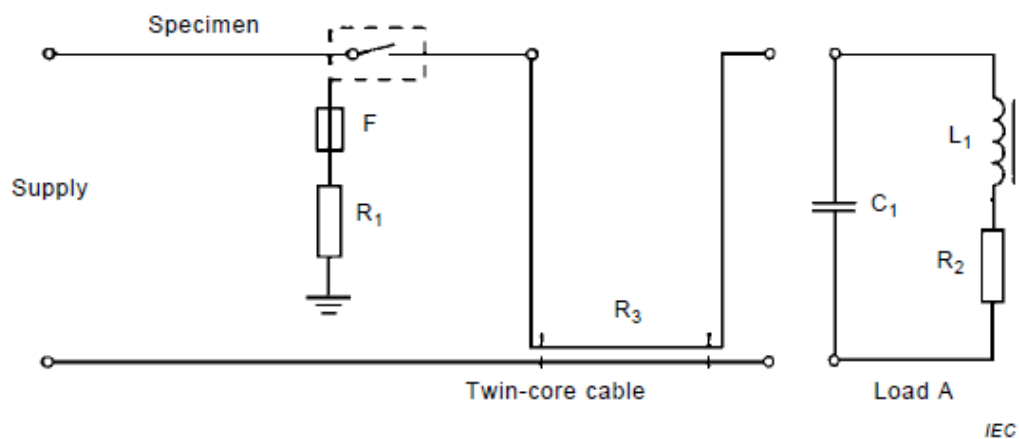


Fig. 16A CIRCUIT DIAGRAMS FOR TESTING SWITCHES FOR USE ON EXTERNALLY BALLASTED LAMP LOADS

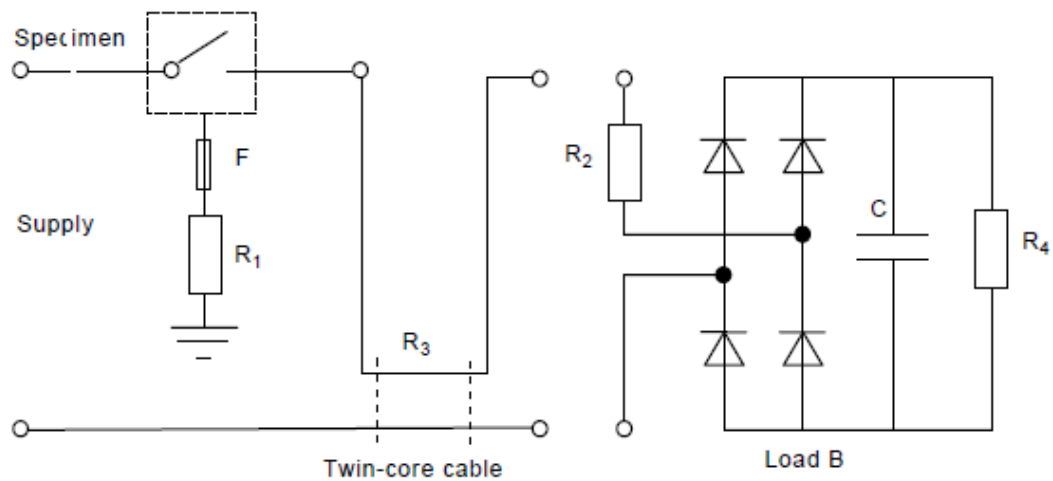


Fig. 16B CIRCUIT DIAGRAMS FOR TESTING SWITCHES ACCORDING TO 19.3

FIG. 16 CIRCUIT DIAGRAMS FOR TESTING SWITCHES

Switches with a rated voltage of 250 V shall be considered as switches for a distribution system 230/400 V.

For other rated voltage and currents the values need to be recalculated.

The values in Table 23 are given for information only. The circuit shall be adjusted to reach the I_{peak} and I^2t values of Table 22.

Table 23 Calculated Circuit Parameters

(Clause 19.3)

Rated Current (A)	230V	
	R_2 [Ω]	C [μ F]
Up to and including 10	1.9	125
Above 10 up to and including 13	1.25	180
Above 13 up to and including 16	0.95	240
Above 16 up to and including 20	0.8	310

Compliance is checked by the following test:

For the test, new specimens are used.

The switches are tested at rated voltage in the apparatus and with the connections specified in **18.2**.

The tolerance of the test voltage is ± 5 percent.

The circuit details and the manner of operation of the selector switch S are as described in **18.2**.

The number of operations is shown in Table 21.

The rate of operation is as specified in **18.2**.

The on-period shall be (25^{+5}_0) percent of the total cycle and the off-period (75^0_{-5}) percent.

For rotary switches of pattern number 5 intended to be operated in either direction, the actuating member is turned in one direction for half the total number of operations and in the reverse direction for the remainder.

For other rotary switches intended to be operated in either direction, three quarters of the total number of operations are effected in the clockwise direction, and the remainder in the reverse direction.

Cord-operated switches shall be tested mounted as in normal use and with a pull of a value adequate to operate the cord-operated switch, but not exceeding 50 N, on the cord throughout the test at $30^\circ \pm 5^\circ$ to the vertical and in a plane perpendicular to the mounting surface.

Switches of pattern number 2 are tested with the poles connected in series.

The two circuits of switches of pattern number 5 are tested as two switches of pattern number 1. If the circuits are identical, only one circuit need be tested.

Switches of pattern numbers 4 and 6 shall be tested for half the number of operations on one way and half the number of operations on the other way.

Switches of pattern number 6/2 are tested as one switch of pattern number 6, if the two pairs of poles are identical. Otherwise, they are tested as two switches of pattern number 6.

For switches of pattern number 7, one side is tested as pattern number 6. The test is repeated on the other side of a new set of specimens if the other side is not identical.

The test specimens shall be connected to the test circuit with cables of length (1 ± 0.1) m.

The metal support of the switch, if any, on which the switch is mounted, and the accessible metal parts of the switch, if any, shall be earthed through a wire fuse (F) which shall not blow during the test. The fuse element shall consist of a copper wire of 0.1 mm diameter and not less than 50 mm in length. R_1 is a resistor limiting the current to about 100 A.

During this test, the switch shall be operated so that the test apparatus does not interfere with the normal action of the switch mechanism and the free movement of the actuating member.

There shall be no forced actuation.

No sustained arcing or welding of the contacts shall occur.

Sticking of the contacts which does not prevent the next operation of the switch is not regarded as welding.

Sticking of contacts is permitted if the contacts can be separated with a force applied to the actuator of a value which does not damage the switch mechanically.

For the purpose of this test, it is not permitted to apply blows to open welded contacts.

After the test, the specimens shall withstand an electric strength test as specified in **16.3** and a terminal temperature rise test as specified in **17**.

For the electric strength test, the test voltage of 4 000 V shall be reduced by 1 000 V and the other test voltage

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by 500 V. For the terminal temperature rise test, the test current shall be reduced to the value of the rated current and the only measurement made shall be at the terminals.

After these tests, it shall also be possible to make and break the switch by hand in the test circuit and the specimen shall not show:

- a) Wear impairing further use;
- b) Discrepancy between the position of the actuating member and that of the moving contacts, if the position of the actuating member is indicated;
- c) Deterioration of the enclosures, insulating lining or barriers to such an extent that the switch cannot be further operated or that it no longer complies with the requirements of Clause 10;
- d) Loosening of electrical or mechanical connections;
- e) Seepage of sealing compound; and
- f) Relative displacement of the moving contacts of switches of pattern numbers 2, 3 or 6/2.

Breakage of the replaceable pull cord not involving the part entering the cord-operated switch shall not be considered a failure to pass the test.

20 MECHANICAL STRENGTH

20.1 General

The test of the relevant sub-clauses 20.5 to 20.9 shall be applied according to the type of construction as specified in 13.3.

Accessories, surface mounting boxes, screwed glands and shrouds shall have adequate mechanical strength so as to withstand the stresses imposed during installation and use.

Compliance is checked by the following tests:

all types of switches	20.2
for switches with a base intended to be mounted directly on a surface	20.3
for boxes	20.2
for screwed glands of switches that have an IP code higher than IP20	20.4
for operating members of cord-operated switches	20.10

Combinations of switches or of switches and socket-outlets are tested in the following way:

- a) in the case of one common cover, as a single product; and
- b) in the case of separate covers, as separate products.

20.2 Pendulum Hammer Test

The specimens are checked by applying blows by means of the pendulum hammer test apparatus as described in IS 9000 (Part 7/Sec 7) (test Eha), with an equivalent mass of 250 g.

The specimens are mounted on a sheet of plywood, 8 mm nominal thickness and approximately 175 mm square, secured at its top and bottom edges to a rigid bracket which is part of the mounting support.

The mounting support shall have a mass of (10 ± 1) kg and shall be mounted on a rigid frame by means of pivots. The frame is fixed to a solid wall.

The design of the mounting is such that

- a) the specimen can be so placed that the point of impact lies in the vertical plane through the axis of the pivot;
- b) the specimen can be removed horizontally and turned about an axis perpendicular to the surface of the plywood; and
- c) the plywood can be turned 60° , in both directions, about a vertical axis.

The switches and boxes are mounted on the plywood as in normal use.

Inlet openings which are not provided with knock-outs, are left open; if they are provided with knock-outs, one of them is opened.

For flush-type switches the specimen is mounted in a recess provided in a block of hornbeam or similar material, which is fixed to a sheet of plywood, and not in its relevant mounting box. If wood is used for the block, the direction of the wood fibers shall be perpendicular to the direction of the impact.

Flush-type screw fixing switches shall be fixed by means of screws to lugs recessed in the hornbeam block. Flush-type claw fixing switches shall be fixed to the block by means of the claws.

Before applying the blows, fixing screws of bases and covers are tightened with a torque equal to two-thirds of the relevant value as specified in Table 5.

The specimen are mounted so that the point of impact lies in the vertical plane through the axis of the pivot.

The striking element is allowed to fall from a height which is specified in Table 24:

Table 24 Height of Fall for Impact Test
(Clause 20.2)

Height of Fall mm	Parts of Enclosures to be Subjected to the Impact ^a	
	Switches having an IP code IPX0	Switches having an IP code higher than IPX0
80	A and B	–
120	C	A and B
160	D	C
200	–	D
^a Parts of enclosures are defined as follows:		
A parts on the front surface, including the parts which are recessed;		
B parts which do not project more than 15 mm from the mounting surface (distance from the wall) after mounting as in normal use, with the exception of the above parts A;		
C parts which project more than 15 mm and not more than 25 mm from the mounting surface (distance from the wall) after mounting as in normal use, with the exception of the above parts A;		
D parts which project more than 25 mm from the mounting surface (distance from the wall) after mounting as in normal use, with the exception of the above parts A.		

The impact energy determined by the part of the specimen which projects most from the mounting surface is applied on all parts of the specimen, with the exception of its parts A.

Parts of accessories exclusively intended for mounting in panel board are submitted to impacts which are obtained by allowing the striking element to fall from height of 100 mm, impacts shall only be applied on parts which are accessible after mounting the accessory in the panel board.

The height of fall is the vertical distance between the position of a checking point, when the pendulum is released, and the position of that point at the moment of impact. The checking point is marked on the surface of the striking element where the line through the point of intersection of the axis of the steel tube of the pendulum and the striking element and perpendicular to the plane through both axes meets the surface.

The specimen are subjected to nine blows, which are evenly distributed over the sample. The blows are not applied to knock out areas. The following blows are applied:

- a) for parts A five blows:
 - 1) one blow in the center, after the specimen has been moved horizontally;
 - 2) one each on the unfavorable point between the center and the edges; and

- 3) and then, after the specimen has been turned 90° about its axis perpendicular to the plywood, one each on similar points.

- b) for parts B (as far as applicable), C and D, four blows:

- 1) two blows on each of the two sides of the specimen on which blows can be applied after the plywood sheet has been turned 60° in each of the opposite directions; and
- 2) two blows on each of the other two sides of the specimen on which blows can be applied after the specimen has been turned 90° about its axis perpendicular to the plywood sheet and the plywood sheet has been turned 60° in each of the opposite directions.

If inlet openings are provided, the specimen is so mounted that the two lines of blows are as nearly as possible equidistant from these openings.

Cover plates and other covers of multiple switches are treated as cover plates or covers of single switches.

For switches that have an IP code higher than IPX0 the test is made with the lids closed and, in addition, the appropriate number of blows is applied to those parts which are exposed when the lids are open.

After the test, the specimen shall show no damage within the meaning of this standard. In particular live parts shall not become accessible.

After the test on a lens (window) for pilot lights the lens may be cracked and/or dislodged, but it shall not be possible to touch live parts with:

- i) the probe B of IS 1401 under the conditions stated in **10.1**; and
- ii) the probe 11 of IS 1401 under the conditions state in **10.1**, but with a force of 10 N.

In case of doubt, it is verified that it is possible to remove and to replace external parts, such as boxes, enclosures, covers and cover plates, without these parts or their insulating lining being broken.

If, however, a cover plate, backed by an inner cover is broken, the test is repeated on the inner cover, which shall remain unbroken.

Damage to the finish, small dents which do not reduce creepage distances below the value specified in **23.1** and small chips which do not adversely affect the protection against electric shock are neglected.

Cracks not visible to the naked eye and surface cracks in fiber-reinforced moulding and the like are ignored.

Cracks or holes in the outer surface of any part of the switch are ignored if the switch complies with this standard even if this part is omitted. If a decorative cover is backed by an inner cover, fracture of the decorative cover is neglected if the inner cover withstands the tests after removal of the decorative cover.

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20.3 Test on the Main Parts of Surface-type Switches

The fixing screws of the main parts are surface-type switches are first fixed to a cylinder of rigid steel sheet having a radius equal to 4.5 times the distance between fixing holes, but in any case not lower than 200 mm

The axes of the holes are in a plane perpendicular to the axis of the cylinder and parallel to the radius through the center of the distance between the holes.

The fixing screws of the main part are gradually tightened, the maximum torque applied being 0.5 Nm for screws having a thread diameter up to and including 3 mm and 1.2 Nm for screws having a larger thread diameter.

The main parts are then fixed in a similar manner to a flat steel sheet.

During and after the tests, the bases shall show no damage impairing their further use.

20.4 Screwed Glands

Screwed glands are fitted with a cylindrical metal rod having a diameter, in millimeters, equal to the nearest whole number below the internal diameter in millimeters, of the packing.

The glands are then tightened by means of a suitable spanner the torque shown in Table 16 being applied to the spanner for 1 min.

After the test, the glands and the enclosures of the specimens shall show no damage within the meaning of this standard.

20.5 Covers, Cover Plates or Actuating Members – Accessibility to Live Parts

20.5.1 General

When testing the force necessary for covers, cover plates or actuating members to come off or not come off, the switches are mounted as for normal use. Flush-type switches are fixed in appropriate mounting boxes, which are installed as for normal use so that the rims of the boxes are flush with the walls, and the covers, cover plates or actuating members are fitted. If they are provided with locking means which can be operated without the aid of a tool, these means are unlocked.

Compliance is then checked by the tests of 20.5.2 and 20.5.3.

20.5.2 Verification of the Non-removal of Covers, Cover Plates or Actuating Members

Forces are gradually applied in directions perpendicular to the mounting surfaces, in such a way that the resulting

force acting on the center of the covers, cover plates, actuating members or parts of them is respectively:

- a) 40 N, for covers, cover plates, actuating members or parts of them complying with the tests of 20.7 and 20.8; or
- b) 80 N, for other covers, cover plates, actuating members or parts of them.

The force is applied for 1 min. The covers, cover plates, or actuating members shall not come off.

The test is then repeated on new specimens, the cover or cover plate being fitted on the wall after a sheet of hard material, 1 mm $\pm$ 0.1 mm thick has been fitted around the supporting frame, as shown in Fig. 17.

NOTE — The sheet of hard material is used to simulate wall paper and may consist of a number of pieces.

After the test the specimens shall show no damage within the meaning of this standard.

20.5.3 Verification of the Removal of Covers, Cover Plates of Actuating Members

A force not exceeding 120 N is gradually applied, in directions perpendicular to the mounting/supporting surfaces, to covers, cover plates, actuating members or parts of them by means of a hook placed in turn in each of the grooves, holes, spaces or the like, provided for removing them.

The covers, cover plates or actuating members shall come off.

The test is made 10 times on each separable part the fixing of which is not dependent on screws (the applicable points are equally distributed as far as practicable), the removal force is applied each time to the different grooves, holes or the like provided for removing the separable part.

The test is then repeated on new specimens, the cover, cover plate, or actuating member being fitted on the wall after a sheet of hard material, (1 $\pm$ 0.1) mm thick has been fitted around the supporting frame, as shown in Fig. 17.

After the test, the specimens shall show no damage within the meaning of this standard.

20.6 Covers, Cover Plates or Actuating Members – Accessibility to Non-earthed Metal Parts Separated from Live Parts

The test is made as described in 20.5, but applying for 20.5.2, the following forces:

- a) 10 N, for covers or cover plates or actuating members complying with the test of 20.8 and 20.9; and
- b) 20 N, for other covers or cover plates and actuating members

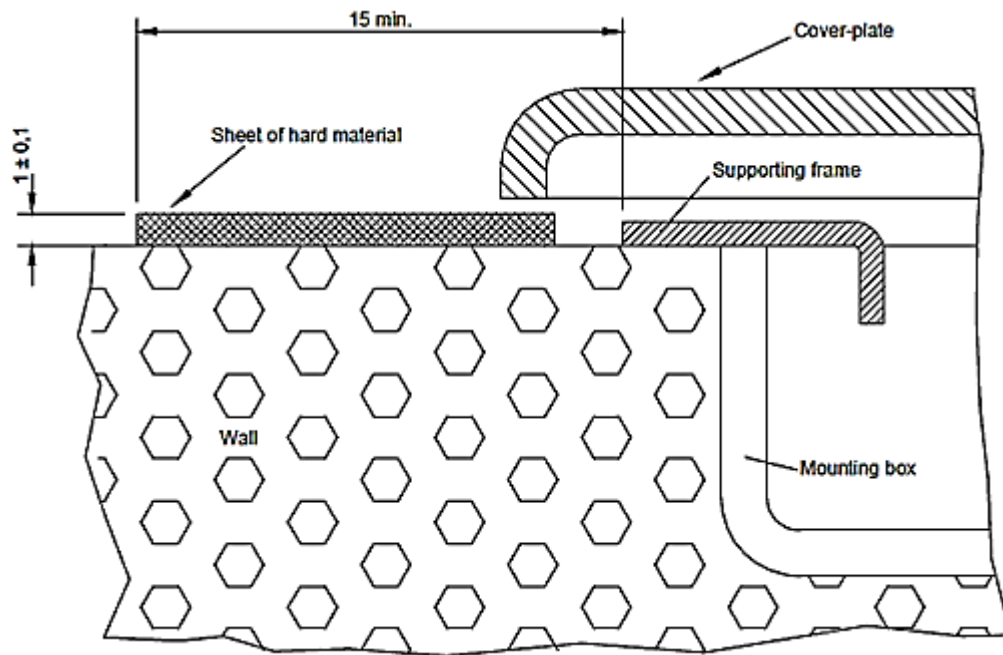


FIG. 17 ARRANGEMENT FOR TEST ON COVERS OR COVER-PLATES

20.7 Covers, Cover Plates or Actuating Members – Accessibility to Insulating Parts, Earthed Metal Parts, the Live Parts of SELV ≤ 25 V AC or Metal Parts Separated from Live Parts

The test is made as described in 20.5, but applying, for 20.5.2, the force of 10 N for all covers, cover plates, or actuating members.

20.8 Covers, Cover Plates or Actuating Members – Application of Gauges

The gauge shown in Fig. 18 is pushed toward each side of each cover, cover plate or actuating member which is fixed without screws on a mounting or supporting surface, as shown in Fig. 19. The face B resting on the mounting/supporting surface, with the face A perpendicular to it the gauge is applied at right angles to each side under test.

In the case of a cover or cover plate fixed without screws to another cover or cover plate or to a mounting box, having the same outline dimensions, the face B of the gauge shall be placed at the same level as the junction; the outline at the cover or cover plate shall not exceed the outline of the supporting surface.

The distances between the face C of the gauge and the outline of the side under test, measured parallel to face B, shall not decrease (with the exception of grooves, holes, reverse tapers or the like, placed at a distance less than 7 mm from a plane including face B and complying

with the test of 20.9) when measurements are repeated starting from point X in the direction of the arrow Y (see Fig. 20).

20.9 Grooves, Holes and Reverse Tapers

A gauge according to Fig. 21, applied with a force of 1 N, shall not enter more than 1.0 mm from the upper part of any groove, hole or reverse taper or the like when the gauge is applied parallel to the mounting/supporting surface and at right angles to the part under test, as shown in Fig. 22.

NOTE — The verification whether the gauge according to Fig. 21 has entered more than 1.0 mm is made with reference to a surface perpendicular to face B and including the upper part of the outline of the grooves, holes, reverse tapers or the like.

20.10 Additional Test for Cord-operated Switch

The operating member of a cord-operated switch shall have adequate strength.

Compliance is checked on a new specimen by the following test:

The switch is mounted on a support as in normal use.

A pull of 100 N is applied for 1 min on the operating member as in normal use, after which a pull of 50 N is applied for 1 min in the most unfavorable direction within a conical surface with the center being the operating cord and the angle not exceeding 80° to the vertical.

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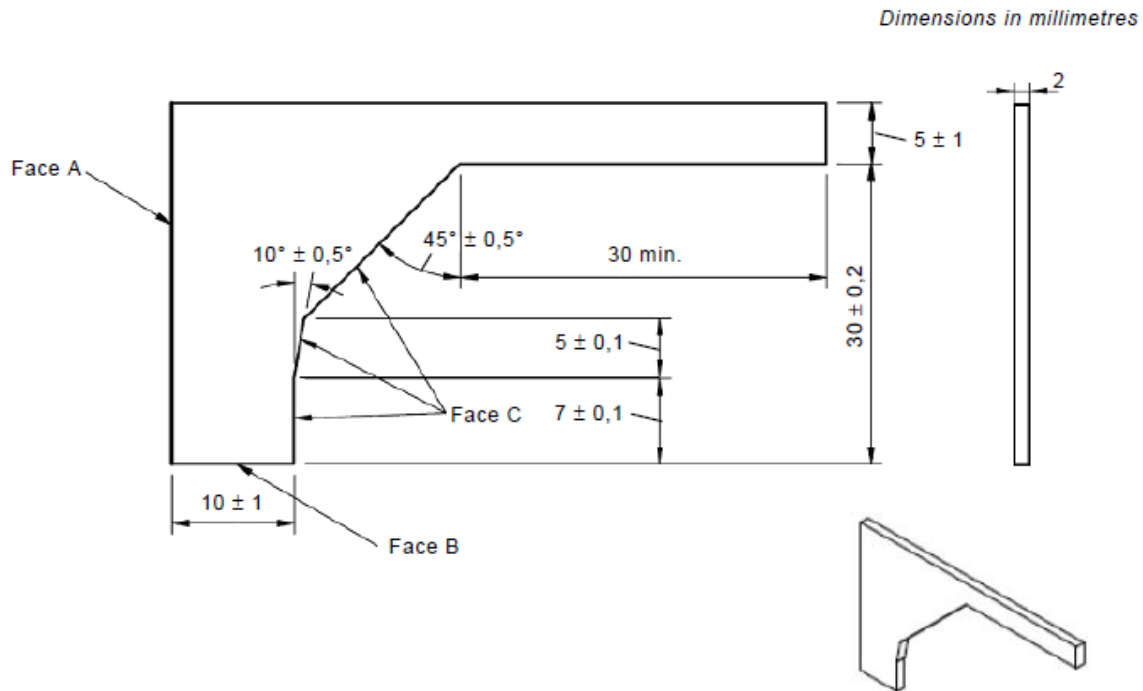


FIG. 18 GAUGE (THICKNESS: ABOUT 2 MM) FOR THE VERIFICATION OF THE
OUTLINE OF COVERS, COVER-PLATES OR ACTUATING MEMBERS

After the test the switch shall show no damage within the meaning of this standard. The operating member shall not have broken and the cord-operated switch shall still operate.

21 RESISTANCE TO HEAT

21.1 General

Switches and boxes shall be sufficiently resistant to heat.

Parts intended for decorative purposes only, such as certain lids, shall be removed if possible and these parts are not subjected to the test.

Compliance is checked by:

- for surface mounting boxes, separable covers, separate cover plates and separate frames by the test of 21.4; and
- for switches, with the exception of the parts, if any, covered by the previous items (a) by the tests of 21.2, 21.3 and with the exception for the switches made from natural or synthetic rubber or a mixture of both by the test of 21.4.

21.2 Basic Heating Test

The samples are kept for 1 h in a heating cabinet at a temperature of $(100 \pm 2) ^\circ\text{C}$.

During the test, they shall not undergo any change impairing their further use and sealing compound, if any, shall not flow to such an extent that live parts are exposed.

After the test and after the specimens have been allowed to cool down to approximately room temperature, there shall be no access to live parts which are normally not accessible when the specimens are mounted as in normal use, even if the standard test finger is applied with a force not exceeding 5 N.

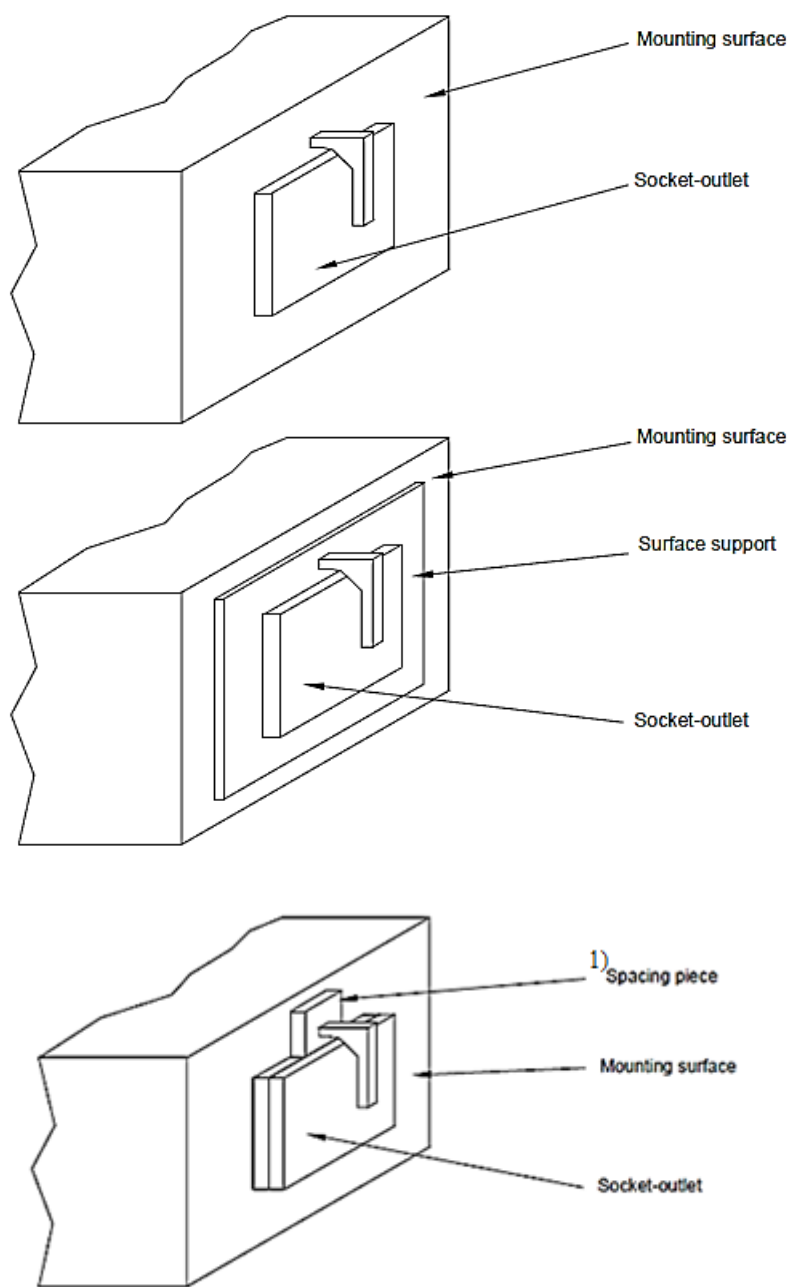
After the test, markings shall still be legible.

Discoloration, blisters or slight displacement of the sealing compound is disregarded provided that safety is not impaired within the meaning of this standard.

21.3 Ball Pressure Test on Parts of Insulating Material Necessary to Retain Current Carrying Parts and Parts of the Earthing Circuit in Position

Parts of insulating material necessary to retain current carrying parts and parts of the earthing circuit in position are subjected to a ball-pressure test by means of the apparatus shown in Fig. 23, except that insulating parts necessary to retain the earthing terminals in a box shall be tested as specified in 21.4.

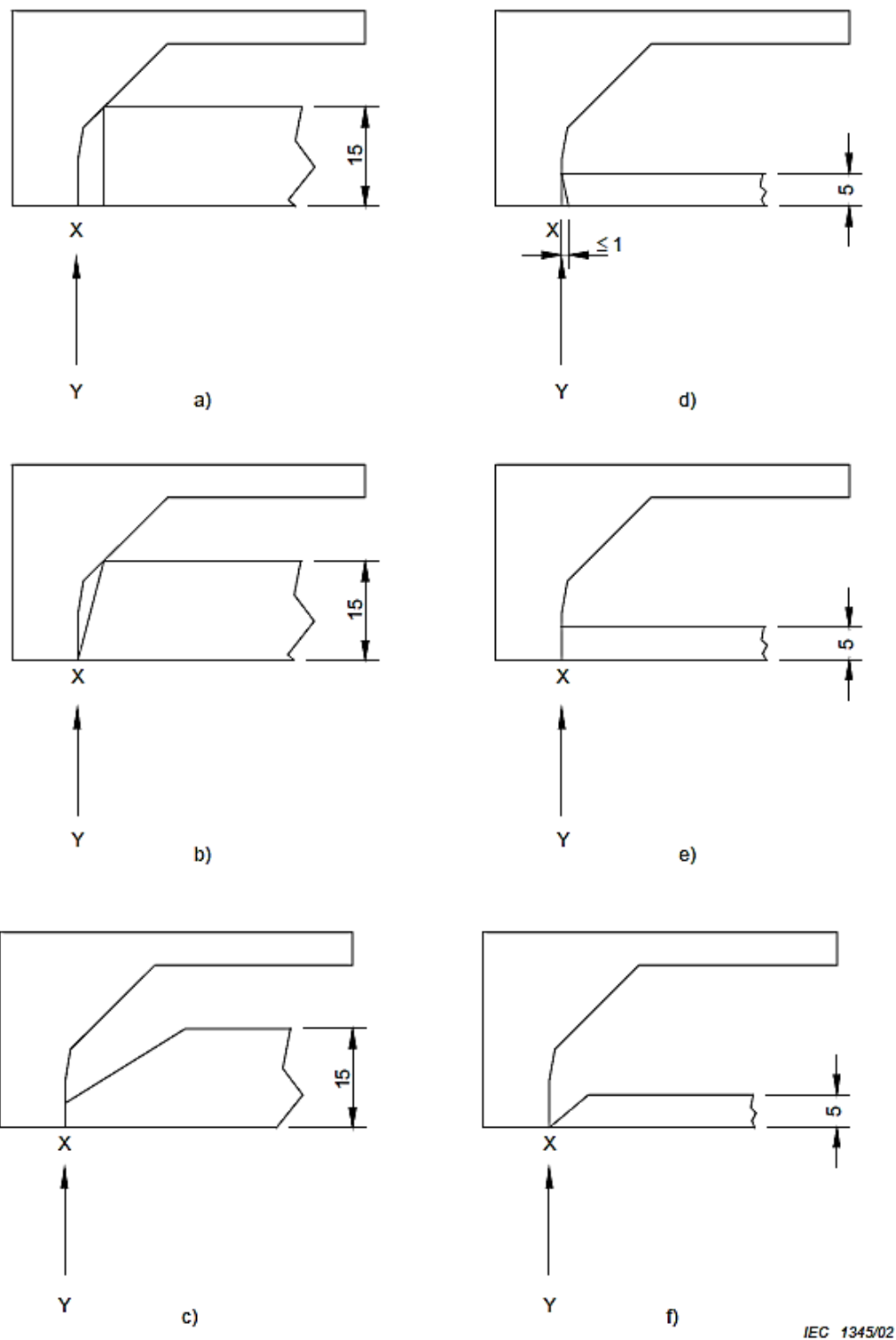
When it is not possible to carry out the test on the specimen, the test should be carried out on a specimen



¹⁾ Spacing piece having the same thickness as that of the supporting part.

FIG. 19 EXAMPLES OF APPLICATION OF THE GAUGE OF FIG. 18 ON COVERS FIXED WITHOUT SCREWS ON A MOUNTING SURFACE OR SUPPORTING SURFACE

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Cases a) and b) do not comply.

Cases c), d), e) and f) comply (compliance shall, however, also be checked with the requirements of **20.8**, using the gauge shown in Fig. 21).

FIG. 20 EXAMPLES OF APPLICATION OF THE GAUGE OF
FIG. 18 IN ACCORDANCE WITH THE REQUIREMENTS OF **20.8**

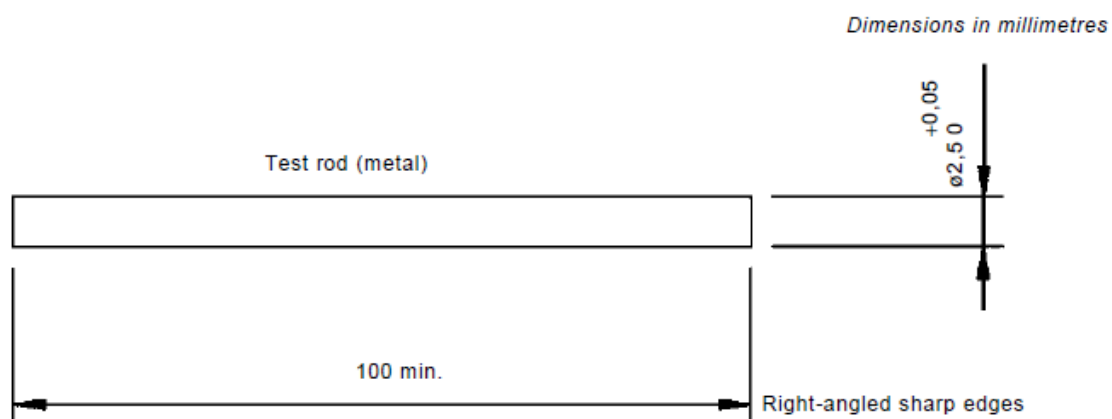


FIG. 21 GAUGE FOR VERIFICATION OF GROOVES, HOLES AND REVERSE TAPERS

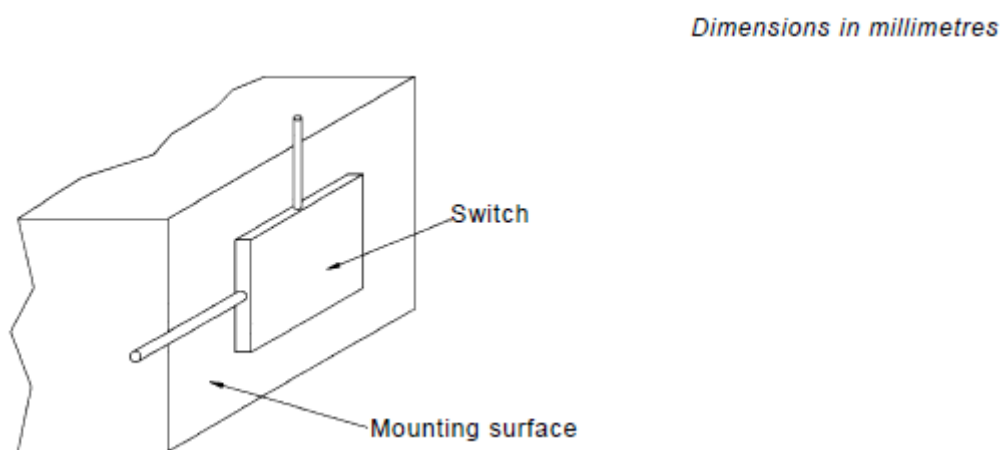


FIG. 22 SKETCH SHOWING THE DIRECTION OF APPLICATION OF THE GAUGE OF FIG. 21

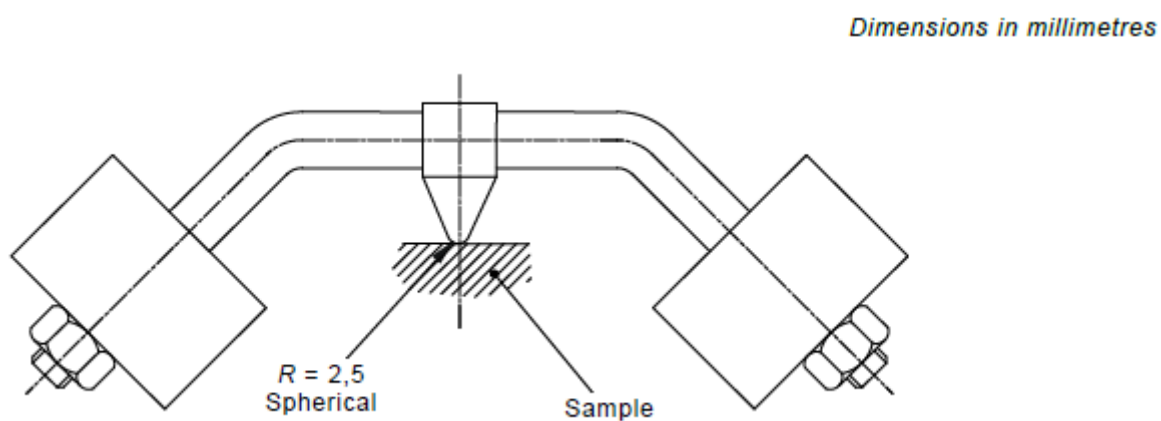


FIG. 23 BALL-PRESSURE APPARATUS

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of the material at least 2 mm thick. If this is not possible, up to and including four layers each, cut out of the same specimen, may be used, in which case the total thickness of the layers should not be less than 2.5 mm.

The surface of the part to be tested is placed in the horizontal position and a steel ball of 5 mm diameter is pressed against the surface with a force of 20 N.

The test load and the supporting means shall be placed within the heating cabinet for a sufficient time to ensure that they have attained the stabilized. Testing temperature before the test commences.

The test is made in heating cabinet at a temperature of $(125 \pm 2)^\circ\text{C}$.

After 1 h, the ball is removed from the specimen which is then cooled down within 10 s to approximately room temperature by immersion in cold water.

The diameter of the impression caused by the ball is measured and shall not exceed 2 mm.

21.4 Ball Pressure Test on Parts of Insulating Material not Necessary to Retain Current Carrying Parts and Parts of the Earthing Circuit in Position

Parts of insulating material not necessary to retain current-carrying parts and parts of the earthing circuit in position, even though they are in contact with them, are subjected to a ball-pressure test in accordance with 21.3, but the test is made at a temperature of $(70 \pm 2)^\circ\text{C}$, or $(40 \pm 2)^\circ\text{C}$, plus the highest temperature rise determined for the relevant part during the test of 17, whichever is the higher.

22 SCREWS, CURRENT-CARRYING PARTS AND CONNECTIONS

22.1 General

Connections, electrical or mechanical shall withstand the mechanical stresses occurring in normal use.

Mechanical connections to be used during installation of accessories may be made using thread-forming screws or thread cutting screws and when the screws are supplied together with the piece in which they are intended to be inserted. In addition thread cutting screws intended to be used during installation shall be captive with the relevant part of accessory.

Screws or nuts which transmit contact pressure shall be of metal and shall be in engagement with a metal thread.

Compliance is checked by inspection and for screws and nuts which are operated when connecting the external conductors and mounting the switch during installations by the following test.

NOTE — The requirements for the verifications of terminals are given in 12.

The screws or nuts are tightened and loosened:

- 10 times for screws in engagement with a thread of insulating material; and
- 5 times in all other cases.

Screws or nuts in engagement with a thread of insulating material are completely removed and reinserted each time.

The test is made by means of a suitable test screw driver or suitable tool, applying a torque as specified in 12.2.5.

The conductor is moved each time the screw or nut is loosened.

During the test, no damage impairing the further use of the screwed connections shall occur, such as breakage of screws or damage to the head slots (rendering the use of an appropriate screwdriver impossible), threads, washers or stirrups.

NOTE — Screws or nuts which are operated when assembling the switch include screws for fixing covers or cover plates, etc, but not connecting means for screwed conduits and screws for fixing the base of a switch.

The shape of the blade of the test screwdriver shall suit the head of the screw to be tested.

The screws and nuts shall be tightened in one smooth and continuous motion.

NOTE — Screwed connections are considered as partially checked by the tests of 19 and 20.

22.2 Correct Insertion of Screws

For Screws in engagement with a thread of insulating material and which are operated when mounting of the switch during installation, their correct introduction into the screw hole or nut shall be ensured.

Compliance is checked by inspection.

NOTE — the requirement with regard to correct introduction is met if introduction of the screw in slanting manner is prevented, for example, by guiding the screw by the part to be fixed by a recess in the female thread or by the use of a screw with the leading thread removed.

22.3 Contact Pressure of Electrical Connections

Electrical connections shall be so designed that contact pressure is not transmitted through insulating material other than ceramic, pure mica or other material with characteristics no less suitable, unless there is sufficient resiliency in the metallic parts to compensate for any possible shrinkage or yielding of the insulating material.

Compliance is checked by inspection, and by manual test.

NOTE — The suitability of the material is considered in respect of the stability of the dimensions.

22.4 Screws and Rivets used both as Electrical and Mechanical Connections

Screws and rivets, which serve as electrical as well as mechanical connections, shall be locked against loosening and/or turning.

Compliance is checked by inspection and by manual test.

NOTES

- 1 spring washers may provide satisfactory locking.
- 2 For rivets, a non-circular shank or an appropriate notch may be sufficient.
- 3 Sealing compound which softens on heating provides satisfactory locking only for screw connections not subjected to torsion in normal use.

22.5 Material of Current Carrying Parts

Current-carrying parts, including those of terminals (also earthing terminals), shall be of a metal having, under the conditions occurring in the accessory, mechanical strength, electrical conductivity and resistance to corrosion adequate for their intended use.

The requirement of 22.5 does not apply to screws, nuts, washers, clamping plates and similar parts of terminals.

Compliance is checked by inspection and, if necessary, by chemical analysis.

Examples of suitable metals, when used within the permissible temperature range and under normal conditions of chemical pollution, are:

- a) Copper;
- b) An alloy containing at least 58 percent copper for parts made from cold rolled sheet, or at least 50 percent copper for other parts; or
- c) Stainless steel containing at least 13 percent chromium and not more than 0.09 percent carbon;
- d) Steel provided with an electroplated coating of zinc according to IS 1573, the coating having a thickness of at least:
 - 1) 5 μm , for switches that have an IP code IPX0,
 - 2) 12.5 μm , for switches that have an IP code IPX4, or
 - 3) 25 μm , for switches that have an IP code IPX5 or IPX6.
- e) Steel provided with an electroplated coating of nickel and chromium according to IS 1068, the coating having a thickness of at least:
 - 1) 20 μm , for switches that have an IP code IPX0,
 - 2) 30 μm , for switches that have an IP code IPX4, or
 - 3) 40 μm , for switches that have an IP code IPX5 or IPX6.
- f) Steel provided with an electroplated coating of tin according to IS 1359, the coating having a thickness of at least:
 - 1) 12 μm , for switches that have an IP code IPX0,
 - 2) 20 μm , for switches that have an IP code IPX4, or
 - 3) 30 μm , for switches that have an IP code IPX5 or IPX6.

Current-carrying parts which may be subjected to mechanical wear shall not be made of steel provided with and electroplated coating.

Under moist conditions metals showing a great difference of electrochemical potential with respect to each other shall not be used in contact with each other.

Compliance is checked by a test which is under consideration.

NOTE — The requirement of this sub-clause does not apply to screws, nuts, washers, clamping plates and similar parts of terminals.

22.6 Contacts Subjected to Sliding Actions

Contacts which are subjected to sliding action in normal use shall be of a metal resistant to corrosion.

Compliance with the requirements of 22.5 and 22.6 is checked by inspection and by chemical analysis.

22.7 Thread-forming and Thread Cutting Screws

Thread-forming screws shall not be used for the connection of current-carrying parts. Thread-forming screws may be used to provide earthing continuity, provided that it is not necessary to disturb the connection in normal use and at least two screws are used for each connection.

Compliance is checked by inspection.

23 CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH SEALING COMPOUND

23.1 General

Creepage distances, clearances and distances through sealing compound shall be not less than the values shown in Table 9.

Sub clause 23.1 does not apply to pilot light units. Requirements for pilot light units are given in 13.16.

Compliance is checked by measurements.

The measurement are made on the switch fitted with conductors of the largest cross-sectional area specified in 12 and also without conductors.

Distances through slots or openings in external parts of insulating material are measured to metal foil in contact with the accessible surface; the foil is pushed into corners and the like by means of the straight unjointed test finger having the same dimensions as the standard test finger but is not pressed into openings.

The conductor shall be inserted into the terminal and so connected that the core insulation touches the metal part of the clamping unit or, if the core insulation is prevented by construction from touching the metal part, the outside of the obstruction.

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For surface-type switches that have an IP code IP 20, the most unfavorable conduit or cable is introduced for a distance of 1 mm into the switch, in accordance with 13.12.

If the metal frame supporting the base of flush-type switches is movable, this frame is placed in the most unfavorable position.

Any metal part in contact with a metal part of the mechanism is considered to be a metal part of the mechanism.

In double-break switches, the creepage distance mentioned under Item 1 or the clearance mentioned under Item 6 in Table 9 is the sum of the creepage distance or clearance between one fixed contact and the moving part, and that between the moving part and the other fixed contact.

The contribution to the creepage distance of any groove less than 1 mm wide is limited to its width.

Any air-gap less than 1 mm wide is ignored in computing the total clearance.

NOTE — The surface on which the base of surface-type switch is mounted includes any surface in contact with the base when the switch is installed. If the base is provided with a metal plate at the back, this plate is not regarded as the mounting surface.

23.2 Insulating Compound

Insulating compound shall not protrude above the edge of the cavity in which it is contained.

Compliance is checked by inspection.

24 RESISTANCE OF INSULATING MATERIAL TO ABNORMAL HEAT, TO FIRE AND TO TRACKING

24.1 Resistance to Abnormal Heat and to Fire

Parts of insulating material which might be exposed to thermal stresses due to electric effects, and the deterioration of which might impair the safety of the accessory, shall not be unduly affected by abnormal heat and by fire.

Compliance is checked by performing the glow-wire test according to IS 11000 (Part 2/Sec 1) under the following conditions:

- For parts of insulating material necessary to retain current-carrying parts, and parts of the earthing circuit, in position, by the test made at a temperature of 850°C with the exception of parts of insulating material needed to retain the earth terminal in position in a box, which shall be tested at a temperature of 650°C; and
- For parts of insulating material not necessary to retain current-carrying parts and parts of the earthing circuit in position, even they are in contact with them, by the test made at a temperature of 650°C.

Current carrying part or a part of the earthing circuit retained by mechanical means is considered to be retained in position. The use of grease or the like is not considered to be mechanical means.

External conductors cannot be considered as retaining the current-carrying parts.

In case of doubt, to determine whether an insulating material is necessary to retain current carrying parts and parts of the earthing circuit in position, the device is examined without conductors while held in positions most likely to cause displacement of the current-carrying parts or parts of the earthing circuit with the insulating material in question removed.

If the tests specified have to be made at more than one place on the same switch, care shall be taken to ensure that any deterioration caused by previous tests does not affect the result of the test to be made.

Small parts, where each surface lies completely within a circle of 15 mm in diameter, or where any part of the surface lies outside a 15 mm diameter circle and it is not possible to fit a circle of 8 mm in diameter on any of the surfaces, are not subjected to the test of 24.1 (see Fig. 24 for diagrammatic representation).

NOTE — When checking a surface, projections on the surfaces and holes which are not greater than 2 mm on the largest dimension are disregarded.

The tests are not made on parts of ceramic material.

If possible, the specimen should be a complete switch.

NOTES

1 The glow-wire test is applied to ensure that an electrically heated test wire under defined test conditions does not cause ignition of insulating parts or to ensure that a part of insulating material, which might be ignited by the heated test wire under defined conditions, has a limited time to burn without spreading fire by flame or burning parts or droplets from the tested part falling down on to the pinewood board covered with tissue paper.

2 If the test cannot be made on a complete switch, a suitable part may be cut from it for the purpose of the test.

The test is made on one specimen. In case of doubt, the test shall be repeated on two further specimens.

The specimen shall be stored for 24 h at standard ambient atmospheric conditions before the test.

The test is made applying the glow-wire once.

The specimen shall be positioned during the test in the most unfavorable position of its intended use (with the surface tested in vertical position). The tip of the glow-wire shall be applied to the specified surface of the specimen taking into account the conditions of the intended use under which a heated or glowing element may come into contact with the switch.

During the application time of the glow-wire and during a period of 30 s from the end of the application time, the sample and the surrounding parts, including the layer under the specimen, shall be observed.

Dimensions in millimetres

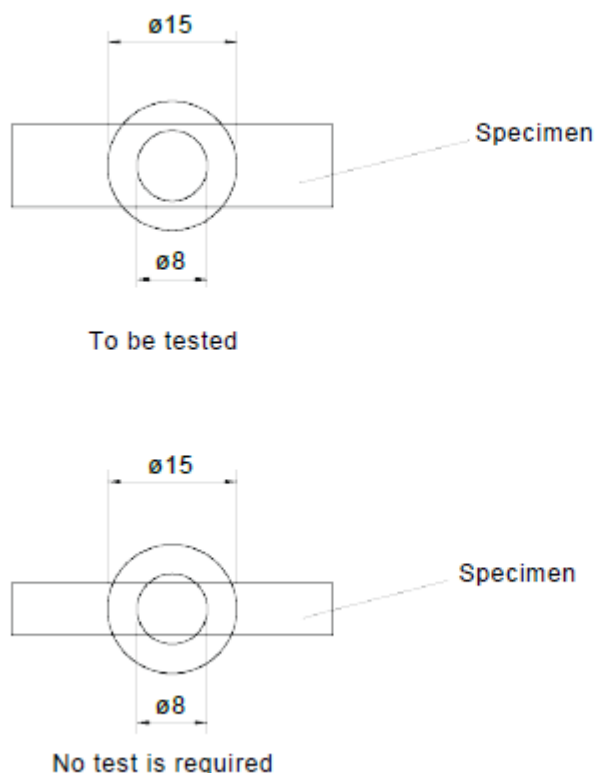


FIG. 24 DETERMINING PARTS OF INSULATING MATERIAL TO BE TESTED –
DIAGRAMMATIC REPRESENTATION (SEE 24.1)

The time when the ignition of the sample occurs and/or the time when the flames extinguish during or after the application time shall be measured and recorded.

The switch is regarded as having passed the glow-wire test if:

- There is no visible flame and sustained glowing; and
- Flames and glowing at the switch extinguish within 30 s after the removal of the glow wire.

There shall be no ignition of the wrapping tissue or scorching of the board.

24.2 Resistance to Tracking

For switches that have an IP code higher than IPX0, parts of insulating material retaining live parts in position shall be of material resistant to tracking.

Compliance is checked according to IS 2824. Ceramic parts are not tested.

A flat surface of the part to be tested at least 15 mm × 15 mm is placed in the horizontal position on the apparatus.

The material under test shall pass a proof-tracking index of 175 V using the solution A with the interval between drops (30 ± 5) s.

No flashover or breakdown between electrodes shall occur before a total of 50 drops have fallen.

25 RESISTANCE TO RUSTING

Ferrous parts, including covers and boxes shall be adequately protected against rusting.

Compliance is checked by the following test:

All grease is removed from the parts to be tested, using a suitable degreasing agent.

The parts are then immersed for 10 min in a 10 percent solution of ammonium chloride in water at a temperature of (27 ± 5) °C.

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Without drying, but after shaking off any drops, the parts are placed for 10 min in a box containing air saturated with moisture at a temperature of $(27 \pm 5)^{\circ}\text{C}$ and a relative humidity not less than 90 percent.

After the parts have been dried for 10 min in heating cabinet at a temperature of $(100 \pm 5)^{\circ}\text{C}$, their surfaces shall show no signs of rust.

NOTES

1 Traces or rust on sharp edges and any yellowish film removable by rubbing is ignored.

2 For small springs and the like, and for inaccessible parts exposed to abrasion, a layer of grease may provide sufficient protection against rusting. Such parts are subjected to the test only if there is doubt about the effectiveness of the grease film, and the test is then made without previous removal of the grease.

26 TESTS

26.1 Type Tests

The following shall be carried out as type tests on selected samples of switches being drawn preferably at random from a regular lot of production:

- a) Rating (*see 6*);
- b) Classification (*see 7*);
- c) Marking (*see 8*);
- d) Checking of dimensions (*see 9*);
- e) Protection against electric shock (*see 10*);
- f) Provision for earthing (*see 11*);
- g) Terminals (*see 12*);
- h) Constructional requirements (*see 13*);
- j) Mechanism (*see 14*);
- k) Resistance to ageing, to harmful ingress of water and to humidity (*see 15*);
- m) Insulation resistance and electric strength (*see 16*);
- n) Temperature rise (*see 17*);
- p) Making and breaking capacity (*see 18*);
- q) Normal operation (*see 19*);
- r) Test for switches intended for self-ballasted lamp loads (*see 19.3*);
- s) Mechanical strength (*see 20*);
- t) Resistance to heat (*see 21*);
- u) Screws, current carrying parts and connections (*see 22*);
- v) Creepage distances, clearances and distance through sealing compound (*see 23*);
- w) Normal operation for fluorescent lamp circuits (*see 19.2*);
- x) Resistance to abnormal heat and fire (*see 24.1*);
- y) Resistance to tracking (*see 24.2*); and
- z) Resistance to rusting (*see 25*)

26.1.1 Number of specimen and the sequence of testing shall be as given in Table 1.

26.1.1.1 Criteria for acceptance

The specimens subjected to the type tests shall pass the test for providing conformity with the requirements of this standard. If one or more failures occurs, the testing authority at its discretion, may call for twice the original number of specimens and subject them to all tests or to those in which the failure occurred. No single failure shall be permitted on the repeat test(s).

26.2 Acceptance Tests

The following shall constitute acceptance tests:

- a) Marking (*see 8*);
- b) Resistance to harmful ingress of water and to humidity (*see 15.2* and *15.3*);
- c) Insulation resistance and electric strength (*see 16*);
- d) Temperature rise (*see 17*); and
- e) Mechanical strength (*see 20*).

26.2.1 A recommended sampling procedure for acceptance test is specified in Annex B.

26.3 Routine Tests

The following shall constitute routine tests:

- a) Marking (*see 8*).

ANNEX A

(Normative)

ADDITIONAL REQUIREMENTS FOR SWITCHES HAVING FACILITIES FOR THE OUTLET AND RETENTION OF FLEXIBLE CABLES

The purpose of Annex A is to bring additional explanations and requirements to the normative text of this document. Therefore the numbering of clauses, sub-clauses, figures and tables follows that of the main body of the text. Only the relevant clauses, sub-clauses, tables and figures are cited.

Any additional tables and figures are numbered starting from A.

3 TERMS AND DEFINITIONS

Add the following definition:

3.23 Flexible Cable Outlet Switch

Switch having provision for a flexible cable outlet

7 CLASSIFICATION

Add the following definition:

7.10 According to the Presence of a Flexible Cable Outlet

- a) without flexible cable outlet; and
- b) with flexible cable outlet.

10 PROTECTION AGAINST ELECTRIC SHOCK

10.1 Prevention of Access to Live Parts

Add the following text at the end of the fourth paragraph:

For flexible cable outlet switches the test is carried out without the flexible cable fitted.

12 TERMINALS

12.2.5 Add the following text after the fifth paragraph:

For flexible cable outlet switches, the test is repeated with flexible cables of the appropriate size (*see 13.15*) following the same procedure.

13 CONSTRUCTIONAL REQUIREMENTS

Add the following sub-clause:

Flexible cable outlet switches shall be so designed that an appropriate flexible cable, complying with IEC 60245-4, code designation 60245 IEC 66 or IEC 60227-5, code designation 60227 IEC 53, or as specified by the manufacturer, may enter the switch through a suitable hole, groove or gland. The entry shall accept the maximum dimensions (outer sheath) of

the appropriate flexible cable, having conductors of the cross-sectional area specified in Table A-1, according to the current rating of the switch, but with a minimum of 1.5 mm² and the entry shall be so shaped as to prevent damage to the flexible cable.

A cable anchorage for the flexible cable shall be provided such that the conductors are relieved from strain, including twisting, where they are connected to the terminals or terminations.

The cable anchorage shall contain the sheath and shall be either of insulating material or, if of metal, shall be provided with an insulating lining fixed to the metal parts.

Cable anchorages shall anchor the flexible cable securely to the switch.

The design shall ensure that:

- a) the cable anchorage cannot be released from the outside; and
- b) clamping the cable does not require the use of a special purpose tool.

Screws which are used when clamping the flexible cable shall not serve to fix any other component unless the switch is rendered manifestly incomplete if the component is omitted or is replaced in an incorrect position, or the component intended to be fixed cannot be removed without further use of a tool.

Compliance is checked by inspection and by the following tests:

Switches are fitted with a flexible cable complying with IEC 60227-5, code designation 60227 IEC 53, having a nominal conductor cross-sectional area of 1.5 mm² and the number of cores corresponding to the number of poles of the switch.

NOTE — For the purposes of this test, earth is considered as a pole.

The conductors are introduced into the terminals and the terminal screws tightened just sufficiently to prevent the conductors easily changing their position. The cable anchorage is used in the normal way, the clamping screws, if any, being tightened to a torque of two-thirds of that given in Table 5.

After this preparation, it shall not be possible to push the flexible cable into the switch to such an extent as to impair safety or so that the cable anchorage is loosened.

The flexible cable is then subjected 25 times to a pull of 30 N. The pulls are applied without jerks in the most

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Table A-1 Limits of External Dimensions of Flexible Cables

(Clause A-5)

Rated Current A	Cross-sectional Area of Conductors mm ²	Number of Conductors	Limits of External Dimensions of Flexible Cables	
			Minimum mm	Maximum mm
6	0.75 up to and including 1.5			
		2	5.7 or 3.7 × 6.0	11.0
		3		11.9
		4		13.1
		5		14.4
10 and 13	1 up to and including 2.5	2	5.9 or 3.9 × 6.2	13.1
		3		14.0
		4		15.5
		5		17
16	1.5 up to and including 4	2	6.8	15.1
		3		16.2
		4		17.9
		5		19.9
20 to 25	2.5 up to and including 6	2	8.4	16.8
		3		18.0
		4		20.0
		5		22.2

NOTE — The limits of the external diameter of cables specified in this table are based on type 60277 IEC 53 according to IEC 60227-5 and type 60245 IEC 66 according to IEC 60245-4 and are given for information.

unfavorable direction, each time for 1 s. immediately afterwards, the flexible cable is subjected for 1 min to a torque of 0.15 Nm as near as practicable to the cable entry.

The above test is then repeated, the switch being fitted with the appropriate largest diameter flexible cable complying with IEC 60245-4, code designation 60245 IEC 66. The pull is increased to 60 N and the torque is increased to 0.35 Nm.

After the test the flexible cable shall not have been displaced by more than 2 mm.

For the measurement of longitudinal displacement a mark is made on the flexible cable whilst it is subjected to a pull, at a distance of approximately 20 mm from the cable anchorage, before starting the test. After the test the displacement of the mark on the flexible cable in relation to the cable anchorage is measured whilst the flexible cable is again subjected to a pull.

An AC voltage of 2 000 V is applied for 1 min between the conductors and the cord anchorage.

During the test, the insulation of the flexible cable shall not be damaged. Breakdown or flashover is considered to indicate damage to the flexible cable.

ANNEX B

(Clause 26.2.1)

SAMPLING PROCEDURE

B-1 LOT

B-1.1 In any consignment, all samples of the same type, designation, rating and manufactured by the same factory under essentially similar conditions of production shall be grouped together to constitute a lot.

B-1.2 From each lot, a certain number of switches as specified in Table B-1 shall be selected at random and subjected to tests specified in 26.2.

B-2 CRITERION FOR CONFORMITY

B-2.1 In Table B-1, N_1 is the size of the first sample. If the number of failures found in this sample is less than or equal to C_1 , the lot shall be considered to be conforming to this standard and accepted. If the number of failures is greater than or equal to C_2 , the lot shall be rejected. If the number of failures is between C_1 and C_2 , further sample of N_2 pieces shall be taken and subjected to all tests.

B-2.1.1 If the number of failures in the two samples combined is less than C_2 , the lot shall be accepted, otherwise rejected.

Table B-1 Sampling Plan

(Clauses C-1.2 and C-2.1)

Lot Size	N_1	N_2	$(N_1 + N_2)$	C_1	C_2
(1)	(2)	(3)	(4)	(5)	(6)
51 to 100	10	20	30	0	3
101 to 200	13	26	39	0	5
201 to 300	20	40	60	1	5
301 to 500	25	50	75	1	6
501 to 800	35	70	105	2	7
801 to 1 300	50	100	150	3	10
1 300 and above	75	150	225	5	12

NOTE — The plan recommended in this table assures that lots with defectives 4 percent or less would be accepted most of the time, and lots with defectives 25 percent and above would be rejected most of the time. The exact consumers risk depends on the lot size and it would be minimum when the lot size is maximum.

ANNEX C

(Informative)

CIRCUIT DEVELOPMENT (19.3)

C-1 RATIONALE

Due to the phasing out of incandescent lamps and the replacement of these lamps by self-ballasted lamps, a joint forum between IEC technical committees 23B, 34A, 77A and 17B was formed to discuss the technical consequences of this change.

One of the main impacts for switches is the change in inrush currents when switching these energy efficient lamps in comparison to incandescent lamps.

The joint forum fixed the maximum values for one lamp. The source impedance Z_{mains} as shown in Table C-1 was added to be able to test the lamp. This will be covered by the lamp standards (for example IEC 60969).

The value of I_{peak} and I^2t shown in Table C-1 represent the highest value that can occur at the moment the contact mechanism closes.

Table C-1 Lamp

(Clause C-1)

P [W]	V_{mains} [V _{rms}]	I_{peak} [A]	I^2t [A ² s]	Z_{mains}
$P \leq 15$	120	60	0.5	$0.450 \Omega + 100 \mu\text{H}$
$P \leq 15$	230	20	0.08	$0.2 \Omega + 400 \mu\text{H}$
$15 < P \leq 25$	120	60	0.5	$0.450 \Omega + 100 \mu\text{H}$
$15 < P \leq 25$	230	35	0.15	$0.2 \Omega + 400 \mu\text{H}$

C-2 I_{PEAK} AND I^2t FOR NORMAL OPERATION TESTS

C-2.1 General

Where 19.2 in IEC 60669-1 is based on the principle to define an equivalent circuit which results in an I_{peak} and I^2t value, 19.3 in this document uses the values of I_{peak} and I^2t as the reference. This approach has been

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preferred, as within testing environments it is not always possible to reach the ideal parameters of Z_{mains} .

In this approach equivalent circuits can be adjusted to reach the required values within the tolerances given in this document.

C-2.2 Switching a Single Lamp

For the purpose of this test the lamp with a $P = 15 \text{ W}$ has been used. This load results in the higher I_{peak}/W ratio compared to the 25 W lamp and is used for C-2.3 when added to get higher loads.

For the calculation a prospective short circuit current of $3\,000 \text{ A}$ was used which was considered adequate.

The R and C values of the equivalent circuit of the lamps in Fig. C-1 and Fig. C-2 are calculated from Table C-1.

The Z_{mains} of these networks are:

- (120 V; 60 Hz; $I_{\text{pssc}} = 3\,000 \text{ A}$; $\cos \phi = 0.9$):
 $0.036 \, \Omega + 46.25 \, \mu\text{H} + \text{cables } 0.25 \, \Omega + 20 \, \mu\text{H} = 0.286 \, \Omega + 66.25 \, \mu\text{H}$
- (230 V; 50 Hz; $I_{\text{pssc}} = 3\,000 \text{ A}$; $\cos \phi = 0.9$):

$$0.069 \, \Omega + 106 \, \mu\text{H} + \text{cables } 0.25 \, \Omega + 20 \, \mu\text{H} = 0.319 \, \Omega + 126 \, \mu\text{H}$$

C-2.3 Switching Multiple Lamps

Multiple lamp loads, an example of which is shown in Fig. C-3, are obtained by combining single lamps with a power of 15 W .

The I_{peak} and I^2t values (see Fig. C.4) for multiple lamps are based on the following principles:

Up to 60 W (four lamps) the worst conditions apply:

15 W lamps which give the highest I_{peak} and I^2t value as given in Table C-1; or

the network impedance as given in C-2.2.

For higher power lamp loads:

for every three additional lamps 3 m of cables have been added ($75 \text{ m}\Omega + 6 \, \mu\text{H}$);

the bulk capacitor within each additional lamp uses the nominal value and not the worst case as given in Table C-1; or

the values have been interpolated to fit the nominal values of Table 22.

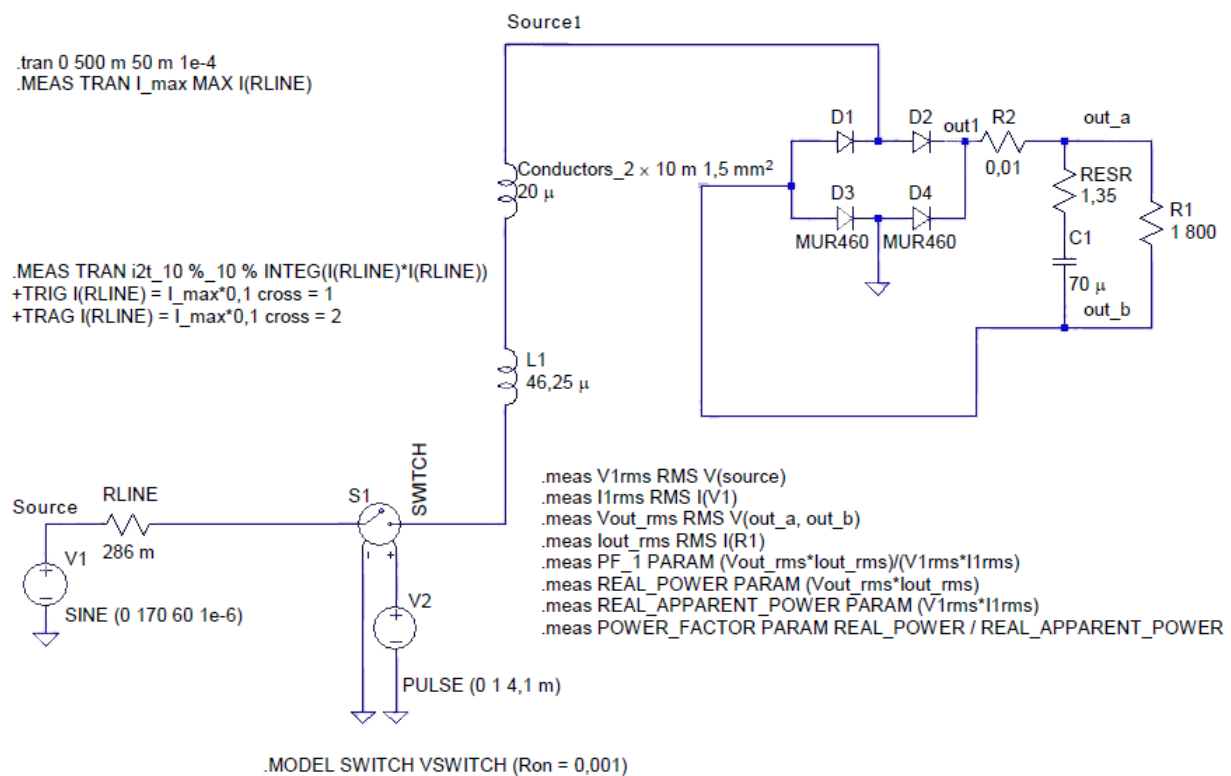
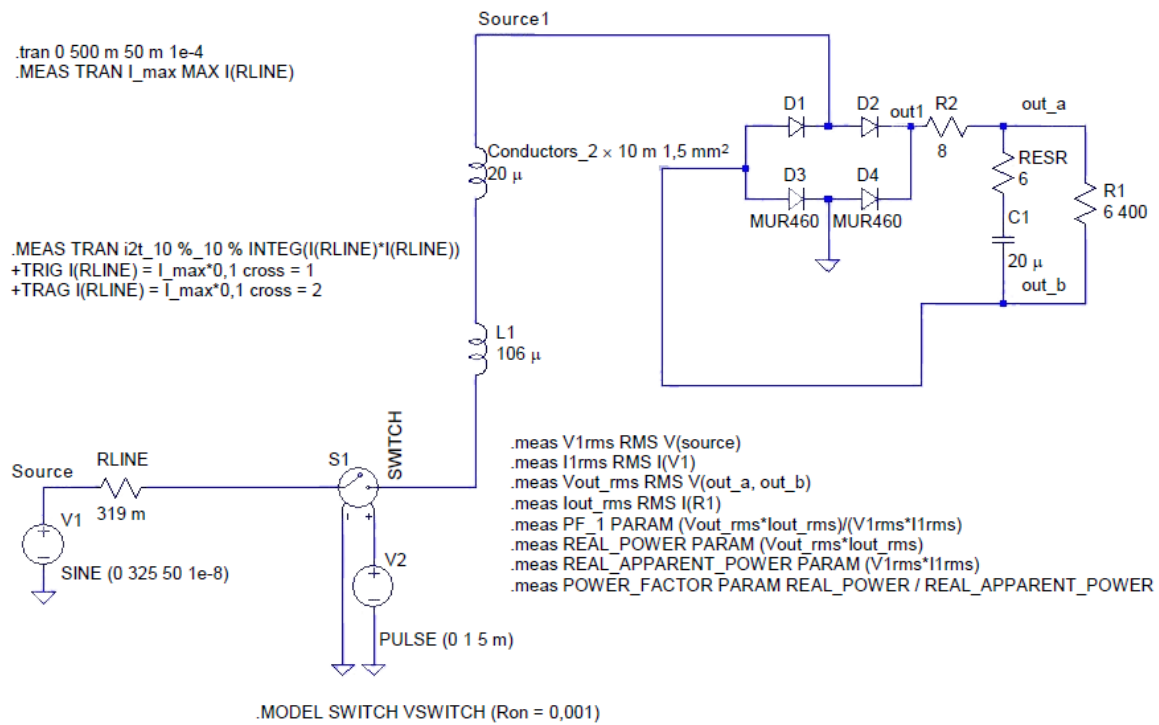


FIG. C-1 120 V 15 W (Lt SPICE MODEL)



NOTE — R1 in Fig. C-1 and C-2 is not the R1 of Fig. 16.

FIG. C-2 230 V 15 W (Lt SPICE MODEL)

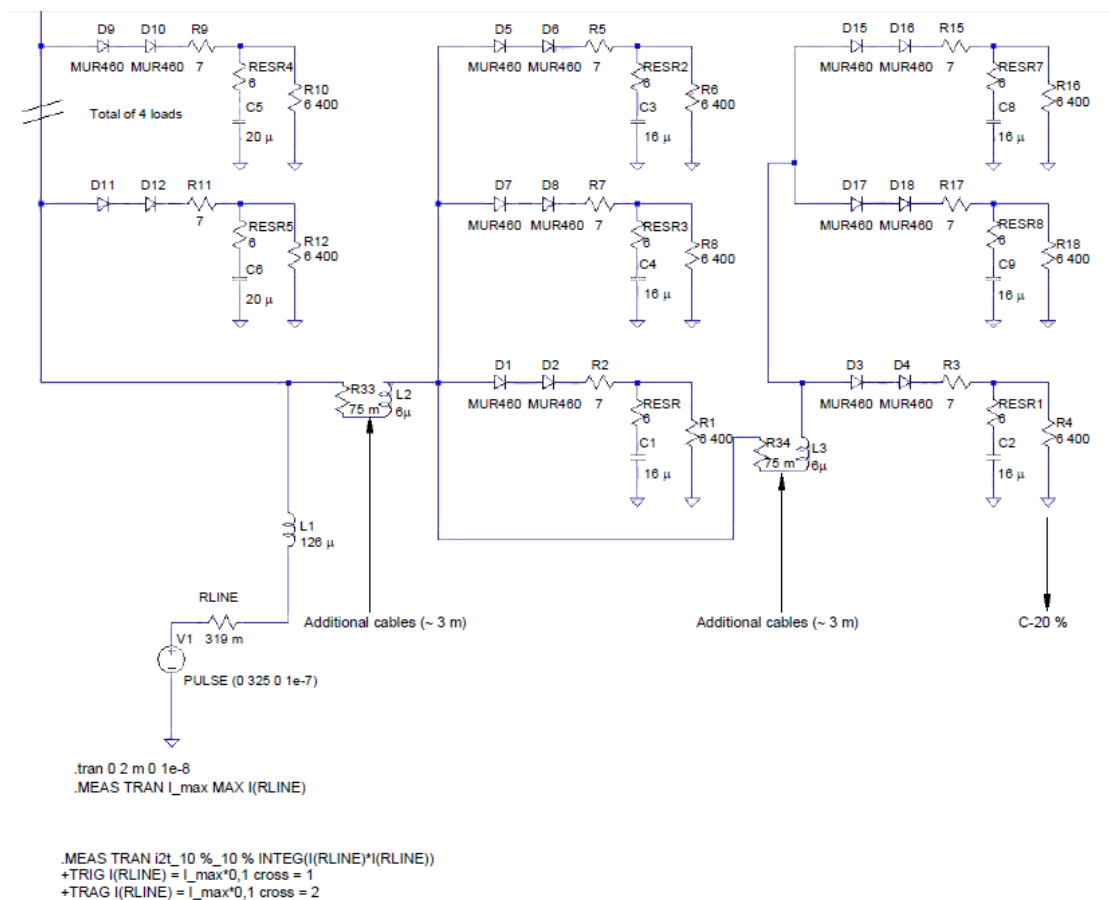


FIG. C-3 MODEL FOR MULTIPLE LAMP LOADS

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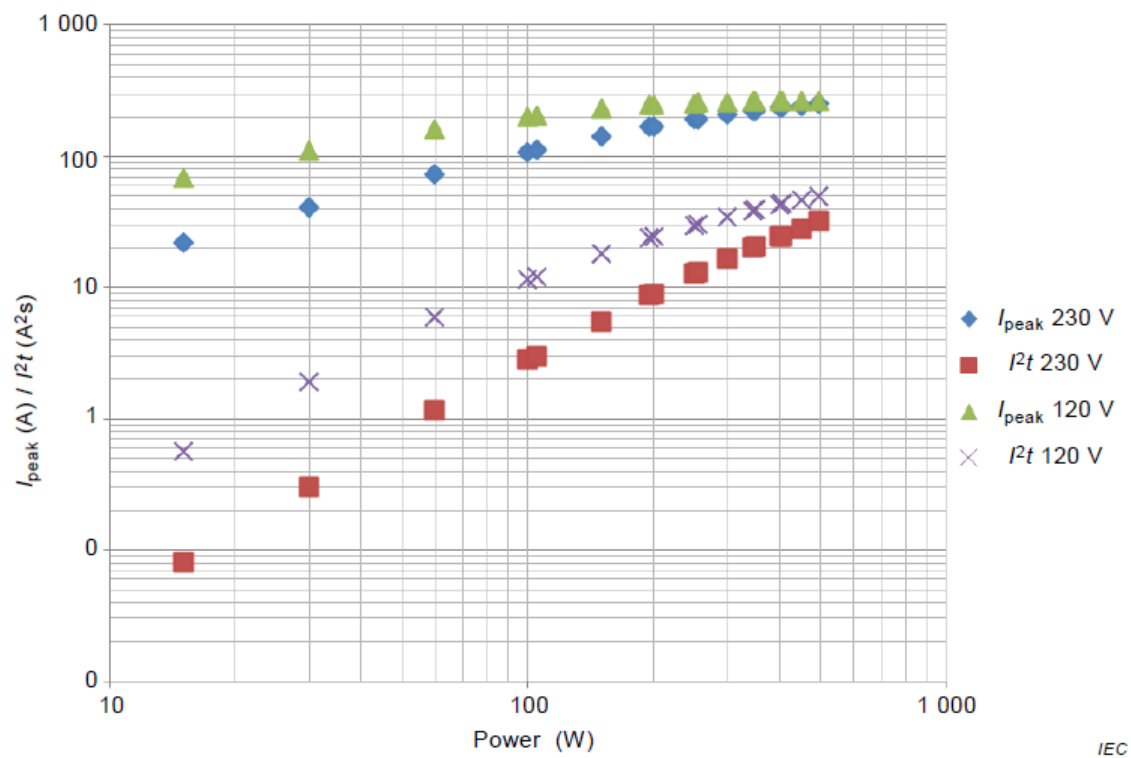


FIG. C-4 I_{peak} AND I^2t FOR MULTIPLE LAMP LOADS

ANNEX D

(Informative)

ADDITIONAL REQUIREMENTS FOR INSULATION-PIERCING TERMINALS

Annex D is provided for information to show the changes planned for the future to introduce requirements for insulation-piercing terminals into this standard.

The purpose of Annex D is to bring additional explanations and requirements to the normative text of this document. Therefore the numbering of clauses, sub-clauses, figures and tables follows that of the main body of the text. Only the relevant clauses, sub-clauses, tables and figures are cited. Any additional tables and figures are numbered starting from D.

Annex D is written in such a way that it will amend the main text, when the requirements for IPT are incorporated.

1 SCOPE

Replace the second paragraph by:

For switches provided with screwless terminals or insulation-piercing terminals, the rated current is limited to 16 A.

3 TERMS AND DEFINITIONS

Add the following new definitions:

3.24 Insulation Piercing Terminal

IPT connecting device for the connection and possible disconnection of one conductor or two or more conductors, the connection being made by piercing, boring through, cutting through, removing, displacing or making ineffective in some other manner the insulation of the conductor(s) without previous stripping.

NOTES

1 The removal of the sheath of the cable, if necessary, is not considered as a previous stripping.

2 Examples of insulation-piercing terminals are given in Fig. D-1.

3.24.1 reusable insulation-piercing terminal
insulation-piercing terminal that can be used more than once.

3.24.2 non-reusable insulation-piercing terminal
insulation-piercing terminal that can be used only once.

NOTE — The accessory can only be removed from the circuit by cutting the conductors or when removing the accessory, it is damaged in such a way that it cannot be reused.

3.25 Clamping Unit

Part(s) of a terminal necessary for the mechanical clamping and the electrical connection of the

conductor(s), including the parts which are necessary to ensure correct contact pressure.

5 GENERAL REMARKS ON TESTS

Add the following Table D.1 at the end of Clause 5:

Table D-1 Specimens Needed for Clause 12 for Insulation-piercing Terminals (IPTs)

Sub-clause	Reusable	Non-reusable
12.4.2	2 new terminals of each type of construction	2 new terminals of each type of construction
12.4.3	4 new terminals of each type of construction	NA
12.4.10	6 new terminals ^a of each type of construction previously submitted to the test of 15.1	6 or 12 new terminals of each type of construction previously submitted to the test of 15.1
12.4.11	12 new terminals of each type of construction	12 new terminals of each type of construction

^aThe terminals of specimens "ABC" shall be used. If more terminals are necessary, new specimens which have been previously submitted to the test of **15.1** may be used.

NOTE — If the IPT is declared as suitable for stripped and unstripped conductors, the specimens needed are twice the number of specimens shown in the table, except for **12.4.3**.

7 CLASSIFICATION

7.8 According to the Type of Terminal

Add the following new list items at the end:

- a) switches with IPTs for rigid conductors only; and
- b) switches with IPTs for rigid and flexible conductors.

Add the following new numbered paragraphs:

7.10 IPT Classification According to the Method of Making the Connection

- a) with a general purpose tool;
- b) with a special tool; only allowed for non-reusable IPT; and
- c) by hand.

7.11 IPT Classification According to Reusability

- a) reusable IPTs; and
- b) non-reusable IPTs.

Switches provided with non-reusable IPTs can only be used in trunking systems according to IEC 61084 (all parts).

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8 MARKING

8.1 General

Add the following new list item after m):

- c) Length of the conductor to be inserted into the IPT, if applicable.

Add the following new sub-clause:

8.9 Manufacturer Information

The following shall be indicated on the manufacturer's documentation for IPTs:

- a) The connection and disconnection procedure, if necessary;
- b) The method of connection according to 7.1.10, if necessary;
- c) For non-reusable IPTs, information that the product shall only be used in cable trunking systems and how the product shall be installed in order to have the possibility to replace it without impairing the safety of the installation;
- d) An indication that the switch is equipped with non-reusable IPTs, if applicable; and
- e) A clear information that the conductor shall not be stripped before connecting it to the terminal, unless the manufacturer has designed the product for this purpose.

12 TERMINALS

12.1 General

Replace the first paragraph by the following:

Switches shall be provided with either terminals having screw clamping, screwless terminals, or insulation-piercing terminals (IPTs).

Replace the third paragraph by the following:

The tests of 12.2.8, 12.3.9 (including the test of 12.3.10), and 12.4.9 (including the test of 12.4.10) shall be carried out after the test of 15.1.

Replace the fourth paragraph by the following:

Compliance is checked by inspection and by the tests of 12.2 or 12.3 or 12.4, as applicable.

Add the following new sub clause:

12.4 IPTs for External Copper Conductors

12.4.1 IPTs shall be of the type suitable for rigid copper conductors only or shall be of the type suitable for both rigid and flexible conductors.

12.4.2 IPTs shall be provided with clamping units which allow the proper connection of rigid or of rigid and flexible copper conductors having nominal cross-sectional areas as shown in Table D-2.

IPTs shall accept insulated conductors having the maximum outer diameter specified in Table D-2 or as specified by the manufacturer for other insulation types.

It shall be possible to connect an unprepared conductor into the IPTs.

The construction of IPTs shall be such that the contact pressure is transmitted by metallic parts. An IPT where the contact pressure is transmitted only through insulating material is not allowed. Compliance is checked for each type of terminal construction by inspection made by measurement and/or by fitting conductors of the smallest and largest cross-sectional areas of all kinds of conductor accepted by the terminal, and when applicable, with unstripped and stripped conductors.

Table D-2 Relationship between Rated Currents and Connectable Cross-Sectional Areas of Copper Conductors for Insulation Piercing Terminals

(Clause D-7.2.2)

Rated Current A	Nominal cross Sectional Areas mm ²	Conductors		Isolated conductor	
		Diameter of the Largest Rigid Conductor	Diameter of the Largest Flexible Conductor	Outer Diameter of the Largest Rigid Stranded Conductor 60227 IEC 01	Outer Diameter of the Largest Flexible Conductor 60227 IEC 08
		mm	mm	mm	mm
Up to and including 4 ^a	0.75 to 1.5	1.7	1.8	3.3	3.4
Above 4 and including 6	1 to 1.5	1.7	1.8	3.3	3.4
Above 6 and including 16 ^b	1.5 to 2.5	2.2	2.4	4.0	4.1
NOTE — These values are based on PVC isolated conductors according to IEC 60227 (all parts); for rubber isolated conductors according to IEC 60245 (all parts) other values can apply.					
^a For special purposes such as ELV applications, conductors from 0,5 mm ² up to 1 mm ² inclusive can be used.					
^b Each switch other than those of pattern numbers 3 and 03 shall be designed to permit the continuity of the supply line. This can be achieved by a terminal with separate independent clamping units for each conductor.					

12.4.3 Reusable IPTs shall be designed in such a way that no insulating material of the conductor remains inside the terminal, impairing their further use if a new conductor is connected and disconnected.

Compliance is checked by connecting and disconnecting five times the same conductor with the largest and smallest cross-sectional area of all kinds of conductor accepted by the terminal as given in Table D-2, rotating the conductor for each application in such a way that it is not connected twice at the same place.

If the insulating material of the conductor remains inside the switch, it shall be possible to withdraw it, and the terminal shall not be damaged in such a way as to impair its further use.

12.4.4 Parts of IPTs mainly intended for carrying current shall be of materials as specified in **22.5**.

Compliance is checked by inspection and, if necessary, by chemical analysis.

NOTE — Additional springs and the like are not considered as parts mainly intended to carry current.

12.4.5 IPTs shall be so designed that they clamp the specified conductors with sufficient contact pressure and without undue damage to the conductor.

The conductor shall be clamped reliably and between metal surfaces.

Compliance is checked by inspection and by the test of **12.4.10**.

12.4.6 The disconnection of a conductor from a reusable IPT shall require an operation other than a pull on the conductor only. It shall be necessary to take deliberate action to disconnect it by hand or with a suitable tool. Dis-connection with the opening intended for the connection of the conductor.

Compliance is checked by inspection and by the test of **12.4.10**.

12.4.7 IPTs which are intended to be used for the interconnection of two or more conductors shall be so designed that:

- a) Each conductor is clamped individually;

NOTE — The clamping of one of the conductors is independent of the clamping of the other conductor(s).

- b) During the connection or for reusable IPT disconnection, the conductors can be connected or disconnected either at the same time or separately; and

- c) Each conductor is introduced in a separate clamping unit (not necessarily in separate holes).

It shall be possible to clamp securely any number of conductors up to the maximum as designed.

Compliance is checked by inspection.

12.4.8 IPTs shall be designed so that adequate insertion of the conductor is obvious and over-insertion is prevented if further insertion is liable to reduce the creepage distances and/or clearances required in Table 9 or to influence the operation of the accessory.

Compliance is checked by inspection.

12.4.9 IPTs shall be properly fixed to the switch.

They shall not work loose when the conductors are connected or disconnected during installation.

Compliance is checked by inspection and by the test of **12.4.10**.

Covering with sealing compound without other means of locking is not sufficient. However, self-hardening resins may be used to fix terminals which are not subject to mechanical stress in normal use.

12.4.10 IPTs shall withstand the mechanical stresses occurring in normal use.

For reusable IPT, compliance is checked by the following tests which are performed on one terminal of each type of terminal construction of the three specimens.

Terminals suitable for rigid conductors only are checked with rigid solid and rigid stranded conductors in a new set of specimens, if existing.

NOTE — “If existing” means that the conductor is available in the market where the product is intended to be sold and installed.

Terminals suitable for rigid and flexible conductors are checked with rigid solid conductors and on a new set of specimens with flexible conductors.

For non-reusable IPTs for rigid conductors, compliance is checked by the following test which is performed on six specimens of each type of terminal: three with the smallest conductor cross-sectional area and three with the largest conductor cross-sectional area. The same number of specimens is also needed if the IPT is also suitable for flexible conductors. If the terminals are also designed to connect stripped conductors, then all tests need to be repeated on a new set of specimens with stripped conductors.

The terminal is placed in the test apparatus according to Fig. 12.

The terminal is fitted first with

- a) the maximum number of conductors of the largest nominal cross-sectional area of the same type;

then the test is repeated with

- b) the maximum number of conductors of the smallest nominal cross-sectional area of the same type,

according to Table D-2.

If the terminal is fitted with more than one conductor, the test is done on each of the conductors consecutively.

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The length of the test conductor shall be at least 75 mm longer than the height H specified in Table 6.

The test conductor is then connected in the clamping unit in the intended manner.

The end of the conductor shall be passed through an appropriate sized bushing in a platen positioned at a height H below the equipment as given in Table 6. The bushing shall be positioned, in a horizontal plane, such that its centre line describes a circle of 75 mm diameter, concentric with the centre of the clamping unit, in the horizontal plane; the platen is then rotated at a rate of (10 ± 2) r/min.

The distance between the mouth of the clamping unit and the upper surface of the bushing shall be within 15 mm of the height in Table 6. The bushing may be lubricated to prevent binding, twisting or rotation of the insulated conductor. Each conductor introduced in the IPT is subjected for 15 min to a circular motion with (10 ± 2) r/min using an apparatus, an example of which is shown in Fig. 12.

The conductor is subjected to a pull having a value shown in Table 7.

After each rotation test, the pulling force given in Table 7 shall be applied to the conductor under test. The force shall be applied in one smooth and continuous application, for 1 min, in the direction of the axis of the conductor.

During the test, the rigid solid conductor, or any wire of rigid stranded, or flexible conductor, shall neither slip out of the clamping unit nor break near the clamping unit.

After these tests, neither the terminals nor the clamping means shall have worked loose and the conductors shall show no deterioration impairing their further use.

In the case of the flexible conductor, the break of individual wires of the conductor shall not be taken into account.

In addition, no lack of the insulating material of the conductor shall be observed, in order to be sure that no piece of insulating material has remained inside the terminal.

12.4.11 IPTs shall withstand the electrical and thermal stresses occurring in normal use. Compliance is checked by the following test, which is carried out on 12 new IPTs which have not been used for any other test.

Terminals suitable for rigid conductors only are checked with rigid solid and rigid stranded conductors, if existing.

NOTE — “If existing” means that the conductor is available in the market where the product is intended to be sold and installed.

Terminals suitable for rigid and flexible conductors are checked with rigid solid conductors and with flexible conductors.

If the terminals are also designed to connect stripped conductors, then all tests need to be repeated on a new set of specimens with stripped conductors.

The test is made with new copper conductors having the minimum and maximum cross-sectional areas according to Table D-2.

Before the test on the reusable IPTs, a rigid solid conductor having the largest cross-sectional area is connected and disconnected once.

Conductors having the smallest cross-sectional area are connected, as in normal use, to each of three IPTs and conductors having the largest cross-sectional area are connected, as in normal use, to each of the other three IPTs.

To ensure normal cooling of terminals, the conductors connected to them shall have a length of at least 1 m.

The use of alternating current is preferable but direct current and extra low voltage are acceptable.

After this test an inspection by the naked eye, with normal or corrected vision, without additional magnification, shall show no changes obviously impairing further use, such as cracks, deformations or the like.

The whole test arrangement including the conductors is placed in a heating cabinet which is initially kept at a temperature of (20 ± 2) °C. Except during the cooling period the test current as defined in Table 10 is applied. The test current shall be applied for the initial 30 min of each cycle.

The IPTs are then subjected to 192 temperature cycles, each cycle having a duration of approximately 1 h, as follows.

The air temperature in the cabinet is raised in approximately 20 min to 40 °C.

It is maintained within ± 5 °C of this value for approximately 10 min. The IPTs are then allowed to cool down in approximately 20 min to a temperature of approximately 30°C, forced cooling being allowed. They are kept at this temperature for approximately 10 min and, if necessary, for measuring the voltage drop, allowed to cool down further to a temperature of (20 ± 2) °C (*see* Fig. D-3).

During the ageing test, the voltage-drop measurement is made in the ambient cool condition to ensure stability.

The voltage drop in the IPTs is measured after the completion of the 24th and 192nd cycles and recorded. The maximum allowable voltage drop of each clamping unit, measured with the current as specified in

Table D-3, shall not exceed the smaller of the two following values:

- either 22.5 mV; or
- 1.5 times the value measured after the 24th cycle.

The measuring points shall be as close as possible to the clamping unit of the IPT. If this is not possible, the measured value shall be reduced by the value of the voltage drop in the conductor between the two measuring points. An example of test-points is shown in Fig. D-2.

The temperature in the heating cabinet shall be measured at a distance of at least 50 mm from the specimens.

12.4.12 A non-reusable IPT shall be so designed and constructed that it is not possible to disconnect the product without destroying it or cutting the conductors. The damage shall be obvious.

Compliance is checked by inspection.

NOTE — An IPT is considered to be permanently destroyed when, for its re-installation, new parts or materials other than the original ones have to be used.

12.4.13 If an IPT uses screws for wire connection, the following test shall be performed before each test of 12.4:

Screws of IPTs are tightened and loosened 5 times by means of an appropriate tool applying the torque as stated in the corresponding column of Table 3. A new conductor end is used each time the screw is loosened and subsequently re-tightened.

Higher values of torque may be used if so stated by the manufacturer of the IPT, when the relevant information is provided.

During the test an IPT shall not be damaged so as to impair its further use, for example, the breakage of screws or damage to the head slots, threads, washers, or stirrups.

The shape of the blade of the test screwdriver shall suit the head of the screw to be tested.

The screws shall be tightened in one smooth and continuous application of the torque.

12.4.14 Screws for making contact-pressure shall not serve to fix any other component, although they may hold the IPT in place or prevent it from turning. Screws shall not be of metal which is soft or liable to creep.

NOTE — The use of aluminium alloy screws in aluminium alloy bodied IPTs requires additional tests, according to IEC 61545.

Compliance is checked by inspection.

Table D-3 Test Current for the Verification of Electrical and Thermal Stresses in Normal Use of Insulation-piercing Terminals
(Clause D-7.2.11)

Rated Current A	Test Current A	Minimum Cross-sectional Area of the Conductor mm ²	Test Current A	Maximum Cross-sectional Area of the Conductor mm ²
Up to and including 4	9	0.75	17.5	1.5
Above 4 and including 6	13.5	1	17.5	1.5
Above 6 and including 16	17.5	1.5	24	2.4

NOTE — For switches having rated currents other than the preferred ones, the test current is determined by interpolation between the next lower and higher preferred rated currents, and the cross-sectional area of the conductor is chosen equal to the one specified for the next higher preferred rated current.

IPT transmitting contact pressure via metal parts

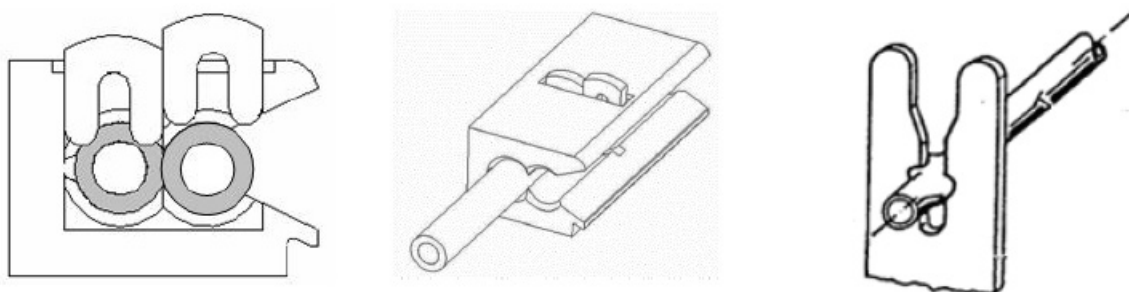
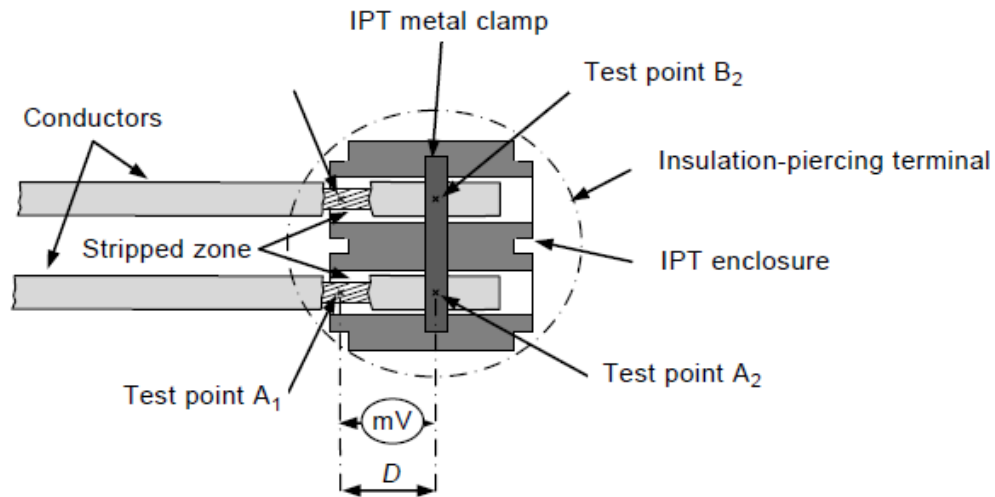


FIG. D-1 EXAMPLE OF INSULATION-PIERCING TERMINALS

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Distance D between two test points shall be as short as possible to avoid including the voltage drop of the conductor when measuring the IPT's voltage drop.

FIG. D-2 EXAMPLE OF TEST-POINTS

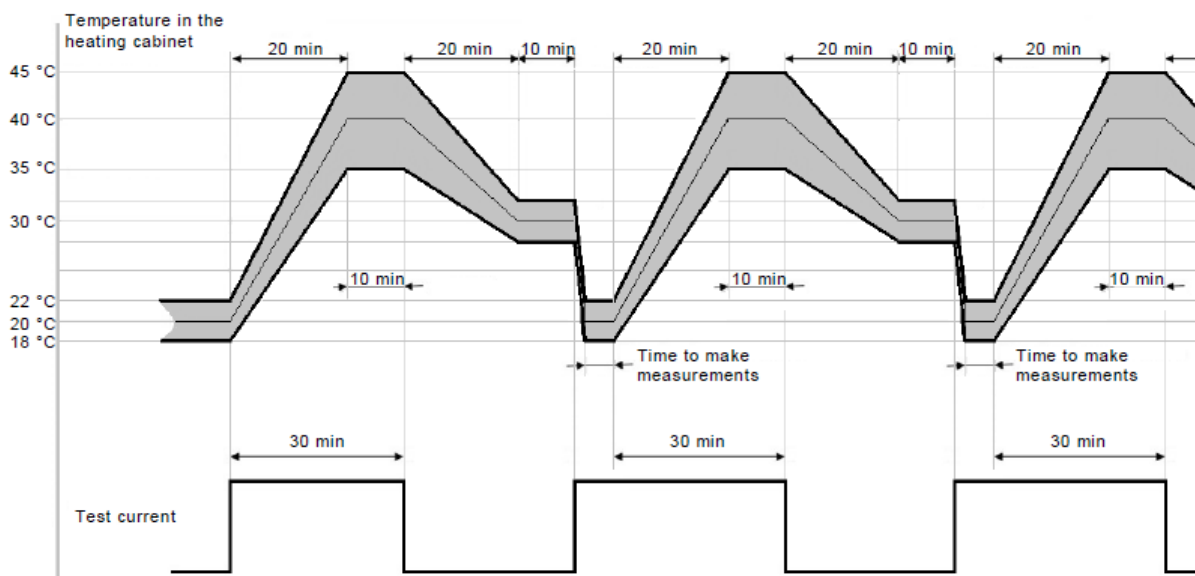


FIG. D-3 TEMPERATURE CYCLE FOR THE VOLTAGE DROP TEST OF 12.4.11

ANNEX E

(Foreword)

COMMITTEE COMPOSITION

Electrical Wiring Accessories Sectional Committee, ETD 14

<i>Organization</i>	<i>Representative(s)</i>
Central Public Works Department, New Delhi	SHRI PRABHAKAR SINGH (Chairman)
All India Plastic Manufacturers Association, Mumbai	SHRI MANOJ GADA
Anchor Electronics and Electricals Pvt Ltd, Mumbai	SHRI YOGESH SONAWANE SHRI JAI BHAGWAN (<i>Alternate I</i>) SHRI ANUSUYA PUROHIT (<i>Alternate II</i>)
Central Electricity Authority, New Delhi	SHRI VIVEK GOYAL SHRIMATI KAVITA JHA (<i>Alternate</i>)
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Delhi Electrical Accessories BIS Licensees Association, New Delhi	SHRI ADARSH KHERA SHRI MANGE RAM GUPTA (<i>Alternate I</i>) SHRI AJIT SINGH (<i>Alternate II</i>)
Ministry of Defence, New Delhi	SHRI Y. A. KAMBLI LT COL K. D. SINGH (<i>Alternate</i>)
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Honeywell Electrical Devices and Systems India Ltd Chennai	SHRI DILIP SINGH SHRI DHIRAJ KUMAR (<i>Alternate</i>)
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Public Works Dept Maharashtra	SHRI HEMANT M. SALI

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Sainico Industries, New Delhi	SHRI YASH SAINI
Schneider Electric India Pvt Ltd, Bangaluru	SHRI SURESH RAJA
Voluntary Organization in Interest of Consumer Education, New Delhi	SHRI K. C. CHAUDHARY
Western Electrical Accessories Manufacturers Association, Mumbai	SHRI KISHORE NANDU SHRI ASHOK MARU (<i>Alternate I</i>) SHRI RAJESH JAGGI (<i>Alternate II</i>)
BIS Directorate General	Ms PRITI BHATNAGAR, SCIENTIST 'F'/SENIOR DIRECTOR AND HEAD (ELECTROTECHNICAL) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]
<i>Member Secretary</i>	
Ms NEHA AGARWAL SCIENTIST 'C'/DEPUTY DIRECTOR (ELECTROTECHNICAL), BIS	

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